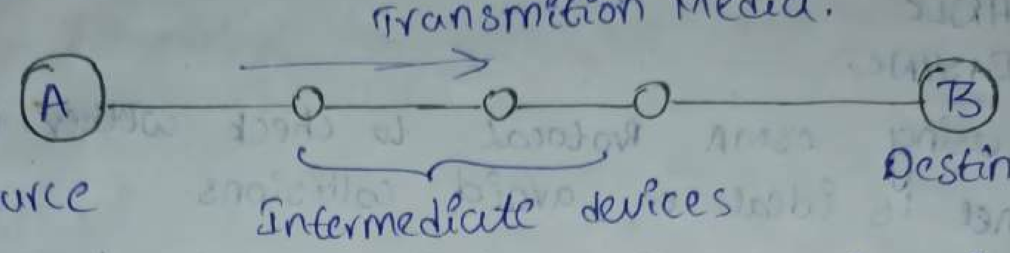
Introduction of Computer Networks

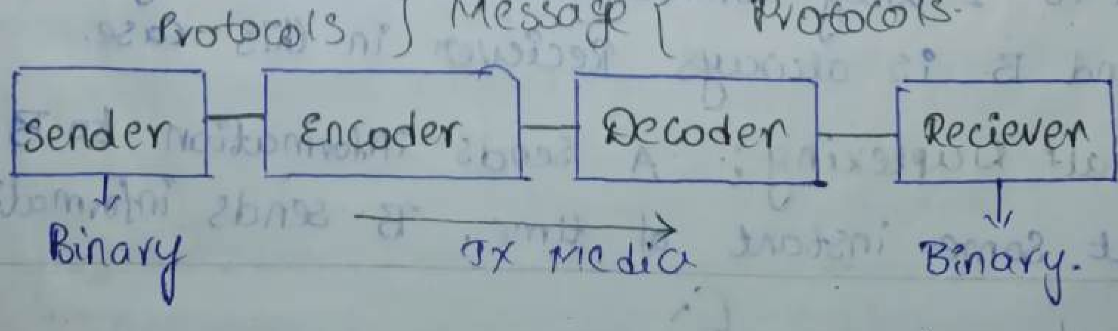
* Interconnection of one or more nodes is known as Network.
* Computer Network is depends on Data communication.



* Tx media is guided line to transfer data from one place to another place. It is of two types.
  1. wireless Tx line :-
     Ex:- Microwaves, Infrared waves.
  2. wired Tx line :-
     Ex:- Parallel cables, Coaxial cables,...
* By following Protocols (set of rules) we avoid errors, loss of data and delays.
* The physical state of the network is known as physical Topology.

Components of Computer Network :-

1. Sender
2. Reciever
3. Tx Media
4. Data/message.
5. Protocol.



Protocols :-

→ What it is communicated.

→ How it is communicated.

→ when it is communicated.

* There are two types of Protocols

1. HDLC
2. BISYNC.

* By using CSMA Protocal to check wether the channel is ideal or avoid collisions.

* Based on protocols, we define computer type.

1. size of data.
2. Timing.
3. formation & Encapsulation.
4. Delivary options.

Ex:- frame 1

D_A	S_A	1	Message

* There are three types of delivery options.

1. Unicast → one to one
2. Multi cast → one to many.
3. Broad cast → one to All.

* There are three types of Data flow.

1. Simplex.
2. Half Duplex.
3. full Duplex.

[Radio]

1. Simplex :- A transfer data & B recieves the data It is 1 way Communication. A is always sender and B is always Reciever in this case.

2. Half Duplexing :- A sends information to B at some instant of time. B sends information

to A at some instance of time.

→ In one instant of time, A is sender and B is reciever.

→ In another instant of time, A is reciever and B is sender

3. Full Duplex :- In this case, Both A and B sends information at the same instant of time. Acknowledge is used.

ACK	MSg

* Layers :-

* There are 7 layears for both Reciever and Transmitt

* 7 layers all together is known as Refrance models

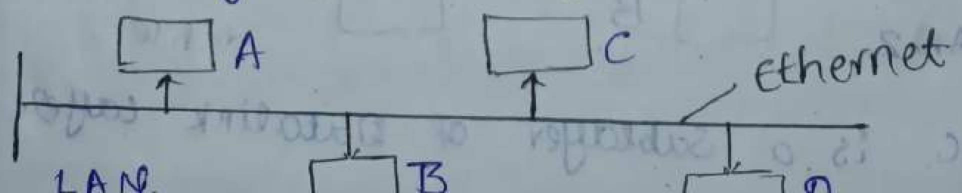
1. Application layer.
2. Presentation Layer
3. Session layer
4. Transport Layer.
5. Network Layer.
6. Datalink layer
7. Physical Layer.

Addressing :-

1. Physical addressing / MAC addressing.
2. Logical addressing / IP address.
3. Port addressing
4. specific addressing.

1. Physical addressing :-

Identification of Individual node in the network is called physical addressing or MAC address.

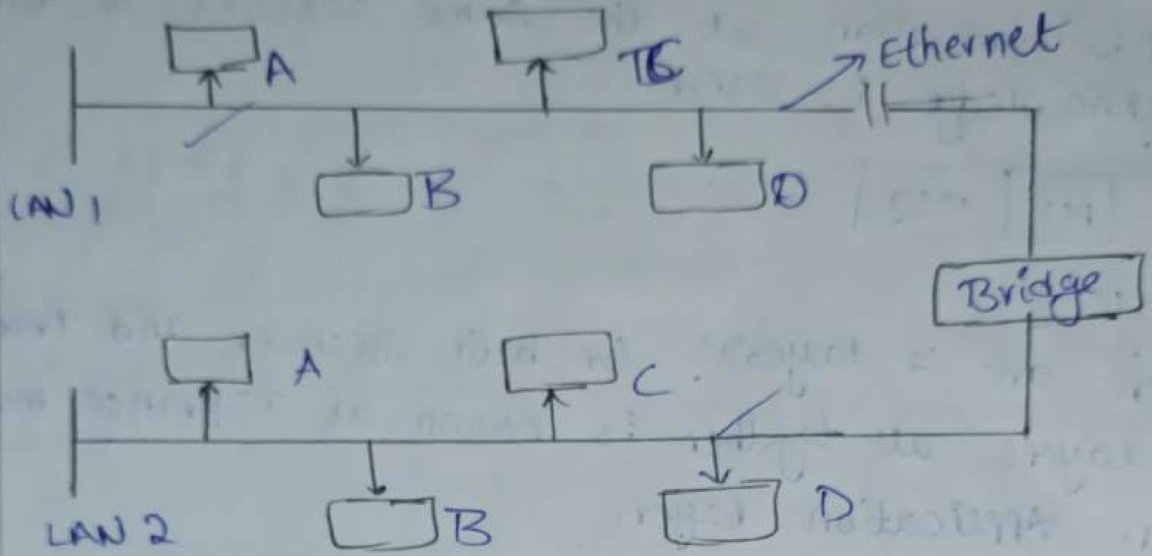


* Logical addressing:-

Identification of network is called logical addressing / IP address - it is also known as Network Address.

S D

LAN1 | A LAN2 | D.

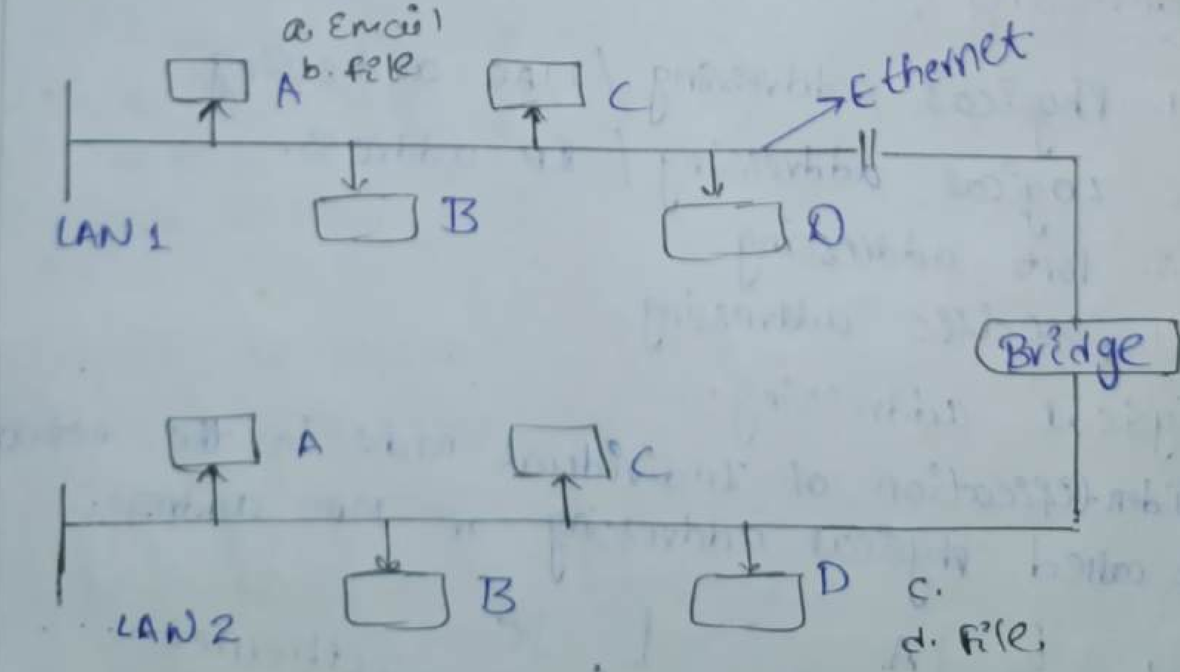


* Port addressing:-

Identification of Task | Process in the node is called Port address.

S D

|b| LAN1| A_d LAN2 | D.



* MAC is a Sublayer of Datalink layer.

Based on area of coverage, networks are classified into three categories.

1. Local Area Network [LAN]
2. Metropolitan Area Network [MAN]
3. wide Area Network [WAN]
4. Personal Area Network [PAN].

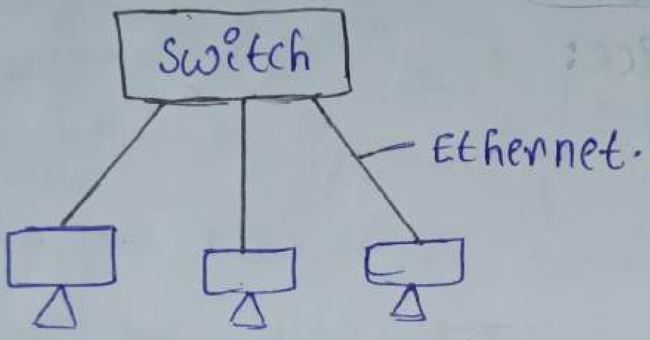
1. Local Area Network :

Computer Network which interconnects Computer within limited area i.e within buildings, offices.

Connecting Device:

1. Ethernet.
2 switch.

Example : wired LAN



Application:

→ Resource utilization (Printer, Scanner)

→ By using Interface application, if one file is present in Node-1, we share that file to all nodes.

wireless LAN.

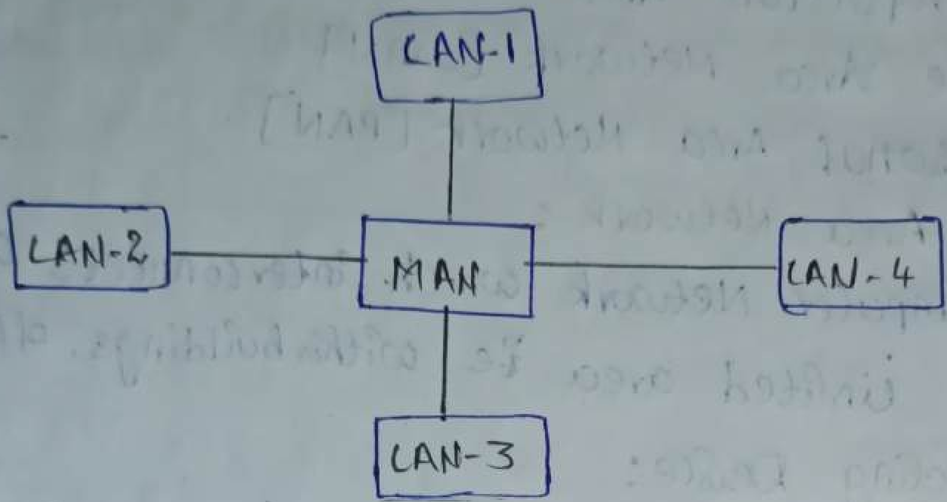
→ WIFI.

There is no physical connection between nodes but they communicate with each other.

2. Metropolitan Area Network:

Computer Network which interconnects within a geographical area i.e within cities.

Example: wired MAN.



Applications:

1. Cable
2. Supermarket.

Connecting Device:

1. Switches.
2. HUB
3. Bridges.

Wireless MAN:

WI-MAX

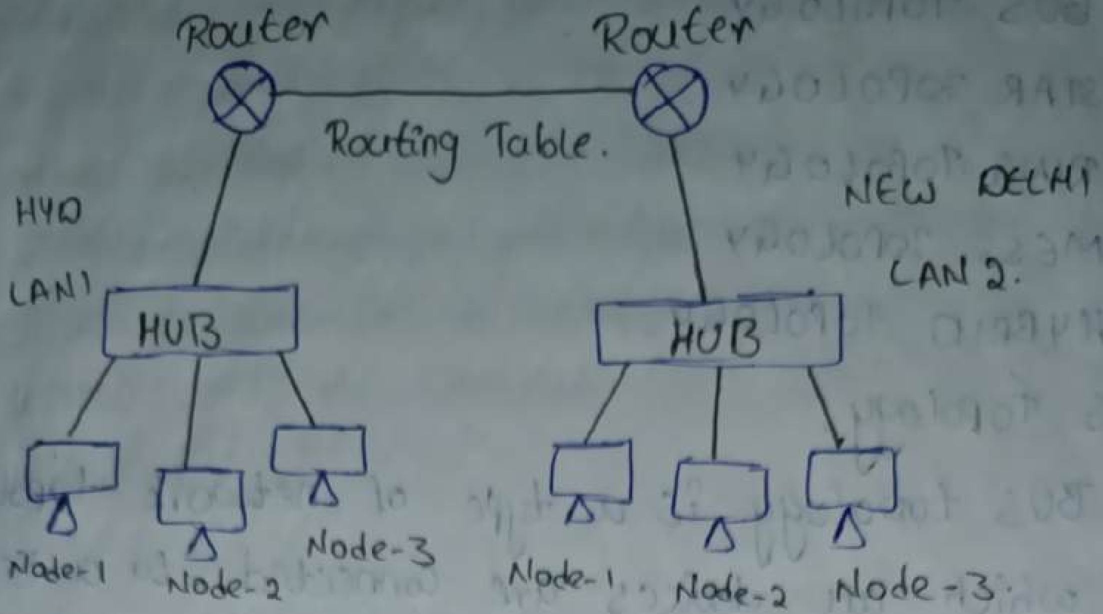
→ It uses frequency hopping spects spectrum technology.

→ frequency hopping = Jumps from 1 frequency range to another frequency range.

3. Wide Area Network:

Computer Networks which interconnects the nodes with in country large geographical area.

Example : wired WAN.



* wifi are connected through Router through ISP.
* It is also known as Network of Network or Internet.

connecting Devices :

Router.

wireless WAN :

3G, 4G, 5G.

4. Personal Area Network :

Computer Network which interconnects the nodes within a small area i.e within room or within 10 meters to 100 meters.

* Point - point connection.

wireless PAN :

1. Bluetooth
2. Zigbee.

* Network Topology :

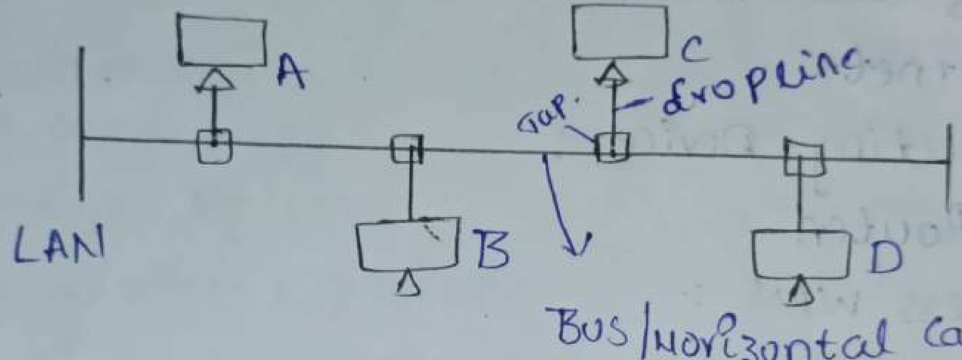
Physical state of network is called physical Topology.

There are 4- types-

1. BUS TOPOLOGY
2. STAR TOPOLOGY
3. RING TOPOLOGY
4. MESS TOPOLOGY
5. HYBRID TOPOLOGY.

1. BUS Topology:

* BUS topology is a type of network topology in which all devices are connected to a single cable called a BUS Topology.
* It is also known as Horizontal Topology



* Flow of data in network is called as logical Topology.
* Data Flows in unidirectional in Bus Topology.
* Based on the destination address, it collects the data.
* Destination address will check at every node.

Advantages:-

1. Easy to install.
2. Low cost.
3. Extension of Network is easy. It is possible by using repeater by adding cable.
4. Scalability.

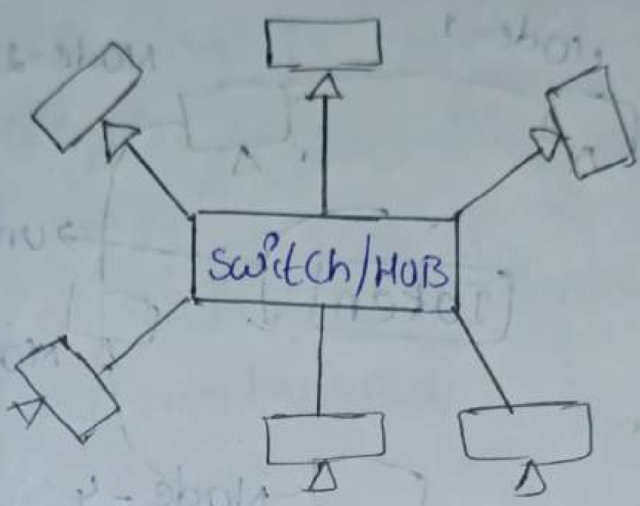
Disadvantages :-

1. Failure of cable/bus affects entire network.

2. No security.

2. STAR Topology :-

Star Topology is a type of network topology in which all the devices or nodes are physically connected to a central node such as a router, switch or HUB.



* Switch/HUB acts like a centralised device through this centeralised device, data travels from source to destination.

Advantanges :-

1. Each and every node have seperate channel due to that chances of collisions are reduced

2. The failure of any one node does not affects entire Network.

Disadvantages :-

1. Cost affective. Because each node have seperate cable.

2. Failure of switch/HUB which affects the entire network.

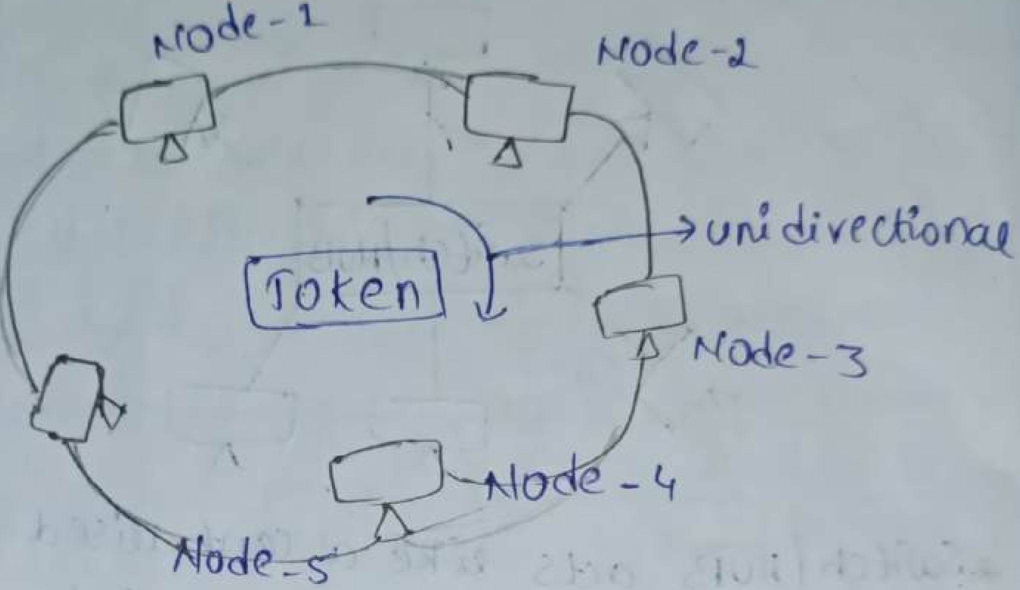
3. If load increases, performance is low.

3. RING Topology:-

The Bus Topology is connected in a ring or circular manner. It is also known as Token RING Topology.

* The data ring topology travels in unidirection with the help of token.

* The data travels through all the nodes in the network.



Advantages:-

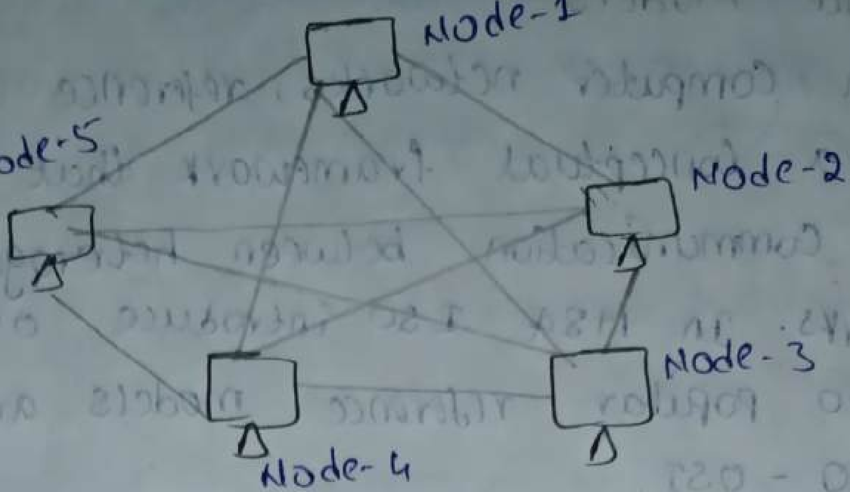
1. Collisions are reduced compared to Bus Topology.
2. Low cost because sing cable is used.
3. Easy to install.

Disadvantages:-

1. The failure of any one node in the Network will affect the entire network.
2. No security for data.

4. Mess Topology:

Mess Topology is a network configuration where devices are inter connected in a decentralised manner.



Advantages :-

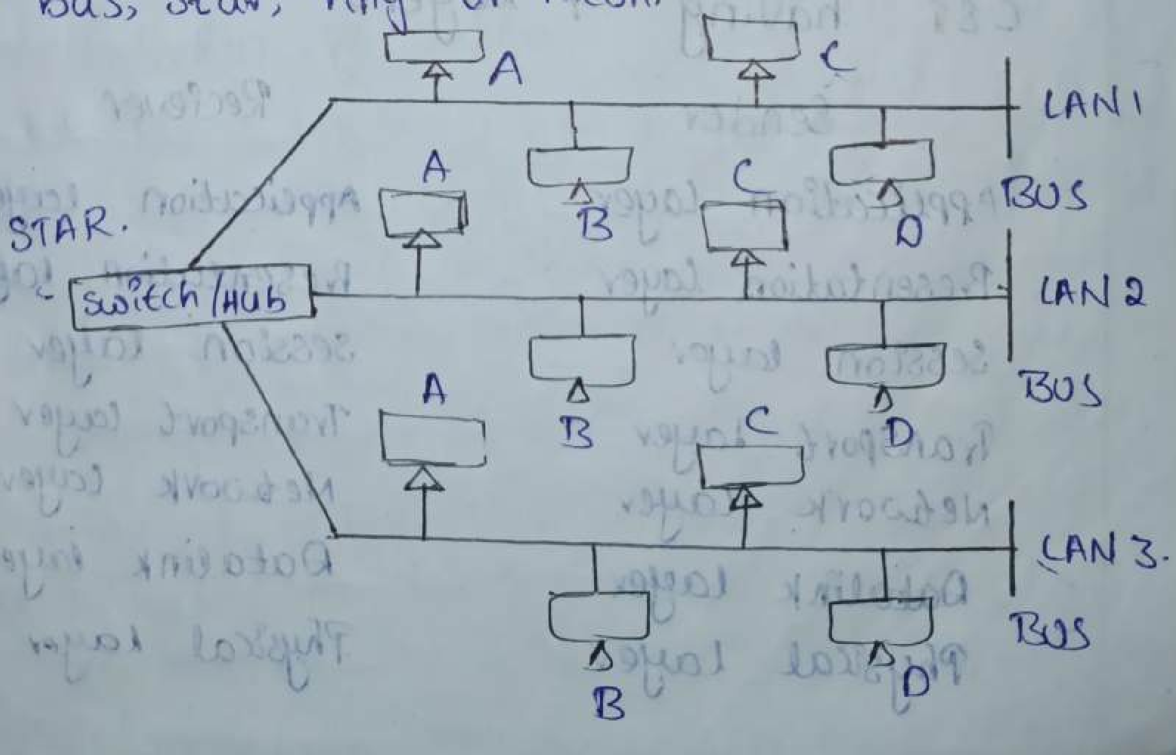
1. Trouble shooting is easy [Identifying failures]
2. Recabicity. Recapability.

Disadvantages :-

1. This is not suitable for broad cast because duplications are happened.
2. It is not support for large networks.
3. cost affect because more cables are required.

5. HYBRID Topology :-

It is a network architecture that combines two or more different network topologies such as bus, Star, ring or mesh.



Reference Models:

In computer networks, reference models gives a conceptual framework that standard-ises communication between heterogeneous networks. In 1984 ISO introduce OSI.

The two popular reference models are:

1. ISO - OST
2. TCP / IP.

1. ISO - OSI :

International standard organisation defines open system Interconnection. where open stands to say non- Proprietary. It is a 7-layer architecture with each layer having specific functionality to perform. All these 7 layers work collaboratively to transmit the data from one person to another across the globe. The OSI reference model was developed by ISO. for standardization in the 1984.

OSI having 7 Layers

Sender	Reciever
Application layer	Application layer
Presentation layer	Presentation layer
Session layer	Session layer
Transport layer	Transport layer
Network layer	Network layer
Datalink layer	Datalink layer
Physical layer	Physical layer.

Its a bottom to top approach

8/7
1. Physical Layer :- It is responsible for transmission of data in the form of bits that is '0' and '1' from source node to destination node. (or) It is responsible for placing the data into physical media.

a. Physical characteristic of transmission media.
b. Representation of data. RPDSPTL
c. Data rate / Transmission rate.
d. Synchronization of bits.
e. Physical Topology.
f. Transmission mode..
g. Line configuration

a. Physical characteristic of transmission media:
It deals with the type of transmission media and mechanic characteristics of transmission media. It is identified the type of connection when we are using wire or wireless connection in transmission.

b. Representation of Data:
The conversion of '0' and '1' into signal at the sender side. and again the signal is converted into '0' and '1' at reciever side. This process is known as representation of data.
Encoding at sender side
Decoding at reciever side.

c. Data Rate / Transmission Rate :-
It defines the data layer. Both sender and reciever maintain same data rate. "No. of bits/sec"

d. Synchronization of bits:-

Both sender and reciever maintain same clock (or) it is simply speed matching mechanism.

e. Physical Topology:.

The physical state of the network is called physical topology.

→ BUS Topology,
→ RING Topology,
→ STAR Topology
→ Mesh Topology.
→ Hybrid Topology.

f. Transmission mode.

It defines flow of data in the network. The type of data flow. It depends on physical layer.

1. Simplex :- ~~Point to point / Peer to peer~~

A transfers the data and B recieves the data.

2. Half Duplex :- A sends information to B at some instant of time. B sends information to A at some instant of time.

3. Full Duplex :- In this case, A sends the information to B and B sends the information to A at same instant of time.

Acknowledge is used.

g. Line configuration:
It defines the type of connection.
→ Point to Point | peer to peer.
→ Point to Multiple connections.

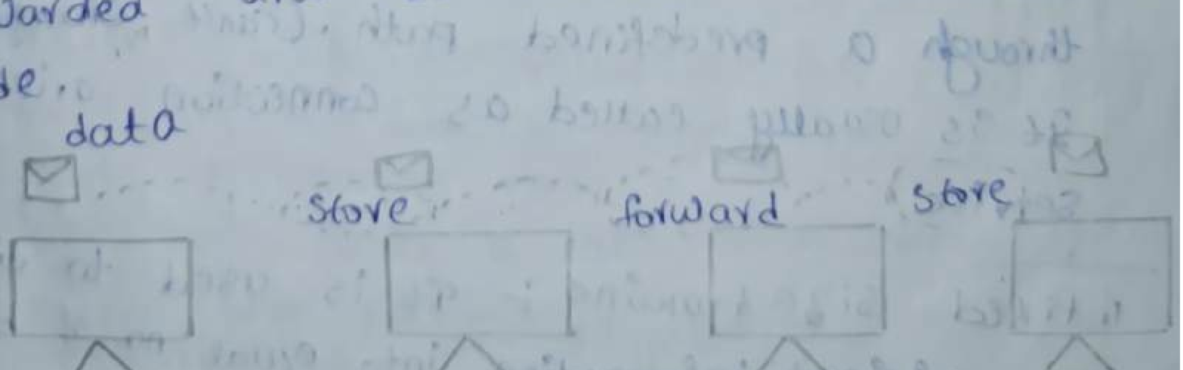
2. Datalink Layer:

19/7/24 Switching Techniques

switching is nothing but to find the optimum path.
They are classified into 3 types.
1. circuit switching.
2. Message switching.
3. Packet switching.

1. circuit switching:- circuit switching is a technique which defines the path before the transmission of data.

2. Message switching: It is simply define as store and forward mechanism. In this switching case, the data is to be transmitted is stores in each device in the path and forwarded until it reaches to the destination mode.



3. Packet switching:- In this packet switching, To transmit the data in a pre-defined path. In this switching technique the data is divided

into multiple chunks based on the available information.

* chunks → group of bits of data.

* The data chunks are divided by group of sequence numbers [identification] along with the sequence number these are transmitted in network path.

* At the reciever end the data chunks are rearranged with the help of (identification) sequence number.

* The packet switching is classified into 2 types.
  1. Datagram switching.
  2. Virtual circuit switching.

1. Datagram switching :- In datagram switching the information or data is to be transmitted based on the availability of media.
   It is usually called as connectionless service why because it chooses the path during the transmission.

2. Virtual circuit switching :- In this technique the data or information is transmitted through a predefined path. [Can't change.]
   It is usually called as connection oriented service.

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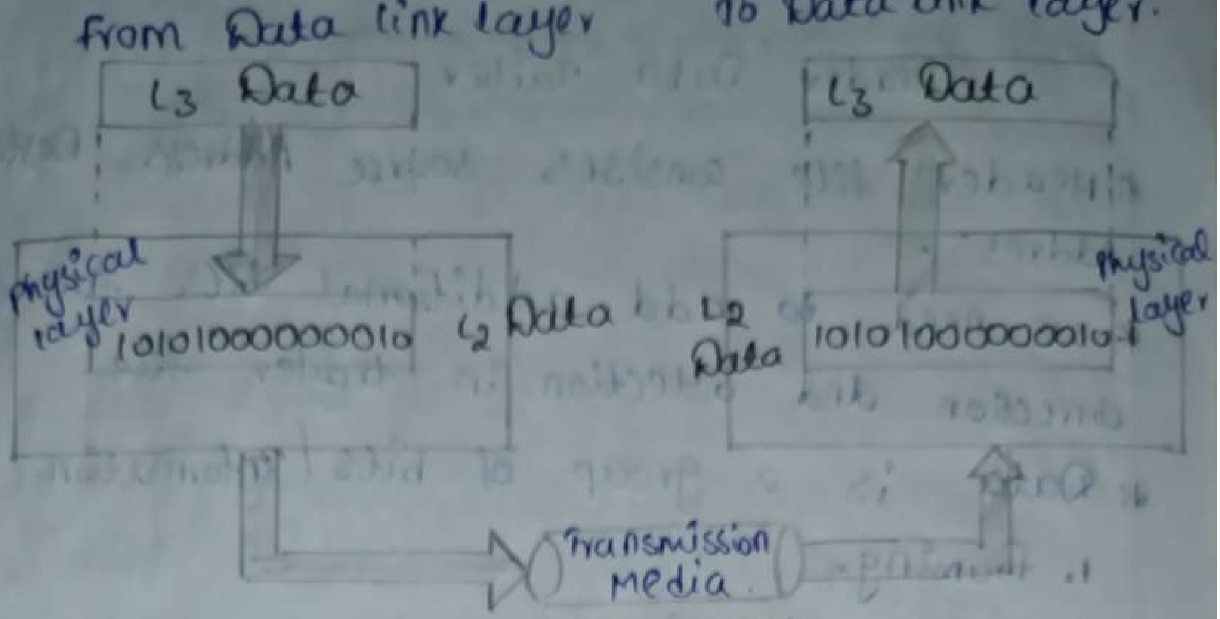
1. fixed size framing :- It is used to divide our original information into equal no. of bits.

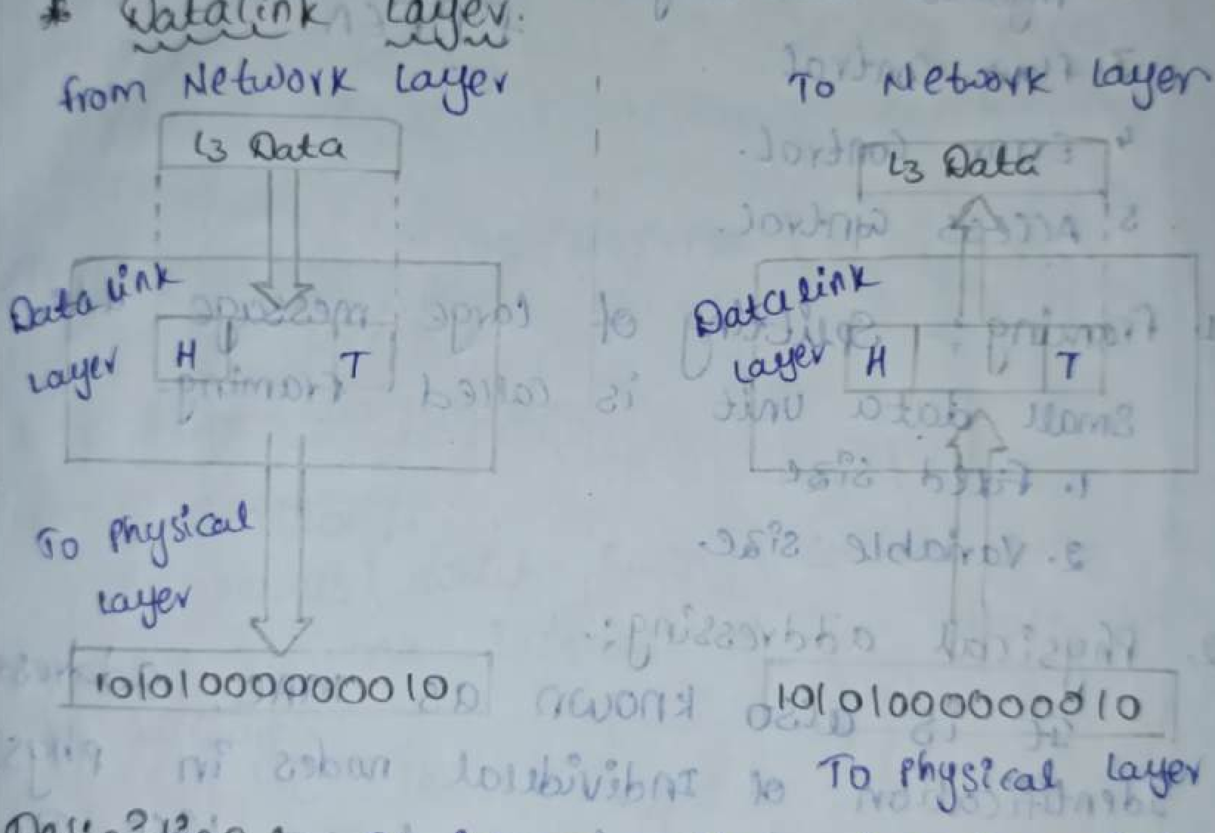
2. Variable size framing :- It is used to divide our original information into unequal no. of bits

Ex:- $\boxed{101|10|10}$ fixed framing $\boxed{10|11|1|01}$ variable framing.

20/7/24 Physical layer



* Datalink Layer.



* Definition :- It is responsible for transmission of data in the form of frames from source node to destination node.
* frame is a block of data divide our original information into parts.
* It is classified into 2 types.

1. fixed size framing.
2. variable size framing.

* frame is the combination of header, trailer, Data.

Header	Data	Tailer

* Header file consists source Address, Destination Address.
* we need to add additional bits for error correction and detection in trailer file.
* Data is a group of bits / information.

1. framing.
2. physical Addressing.
3. flow control.
4. Error control.
5. Access control.

FPFEA

1. Framing :- Splitting of large message into small data unit is called Framing.
   1. fixed size
   2. Variable size.

2. Physical addressing:-
   It is also known as MAC address. Identification of Individual nodes in physical network. It is a length of 48 bits. By using hexadecimal representation we address the data.
   Ex:- FD : A0 : 12 : EB : CA : 01 : AB : 22.
   NIC → Network Interface Card.

Datalink layer

LLC sublayer
MAC sublayer

→ 6 bits (Hexa decimal format)

It deals with MAC sublayer.

3. flow control:-
The continous transmission of data between the sender and reciever without any interrupts.
ARR : Automatic Repeat Request.

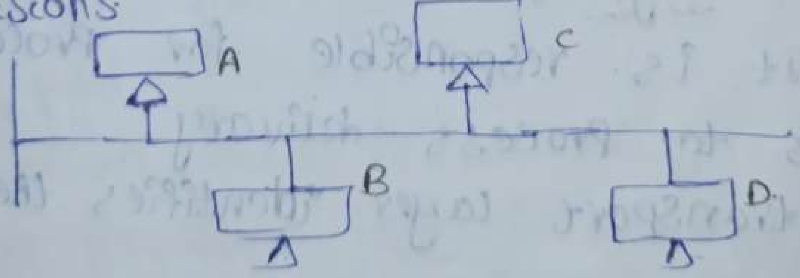
4. Error Control:-
The deviation from the original data

Head	I	Sn	D_A	M	Error	Trailer.

1. VRC → Vertical Redendancy Check.
2. LRC → Longitudnal Ridendancy check.
3. CRC → Cyclic Redendancy check
4. check sum →

they are two types of error.
1. Error detection.
2. Error Confirmation.

5. Access Control:-
The nodes which having control over the channel at one instant of time deals with the collisions



we can access by using time division multiplexing, frame division multiplexing and code division multiplexing.

3. Network Layer :-

It is responsible for transmission of data in the form of Packets from Source node to destination node.

* Packet → small data

1. Logical Addressing
2. Routing.

1. Logical Addressing :- It is nothing but Network address. It is also known as IP address. simply Identification of Network.

2. Routing :- Identification of path or optimal path from Source to destination. collection or router is called Subnet. Router is a connecting device (or) Intermediate device (or) Identify the Path

D.LA	S.L.A	SA	DA	Data	T.

DLC

N/W.

4. Transport Layer :-

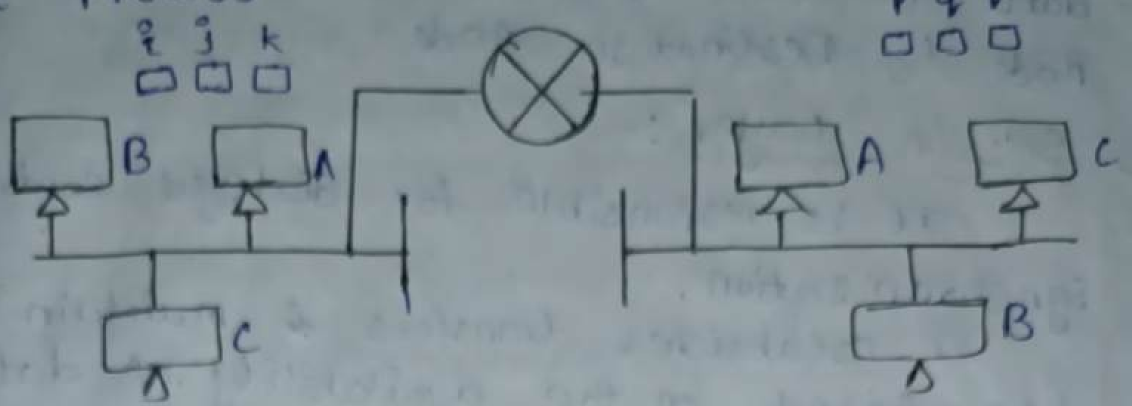
It is responsible for Process to Process to process delivary.

The transport layer identifies the individual tasks

1. Port Addressing.
2. Connection Control.
3. Segmentation and Reassembling.
4. Error Control.

PECS

1. Port Addressing :- Identification of individual tasks of a node the task is nothing but process.



I	PC	L_2	1/A	L_1	message	error.

2. connection control :- Transport layer itself identifies the connection oriented service and connection less service.

→ Connection oriented service :

There is a physical path between sender and reciever.

ones the connection is established it will not change.

→ Connection less oriented service:

There is no physical path between sender and reciever.

3. Segmentation and reassembling : Spliting of large message into small segments is called segmentation. It is done in sender side. The reciever side is reassembled.

4. Error control :- To find whether it is error present or not.

1. Error detection.
2. Error correction.

24/7/24 4. Network Layer :-

It is responsible for transmission of data in the form of packets from source node to Destination node.

5. Session Layer :-

It is responsible for dialogue control synchronization.

It establishes, transfers & maintain the data based on the availability of data/media.

Responsibilities

1. Dialogue Control.
2. Synchronization.

1. Dialogue Control : How the data Exchanges between sender and reciever.
→ one to one.
→ one to many.
→ one to all.

2. Synchronization : Both the sender and reciever have same clock. sender adds synchronization points to data (sink points).

6. Presentation Layer :

It is responsible for data transmission, translation, data compression and data Encryption

1. Data Translation.
2. Data Compression.
3. Data Encryption.

1. Data Translation :

2. Data Compression :-
Compresses the size of the data. It reduces the utilization of Bandwidth.

3. Data Encryption :-
we are providing security to the data by using Encryption Techniques.

7. Application Layer :
It provides all user level applications.

Responsibilities:
1. Email services → SMTP, POP.
2. Domain Name service.
3. Directory services.
4. File Transfer services → FTP.

* Specific address is performed in Application layer.

* TCP/IP Reference Model
→ 1970's by Department of Defence [DOD]
→ VINTCERF & BOBKHAN.
→ 4-layers approach / 5-layers Approach.
→ No Documentation.

Application.L Presentation.L Session.L	Application Layer
Transport Layer	
Network Layer	
Data Link.L Physical.L	Host to Network layer.

1. Host to Network layer:

The Host to network layer deals with the characteristics of a transmission media and it establishes the network and provides interference with upper layers.

2. Internet layer / Network layer:

In internet layer, it provides root from source to destination and also deals with IP addressing (or) logical addressing.

3. Transport layer:

It provides connection oriented and connectionless services and performs process to process delivery.

4. Application layer:

It provides all user level applications.

IP

IPv4	IPv6
→ 32 bits	→ 128 bits.
→ 2^{32} combination	→ 2^{128} combinations.

Ip Config / All.

ARP - Physical Address

RARP - IP Address.

Protocol suit of TCP/IP :-

Application Layer	BOOSTP DACP TELNET	SMTP POP IMAP	SNMP WWW HTTP	NFS DNS RPC	FTP TFTP FTP
Transport Layer	TCP		UDP		
Network Layer	IP ARP NAT	ICMP IGMP	RIP RARP	OSPF	BGP
Host to Network Layer	ARPANET	LAN	ETHERNET	INTERFACING Device.	

ARPANET : Advanced Research Projects Agency Network.

→ Reasearch oriented network utilized by NASA

Interfacing Device :
Switch, HUB.

Internet Layer :
IP, ARP, NAT, ICMP are used for logical addressing and RIP, OSPF and BGP are used for routing.

IP → Internet Protocall $\left[\begin{array}{l} IPv_4 - 32 - 2^{32} \\ IPv_6 - 128 - 2^{128}. \end{array}\right.$

ARP → Address Resolution Protocall [Request]

NAT → Network Address Translation.

ICMP → Internet control Message Protocal [Reply]

* The major Protocal is IP and
* supporting Protocals are ARP, NAT, ICMP.

IGMP → Internet Group Management Protocol.

RARP → Reverse Address Resolution Protocol.

Routing Protocal:-

RIP :- Routing Information Protocal

OSPF :- Open shortest Path first Protocal.

BGP :- Border Gateway Protocol.

* The purpose of these protocalls to maintain Routing tables.
* RIP and OSPF are supports Interior Network
* BGP supports Exterior Network
* To identify optimum path (shortest, Best Path)

Transport Layer:-

1. TCP :- Transmission Control Protocol / Transfer Control Protocall. [virtual circuit Packets]
* It supports connection-oriented service.

2. UDP :- User Datagram Protocal
* It supports connection less - services.

Application Layer:-

* The main services of application Layer is SMTP, POP, IMAP.
* The web browsing techniques are WWW, HTTP.
* The domain services are DNS.
* The file name services are FTP, TFTP. one is of sender side and other is reciever side.

BOOSTP :- Boot Straping Protocol.

DHCP :- Dynamic Host Control Protocol

SMTP :- simple mail Transfer protocol. [Sending Emails]

POP :- Post office protocol.

IMAP :- Internet Message Access Protocal.

IMAP :- Internet Message Access Protocol.
WWW :- world wide web
HTTP :- Hyper Text Transfer Protocol.
HTTP'S :- 's' Provides security.
DNS :- Domain Name service [search bar]. brows̶ing.
It Provides naming services [Name ⟶ IP add]
FTP :- file Transfer Protocol [file transfering of
TFTP :- Trivial file transfer Protocol.
SNMP :- Simple Network Management Protocol.
It is used for collecting, organising and managing device in a network.
TELNET :- Terminal Network.
RPC :- Remote Processor Call.
NFS :- Network file System [Store the Address].

26/7/24 Addressing :-

1. Physical Addressing.
2. Logical Addressing.
3. Port Addressing.
4. specific Addressing.

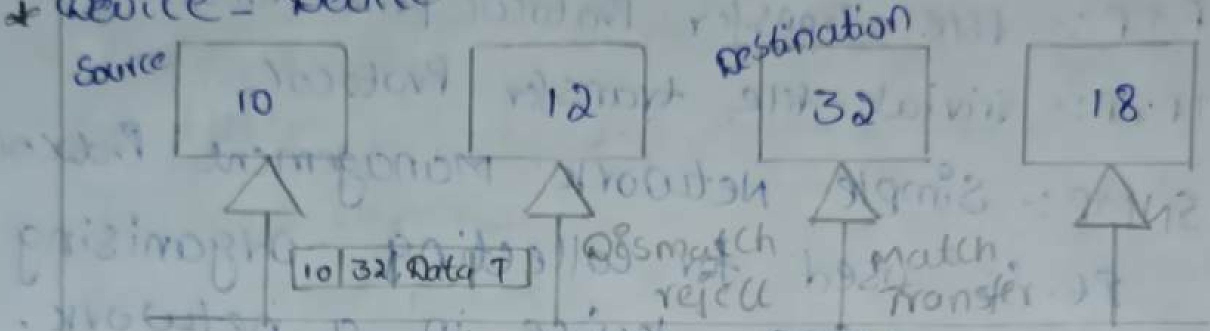
1. Physical Addressing :- [Person Name - MAC Add]
* Physical address doesnot changes throughout the network.
* Identification of Physical node in a network is called Physical addressing.
* It consists of 6 bytes.
* The length of physical address is 48 bits.
* It is represented in hexadecimal format.

* It is also known as MAC address.
* each and Every bit can be seperated with colon [:], coma (,) and underscore [_]

Ex:- AB : 12 : EC : 09 : 89 : 32. IP config/All

Source Physical Add.	Destination Physical Add	Data	Trail.

* Device - Device



2. Logical Addressing :- [Location of Person - IP Add]
* Identification of network is called logical Addressing.
* IP Address changes and Varies.
* It consist of 4 bytes.
* The length of IP Address is 32 bits.
* It is represented in Decimal format.
* IP Address is also known as Network Address.
* It is again classified into 2 types.
1. Static Address.
2. Dynamic Address.

1. Static Address :- In this case, IP Address is fixed. It is like domain Name service.

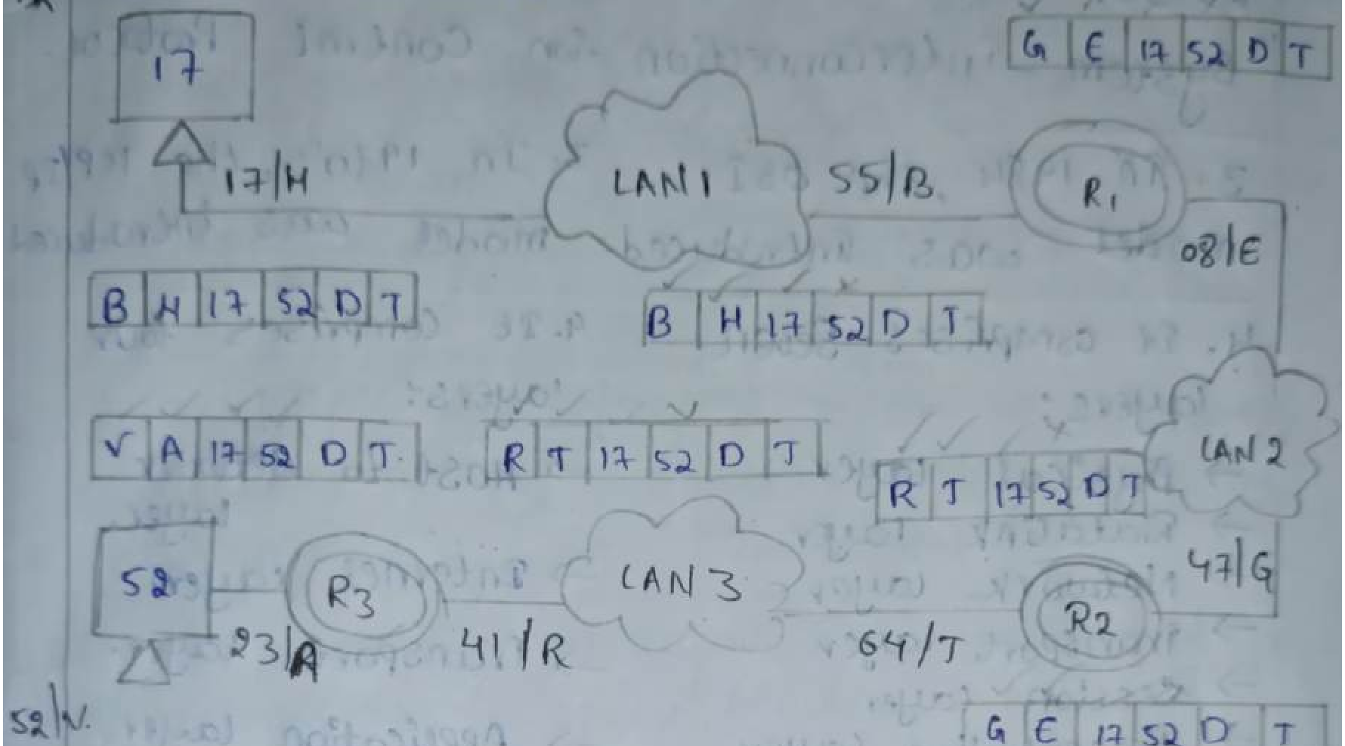
Ex:- Google, BVCE..

2. Dynamic Address :- In this case, the IP. Address is changed

Ex: If we connect with in 1 place it shows 1 address. and move for another place, it shows another address.

Destination IP Address	Source IP Address	Source Phy Add	Destination Phy Add	Data	Trail

* Network - Network.

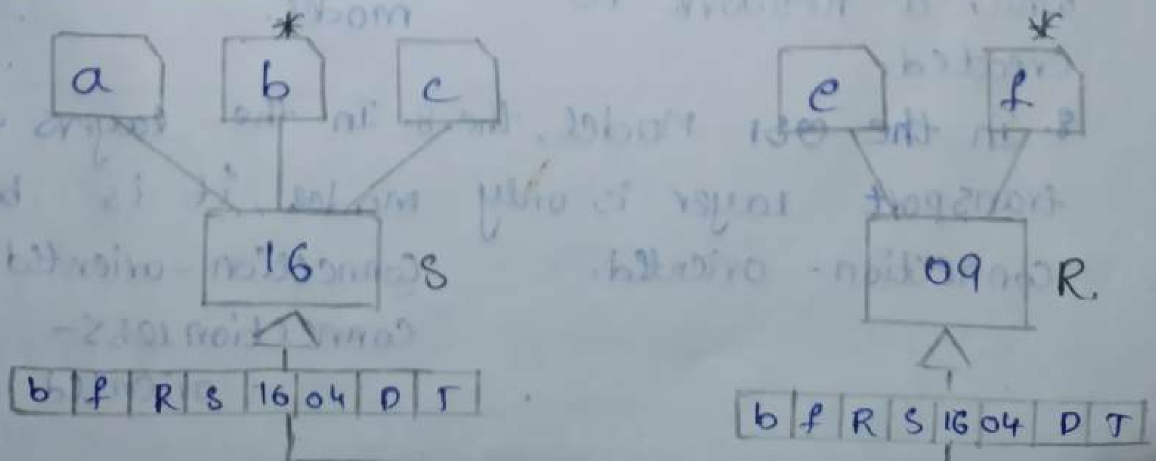


3. Port Addressing :-

* Identification of individual task or Process is nothing but Port Addressing.
* It varies from 0 to 65536.

Source Port Add	Desti Port Add	Desti IP Add	Source IP Add	Source Phy Add	Desti Phy Add.	Data	T

* Proces - Process.



OSI Model	TCP/IP Model
1. It is developed by ISO [International standard organisation]	1. It is developed by DoD [Department of Defence
2. OSI refers to open system Interconnection	2. TCP refers to Transmission Control Protocol.
3. In 1984, the OSI model was Introduced.	3. In 1970's, the TCP/IP model was introduced
4. It comprises seven layers: → Physical layer → Datalink layer → Network layer → Transport layer → Session layer → Presentation layer → Application layer.	4. It comprises four layers: → Host to Network layer. → Internet layer → Transport layer. → Application layer.
5. OSI follows a vertical approach.	5. TCP/IP follows a horizontal approach.
6. It is Protocol-in-dependent.	6. It is Protocol-dependent.
7. An OSI model is a reference model, based on which a network is created.	7. TCP/IP is the implementation of OSI model.
8. In the OSI Model, the transport layer is only Connection-oriented.	8. In the layers of TCP/IP model, it is both Connection-oriented & Connectionless-oriented.

9. In the OSI Model, the datalink layer and Physical layers are seperate layers.	9. In the TCP/IP model, datalink and physical layers are combined as single host-to-network layer.
10. Session and Presentation layers are a part of the OSI Model.	10. There is no session and presentation layer in TCP Model.
11. The minimum size of OSI header is 5 bytes.	11. The minimum header size is 20 bytes.
12. OSI Model was defined before implementation takes place.	12. TCP/IP Model defines after protocol were implemented.
13. It is based on the three concepts i.e service interface and Protocol.	13. It didn't distinguish between service, interface and protocol.
14. It gives guarantee of reliable delivery of packet.	14. It does not give guarantee of reliable delivery of packet.

29/7/24

Unit - II

Part-1 : Physical Layer

Part-11 : Datalink Layer.

Part-1 : Physical Layer.

Responsibilities :-

1. Physical characteristics of transmission Media:-
It deals with the type of Transmission media and mechanic characteristics of Transmission media. It identifies whether the connection is wired or wireless.

2. Representation of Data :-
The conversion of 0's and 1's into signal at sender side and again converts signal into 0's & 1's at reciever side.

3. Data Rate :-
No.of bits per second. The data rate is same at both sender and reciever sides.

4. Synchronization of Bits :-
Both sender & reciever maintain same clock or speed matching mechanism.

5. Transmission Mode :-
It defines how the data will flow in network.
1. simplex :- one way transmition.
2. Half Duplex :- Two way communication But not at a time
3. full Duplex :- It is Two way communication at a time.

6. Line configuration :-
It defines the type of connection i.e Point - Point connection or Point - Multipoint Connection.

7. Physical Topology :-
The physical state of network is called Physical Topology i.e, BUS, RING, MESS, STAR.

29/7/24

UNIT - II

Part - I : Physical Layer
Part - II : Datalink Layer.

Part - I : Physical Layer.

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Media

- Guided Transmition media
  - Twisted-Pair Cable
    - S.T.P
    - U.T.P
  - Co-axial cable
  - optical-fiber Cable
    - Single Mode
    - Multimode
      - STCP Index
      - Graded Index
- Un Guided Transmition Media.
  - Radio waves
  - Micro waves
  - Infrare rays

* Guided Transmition Media:-

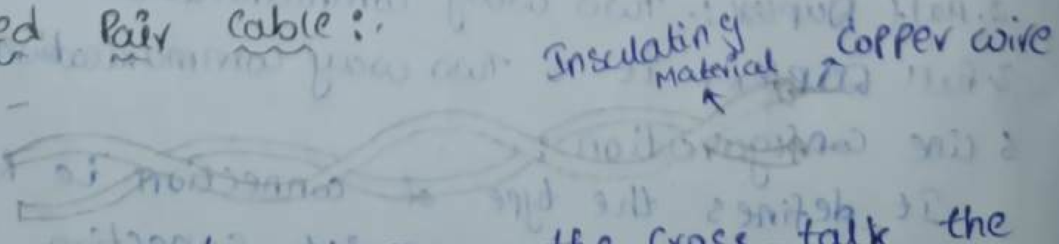
→ In the guided transmision media, there is a physical path is Established between sender & reciever.

→ It is visible transmition media. Cables are used.

→ we use three types of cables. They are

1. Twisted Pair Cable
2. Co-axial Cable.
3. Optical fibre Cable.

1. Twisted Pair Cable:-



→ In order to overcome the Cross-talk the cables are twisted.

→ In a plane wire, along the electric field, magnetic field is setup. Because of magnetic field, we loss some information, so to overcome this problem we go for twisted pair of cables.

→ Twisted Pair cables carry information in the form of electrical signal.

→ Twisted pair cables are also known as copper cables.

Applications :-

→ LAN (Local Area Network) cable purpose.

→ If No. of twists are increases, the efficiency of output is also increases.

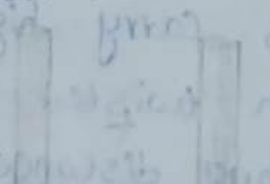
→ It is used for short - distance communication.

* They are classified into 2 types:
  1. Unshield Twisted Pair.
  2. Shielded Twisted Pair.

* It has 8 wires. so, we have 4 twisted pair.

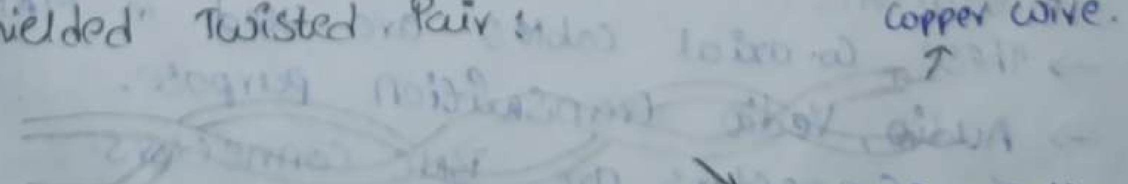
Colours :-
1. Green : Recieving and forwarding.
2. orange : sending movies and photos.
3. Blue : } Bandwidth { 1mbp/s - sending - B.W
4. Brown : } 890kB/s - recieving . speed }

Ex:- green Twisted pair



Solid Green + white Stripped Green = Twisted Pair

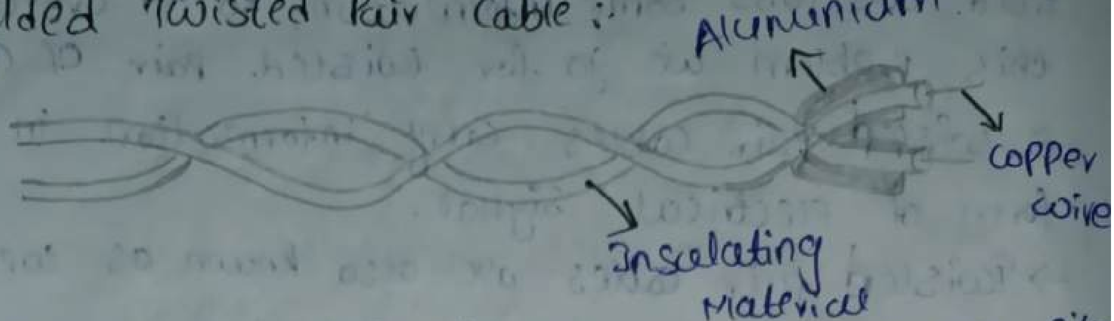
1. Unshielded Twisted Pair :-



→ Unshielded twisted pair acts like a Twisted Pair cable. we don't have any shielding.

→ Normal twisted pair cables are known as unshielded cables.

2. Shielded Twisted Pair Cable :



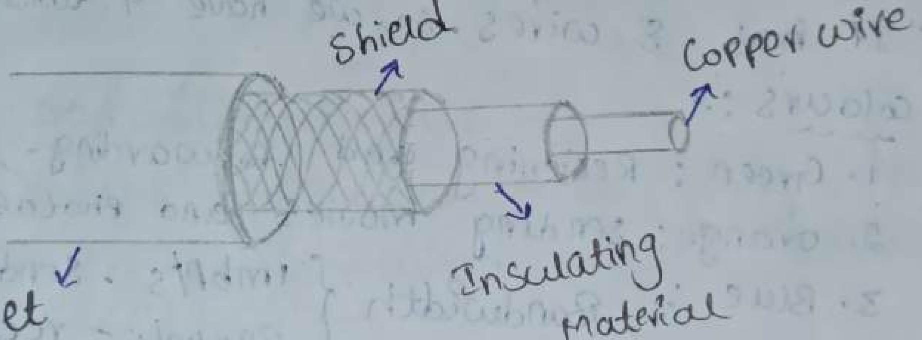
→ In shielded twisted pair cables, we use silver and aluminium for shielding.

→ Silver is high in cost. so, we use aluminium

→ The purpose of shielding is to reduce more noise.

→ shield Twisted Pair cables are used for long distance communication compared to unshielded twisted pair cable.

II Coaxial Cable :



The co-axial cables are also comes under copper cable. co-axial cables carry high frequency signals. when compared with twisted pair cable. They are also used for long distance. communicati

Applications :

→ These co-axial cables are used in MAN.

→ Audio, Vedio transmition purpose.

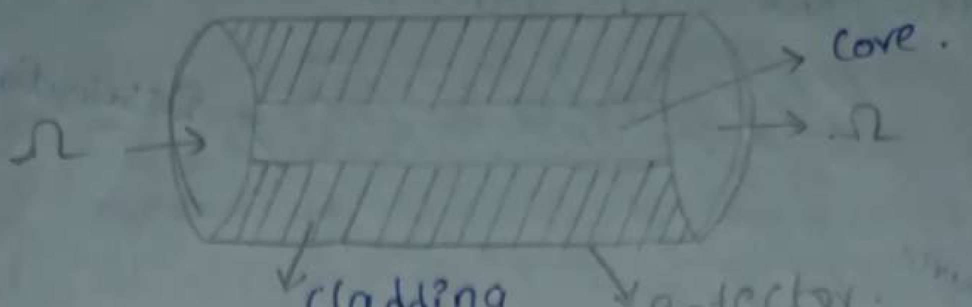
→ The connectors are BNC connectors.

BNC - Beyonet Neil - calamen.

3. Optical fiber cable :-

→ These are used for long distance communicati compared with co-axial and twisted pair cable.

→ Here, the data travels in the form of light



Core → Its diameter is less, It transmits light wave. It is in transparent.

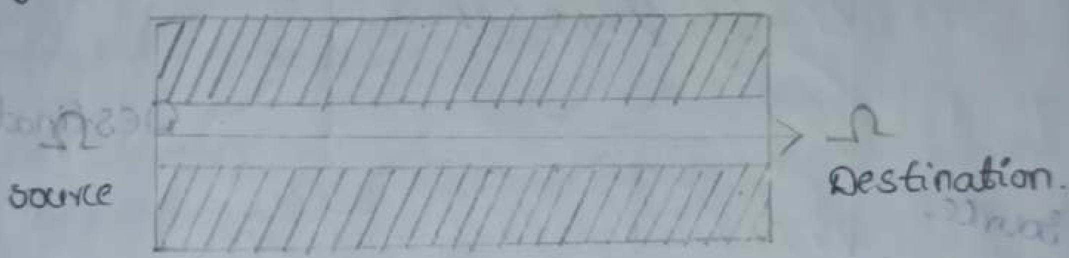
cladding → It is a body of light wave.

Protector → It is in black film.

optical fibers are classified into 2 types.

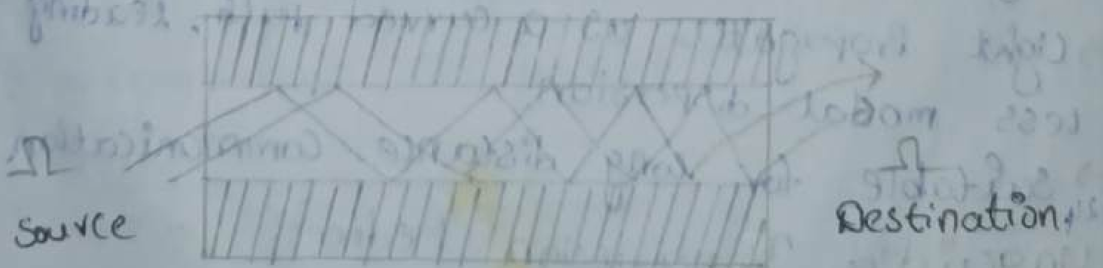
1. Single Mode.
2. Multi-mode.

1. single Mode :-



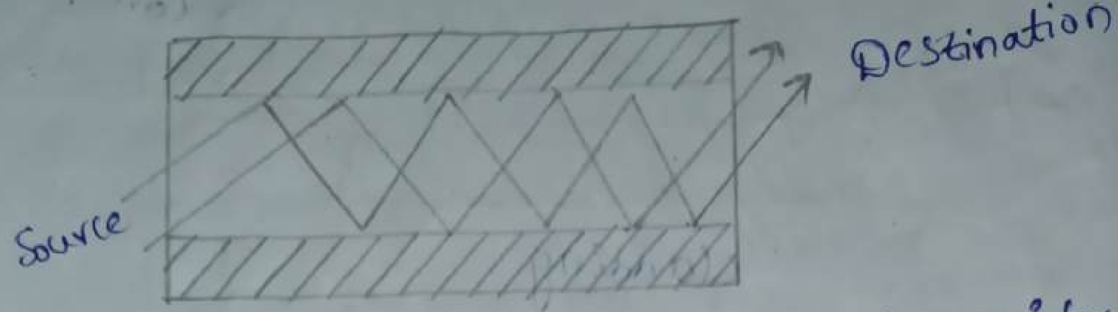
* The core diameter is minimum. It carries single signal at a time.
* It travels like a Horizontal line.

2. Multi-Mode :-



It carries multiple signals with a difference of 90°.

Multi-Mode Step Index :-



→ The refractive index of the core is uniform throughout and undergoes an abrupt change at the core cladding boundary.
→ lower Bandwidth Compared to Graded-Index
→ Light Propagates straight, leading to modal dispersion.
→ suitable for short-distance communication.

Multimode Graded Index :-

→ The refractive index of the core is made to vary gradualy such that it is maximum at center of core.
→ High Bandwidth Compared to step-Index.
→ Light Propagates as a curved path, leading to less modal dispersion.
31/7/24 → suitable for long distance communication

2. Unguided Transmission Media :-

There is no physical path and physical connection between source and Destination. unguided transmission media is also known as wireless communication.

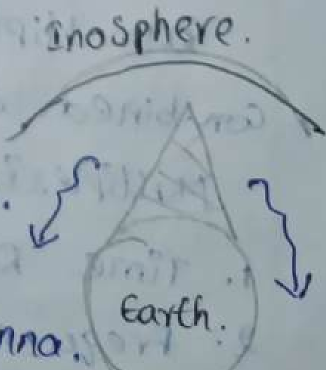
They are classified into 3 types:
1. Radio waves.
2. Micro waves.
3. Infrared waves.

1. Radio waves:-
These are a type of electromagnetic radiation that can be used to transmit data.
Frequency Range: 3KHz to 1GHz.
Application: Bluetooth, wifi.

2. Micro waves:-
These are used for fast and reliable data transmission, voice and video streaming over short distances. [Point - Point / Point - Multipoint]
Frequency Range: 1GHz to 300 GHz.
Application: cellular Mobile Communication.

3. Infrared waves:-
Infrared waves are electromagnetic waves that can be used to transmit data. They are light waves in infrared spectrum.
Frequency Range: 300 GHz to 400 GHz.
Application: TV Remote control.

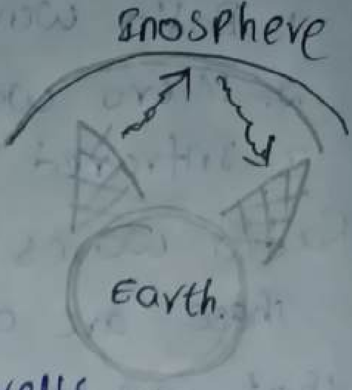
* There are three types of wave propagations:-
1. Groundwave Propagation.
2. Sky wave Propagation.
3. Line of sight Propagation.



1. Groundwave Propagation:-
We connect one antenna to Earth. That antenna radiate waves in uni direction. i.e omni directional antenna. The waves didn't cross inosphere & radiates aroun

earth only.

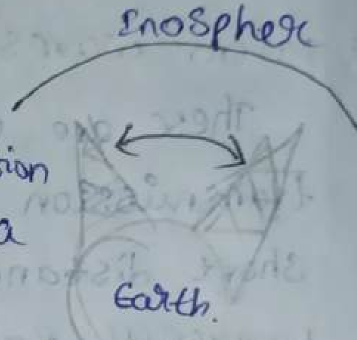
2. Sky wave Propagation :-



Here, we connect 2 Antennas to Earth. one is transmitting antenna and another one is recieving antenna.

The transmitted waves travells long distance and touches inosphere and they reflected back and those waves are recieved by recieving antenna.

3. Line of sight Propagation :-



There is a direct communication between transmitting antenna and recievong antenna.
[OBSERVE Antenna & TARGET Antenna]

* Multiplexing :-

It is a technique which is used to combining of different data and make the data capable of Transmitting in a single Transmition media.

After Transmitting of different signals by using multiplexing technique. At the reciever end, we need to use the technique called demultiplexing.

Demultiplexing is nothing but seperating the combined data.

Multiplexing is classified into 3 types.

1. Time Division Multiplexing.
2. Frequency Division Multiplexing.
3. Code Division Multiplexing.

1. Time Division Multiplexing :-

Transmition of multiple signals into a single channel is called Multiplexing.

→ The available Time is going to divided into individual slots. Each slot is assigned to individual station.

→ The data available in the station is transmitted only when specified time slot.

→ The Entire channel is utilized by single node at one instant of time.

→ The main advantage of TDM is Efficient utilis-ation of Bandwidth.

2. Frequency Division Multiplexing :-

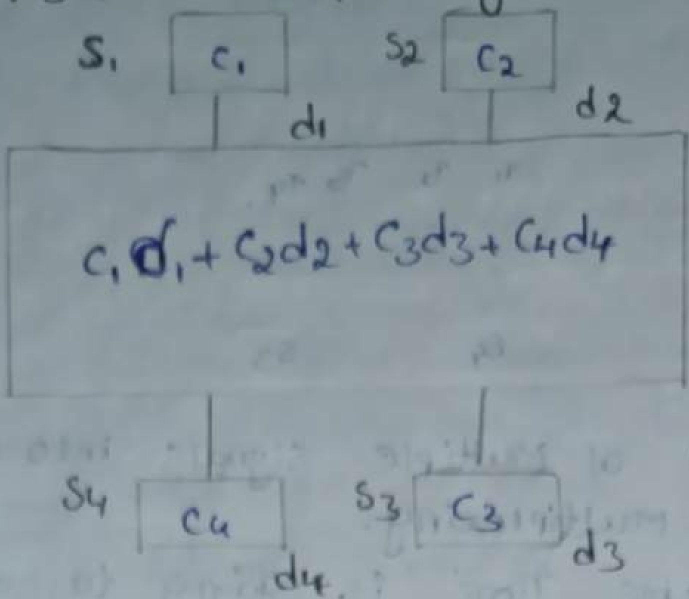
→ FDM is a technique in which the available bandwidth is seperated into different bands.

→ Each Band is allocated to individual stations.

→ The Bands are seperated by small range of frequencies are called Guard Band.

→ The Entire bandwidth doesnot utilised by chann

3. code Division Multiplexing :-



→ It is a technique which we assign a unique code to each station.

→ Whenever the station is capable of transmittin some data, the data is multiplied with the unique code. and then it fed to the medium or communication link.

→ c_1, c_2, c_3, c_4 → codes

→ d_1, d_2, d_3, d_4 → Data.

There are 2 rules we have to follow to assign code.

① $c_n \cdot c_i = 0$.

Multiply any code with another code. we must give '0'.

② $c_n \cdot c_n = 1$.

Multiply any code with itself code. we must give '1'.

example :-

If we select state - 2.

$c_2 (c_1 d_1 + c_2 d_2 + c_3 d_3 + c_4 d_4)$

$\Rightarrow c_1 c_2 d_1 + c_2 \cdot c_2 d_2 + c_2 c_3 d_3 + c_2 c_4 d_4$

$\Rightarrow (0) d_1 + (1) d_2 + (0) d_3 + (0) d_4$

$\Rightarrow d_2$.

So, we recieve d_2 data from station-2.

Part - II : DataLink Layer.

Responsibilities :-

1. framing :- Splitting of large message into small data unit is called framing.
   1. fixed size
   2. variable size.
2. Physical Addressing :- It is also known as MAC address. Identification of individual nodes in physical network. It is a length of 48 bits. By using hexadecimal representation we address the data.
   1. LLC sublayer
   2. MAC Sublayer.
3. flow control :- The continous transmission of data between the sender and reciever without any interrupts.

   ARR : Automatic Repeat Request.
4. error Control :- The deviation from the origina data
   1. Error detection.
   2. Error confirmation.
5. Access Control : The nodes which having control over the channel at one instant of time deals with the collisions.

   we can access by using TDM, FDM and CDM.

Datalink layer :- It transfers the data from Source to Destination in the form of frames. Data is converted into frames.
It has 2 Sub layers:
1. uppermost layer → Network layer
2. lowermost layer → Physical layer.

Network Layer	
Data link layer	LLC sublayer MAC sublayer
Physical Layer	

Logical link control sublayer :-
It is used for sharing the data between the upper and lower layer i.e. Network Layer and Physical layer.
Media Access control sublayer :-
It is used for address / Assigning the address to the frames.
services Provided by Datalink :-
1. Physical Addressing.
2. framing.
3. Error control.
4. Flow control.
5. Access control.
services provided ~~by~~ Network layer:-
Acknowledgement :- It is nothing but delivary status of frames.

Based on the Acknowledgement, we have 3 services.

1. unacknowledged connectionless service.
2. Acknowledged connectionless service.
3. Acknowledged connection-oriented service.

1. Unacknowledge connectionless service:-

The sender does not recieve any confirmation or acknowledgement from the reciever that the data packets were successfully recieved. If a frame is lost or corrupted during transmission..

There is no need to set up a dedicted path or session between the sender and reciever before transmission. Each packet is sent independ-ent of other.

2. Acknowledged connectionless service:.

An acknowledged connectionless service sends data packets independently without a dedicated connection but requires acknowledgement for each packet. This ensure the sender is informed of successful delivery and can transmit if necessary. It combines aspects of both connectionless and reliable communication.

3. Acknowledged connection-oriented service:.

An acknowledged connection-oriented service involves establishing a dedicated connection between sender and reciever before data transmission. Data packets are sent in a reliable, ordered manner with acknowledgements confirming successful reciept. This ensures reliable communication with error detection and recovery.

2/8/24 Framing classification
1. fixed size framing.
2. variable size framing.

1. Fixed size framing :- The total information is seperated into frames [Group of bits]. So in fixed size framing, all the frames have same length.

2. Variable size framing :- The total information is splited into frames. So in variable size framing all the frames have different length.

* There are two ways of framing Approaches
1. Bit - oriented Approach
2. Byte - oriented Approach.

1. Bit oriented Approach :-
Bit means either '0' or '1'. The frame is represented in the form of bits.
Bit - oriented approach follows HLDC Protocol [High - level Datalink control protocol.

8 bit	16 bit	//	46 bit	8 bit
Beginning of frame	Header	Data	CRC	Ending of frame

* Beggining of Frame (or) Ending of Frame :-
It defines starting and Ending of the frame with the unique sequence "01111110". It is of 8 bit length.

* Header field :-
The next of this frame format is header

field. This field Contains 2 sub fields.
1. Address field. It is of length 16 bits.
2. Control field.

1. Address field :- It defines the addresses of the Source and Destination.

2. Control field :- It defines the type of frame.

There are 3 control frames.

1. Information Frame [I-Frame]
2. Supervisary / Control Frame [SP-Frame]
3. Management / Un-Name Frame [U-Frame].

1. Information frame :-
In control field, the first bit is started with '0'. then that frame is said to be [I-Frame]. Ex :- "00", "01". It consists of "data".

2. Supervisary frame :-
The first two bits of control field starts with "10" then it is said to be [SP-frame]. It is also known as control frame, it does not carries any information. It gives commands to the node in the network.
Acknowledged frame is used.

3. Un-Name frame :-
The first two bits of Control field starts with '11' then it is said to be [U-frame]. It sends the agreement between source and Destination.

* Data :-
It is also known as pay load. It is the third field of this frame format.
It is a variable length field, which transmits

the original data in the form of bits.

* CRC

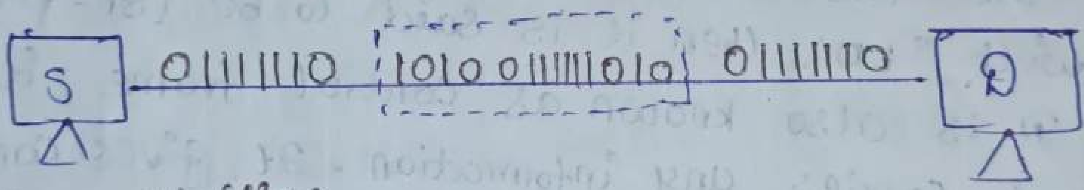
CRC means cyclic Redundancy check. It is used for Error control. which contains the group of bits used for Error detection.

In order to overcome the errors in CRC, we are using Bit stuffing Technique.

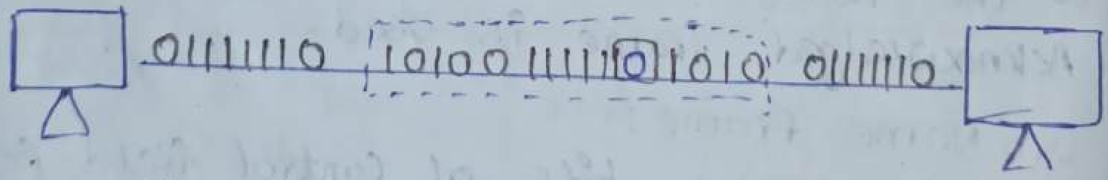
Bit stuffing :-

whenever the sequence of starting of frame and Ending of frame will appear in the data. It consider the data as End of the frame After that information will be loss. So in that case we add '0' in data after 5 consecutive 1's are appeared.

Before stuffing.



After stuffing



2. Byte-oriented Approach :

It is also known as Character oriented approach. It transmits the frames in the form of bytes.

Byte-oriented approach follows BI.SYNC Protocol to transmit the frames.

Binary Synchronous Communication Protocol.

1 byte	1 byte	1 byte	2 bytes	1 byte		1 byte	2 bytes
8 bit	8 bit	8 bit	16 bits	8 bit		8 bit	16 bit
SYN	SYN	SOH	Header	STX	Data	ETX	CRC.

SYN :-

The first two fields are synchronization bits whose length is 8 bits each. These fields are used for synchronizing the destination To allert the reciever, The next to the starting of the frame and to maintain the clock.

SOH :-

SOH means Start of Header. It is length of 8 bits. It is a control character used to indicate the beggining of a header segment in a data packet.

STX :-

Start ~~op~~ Text. It defines the stating of data. It is a 8 bit length.

ETX:-

End of Text. It defines the Ending of data It is a 8 bit inlength.

CRC :-

cyclic Redundancy check. whose length is 16 bit. It is used for Error control. which contains the bytes used for Error detection.

Byte Stuffing:-

It is the concept of adding an External byte to the data when source identifies the sequence in the data.

DLE - Data link Escape.

Before Byte Stuffing:

A	Data	B

→

A	STX	Data	B

After Byte stuffing.

A	STX	B

→

A	DLE	STX	B.

5/8/24

Error Control:-

It is the concept of finding the recieved data is correct or incorrect. Basically we are having error detection and Error correction Techniques to find the Error

→ They are 2 types of Errors-

1. Bit Error.
2. Burst Error.

1. Bit Error:-

The change of single bit is called bit Error

sender	1	0	0	1	1	0	1
Reciever	1	0	1	1	1	0	1

2. Burst Error:-

The change of more than one bit is called Burst Error.

sender	0	1	0	1	1	0	1	0
Reciever	0	0	0	1	0	0	1	0

Error Detection :-

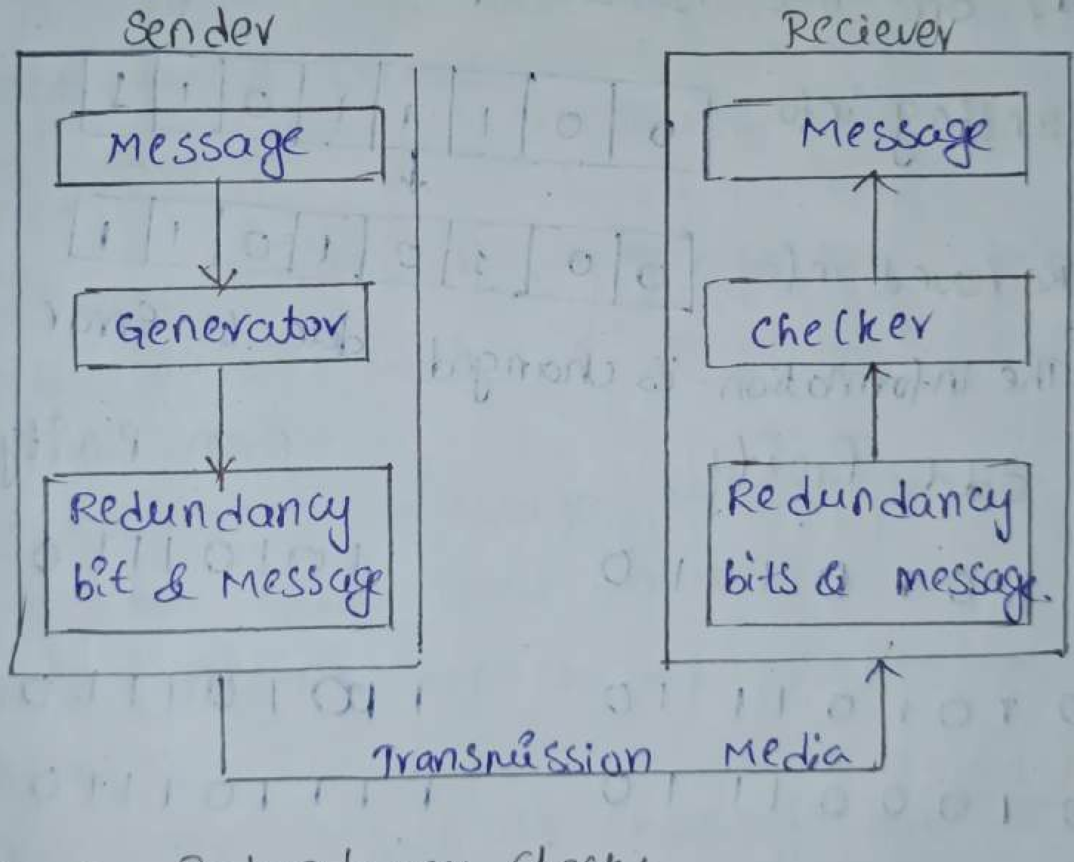
It is the concept of detecting the error by using some techniques.

There are 4 Techniques

1. VRC → Vertical Redundancy check
2. LRC → Longitudnal Redundancy check.
3. checksum.
4. CRC → Cyclic Redundancy check.

Redundancy :-

Adding Extra bits to the original message for the purpose of Error control.



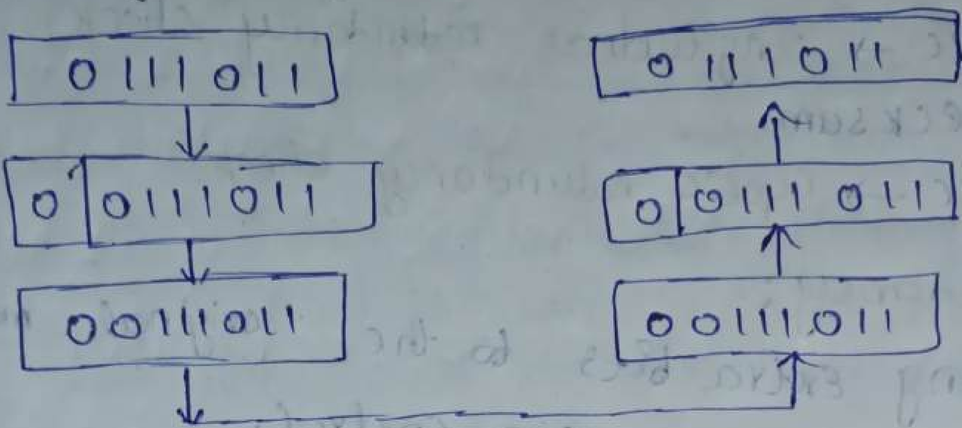
1. Vertical Redundancy Check :-

It is used to find single bit Errors. In this VRC, parity generator is used to create a parity bit based on even parity or odd parity.

At destination End, the parity checker will [illegible] the parity and it compares with the recieved Parity bit. If both bits are matched together then it will accept the data otherwise it rejects the data.

Odd Parity.

Message :- 0111011



If one bit is corrupted during transmission

sending info	0	0	1	1	1	0	1	1
Recieved info.	0	0	1	0	1	0	1	1

The information is changed due to Error.

2. Odd Parity — Even Parity.

Odd Parity	Even Parity
1 0 1 0 1 1 1 0	1 0 1 0 1 1 1 0.
0 1 0 1 0 1 1 1 0	1 1 0 1 0 1 1 1 0.
× 0 1 0 0 0 0 1 1 1 0	1 1 1 1 0 1 1 1 0 ×
✓ 0 1 0 0 1 1 1 1 1 0	1 1 0 0 1 1 1 1 0 ✓

Message	Even Parity	odd Parity.
10011001	010011001	110011001
11001100	011001100	111001100
11100111	011100111	111100111
01000101	101000101	001000101
00001101	100001101	000001101.

7/8/24 2. Longitudinal Redundancy check :-

The group of data arranged in columns and rows and then find parity bit using even or odd parities for each column.

Transmit the entire blocks of data along with this generated parity bit.

At the reciever end, calculate the parity bits for entire data. If the parity bits are matches it accepts the data otherwise it rejects the data and the reciever re-transmit the data to source.

It detects single bit errors and also detect burst errors.

Procedure :-

step-1 :- Divide the message into 'k' no. of data blocks with 'n' no. of bits in each block.

step-2 :- Arrange message blocks in the form of rows and columns.

step-3 :- Perform even or odd parity for each column.

step-4 :- The result is nothing but LRC.

1. Calculate CRC for block of data

10110000, 00001000, 1111110, 10101101.

(with d_1, d_2, d_3, d_4 marked above the four blocks)

and then transmit to reciever.

i. Assume in the time of transmission, the data 4 ie d_4 is corrupted by 4 bit.

ii. The data 4 ie d_4 is corrupted by 7th bit and data 2 ie d_2 is corrupted by 7th bit

Sol step-1 :- sender side.

10110000, 00001000, 1111110, 10101101

step-2 :-

1 0 1 1 0 0 0 0
0 0 0 0 1 0 0 0
1 1 1 1 1 1 1 0
1 0 1 0 1 1 0 1

step-3 :- Assume Even Parity.

1 0 1 1 0 0 0 0
0 0 0 0 1 0 0 0
1 1 1 1 1 1 1 0
1 0 1 0 1 1 0 1
---
1 1 1 0 1 0 1 1 → Even Parity for Each colum

step-4 :- 11101011

11101011 + 10110000 00001000 1111110 10101101.

i, ~~when d~~ Reciever Side

10110000
00001000
1111110
10101101
11101011

00000000 → Even Parity.

when data is recieved without Errors means it recieves accurate information.

i, when data is corrupted by 4th bit in data-4 i.e d_4.

```
 10110000
 00001000
 11111110
 10100101
 11101011
 ________
 00001000 - Error is detected at 4th bit
```

ii, when data is corrupted by 7th bit of d_4 and also 7th bit of d_2.

```
 10110000
 01001000
 11111110
 11101101
 11101011
 ________
 00000000 - Error is not detected.
```

The LRC doesnot detects the error at same positions of differend data.

2. Calculate LRC for block of data

00011100, 10101010, 11001100, 00110101
d_1, d_2, d_3, d_4

and then transmit to reciever.

i, Assume in the time of transmission, the data d_1 is corrupted by 5th bit and data d_4 is corrupted by 7th bit

ii) Assume in the time of transmission, the data d_2 is corrupted by 4th bit and data d_3 is corrupted by 4th bit

Step-1 :- Sender side

00011100, 10101010, 11001100, 00110101

Step-2 :-

00011100
10101010
11001100
00110101

Step-3 :- Apply Even parity.

00011100
10101010
11001100
00110101

01001111 → Even parity for Each column.

Step-4 :-

01001111 + 00011100 10101010 11001100 00110101

Reciever side.

00011100
10101010
11001100
00110101
01001111

00000000 → No Error is detected.

i, when data is corrupted by 5th bit of d_1 and by 7th of d_4.

```
  00001100
  10101010
  11001100
  01110101
  01001111
  --------
  01010000 → It detects 2 Errors.
```

ii, when data is corrupted by 4th bit of d_2 and also corrupted by 4th bit of d_3.

```
  00011100
  10100010
  11000100
  00110101
  01001111
  --------
  00000000 → Error is not
                   detected.
```

8/8/24 **Checksum :-**

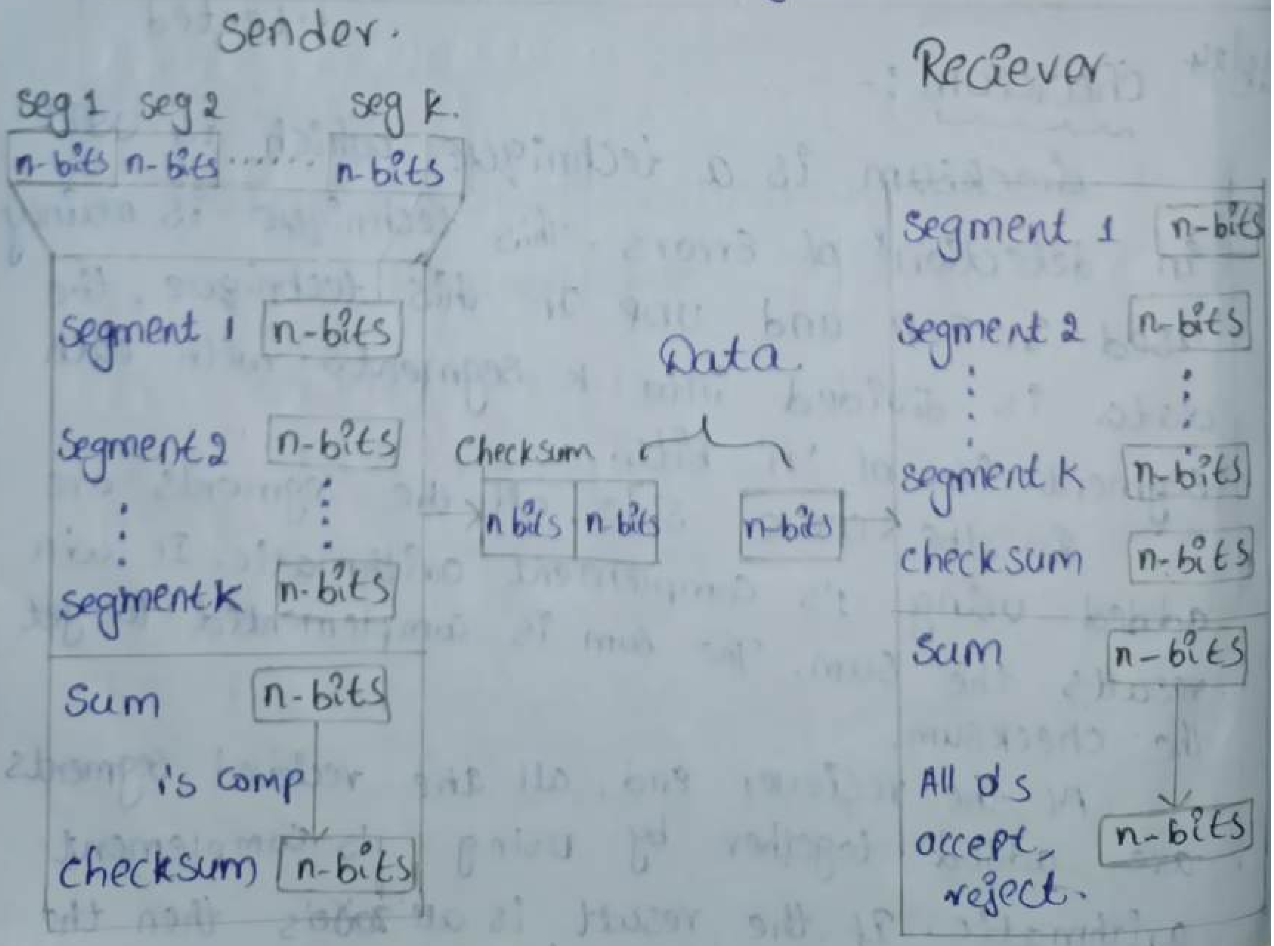
checksum is a technique which is used in detection of Errors. This technique is mainly used in TCP and UDP. In this technique, the data is divided into k segments with each segment is of 'n' bits.

In the sender side, all the segments are added using 1's complement arithmatic. It will results the sum. The sum is complemented to get the checksum.

At the reciever end, all the recieved segments are added together by using 1's complement arithmatic. If the result is all zero's then the data is accepted by the reciever otherwise it rejects the data.

Procedure for calculating checksum.

Sender side	Reciever side.
① The data is divided into number of segments [k] with each segment having 'n' bits.	① The recieved data is divided into number of segments [k] with each segment having 'n' bits.
② All the segments are added by using 1's complement arithmatic	② All the sections are added by using 1's complement arithmatic.
③ The sum is bitwise complemented to get the checksum. These checksum is transmitted along with the data.	③ The generated sum is complemented and the result is all 0's then it will accept the data otherwise rejects.



1. calculate checksum for given data.

10001000 10101010 1110111 00001111.

by using checksum ?: Transmit the data along with checksum and find if there are any errors at reciever side?

<u>Sender Side</u>

10001000	1010 1010	1110111,	00001111
d_1	d_2	d_3	d_4.

```
    1 0 0 0 1 0 0 0
    1 0 1 0 1 0 1 0
    1 1 1 0 1 1 1 1
    0 0 0 0 1 1 1 1
   ----------------
10 0 0 1 1 0 0 0 0
                1 0
   ----------------
   0 0 1 1 0 0 1 0 → sum.
   1 1 0 0 1 1 0 1 → 1's Comp of Sum
                     ↓
                     Checksum.
```

<u>Reciever Side</u>

10001000	10101010	1110111	00001111	11001101

```
    1 0 0 0 1 0 0 0
    1 0 1 0 1 0 1 0
    1 1 1 0 1 1 1 1
    0 0 0 0 1 1 1 1
    1 1 0 0 1 1 0 1
   ----------------
10  1 1 1 1 1 1 0 1
                1 0
   ----------------
    1 1 1 1 1 1 1 1 → sum
    0 0 0 0 0 0 0 0 → 1's compl of sum
```

At the reciever side, the result is all 0's that means there is no Error in the data, it accepts the data.

* when data 2 d_2 is corrupted at 6, 7 th positions.

```
 10  1 1  11 10 10 1
    1 0 0 0 1 0 0 0
    1 1 0 0 1 0 1 0
    1 1 1 0 1 1 1 1
    0 0 0 0 1 1 1 1
    1 1 0 0 1 1 0 1
   -----------------
 11|0 0 0 1 1 1 0 1
                1 1
   -----------------
    0 0 1 0 0 0 0 0  → sum
                     → 1's
    1 0 1 1 1 1 1 1  → reject.
```

9/8/24

Cyclic Redundancy check :-

It is a type of detection technique used to find the Errors in recieved data.

The CRC is also known as polynomial code.

Ex:- $x^5 + x^3 + x^2 + x$

1 0 1 1 1

Procedure for CRC:-

1. Appended the 0's to the original data based on the Expression (r-1) at end of data.
2. By using modulo 2 operation and find the remainder.

3. Consider the length of the CRC code from the remainder based on the appended 0's. (from LSB).

4. Add this code with original data in the place of appended 0's and transmit through the medium.

5. At the reciever side, it performs the same operation on recieved data with the same divisor. If the remainder is 0, then the data should be accepted. otherwise reject.

6. Neglect the MSB bit if it is '0'. during Every operation in division.

1. find the CRC code for the following message data is 1000101 with the divisor of 1011. At the reciever end, assume the single bit data is corrupted and then calculate CRC code?

M = 100101

r = 1011

Step-1 :- find the appended zero's and add to the data at LSB.

r-1 ⇒ length of r - 1

⇒ 4-1

⇒ 3 zeros.

Now, M = 100101 $\underbrace{000}_{\text{CRC Code.}}$

X-OR

Sender side

```
1011 | 1000101000
       1011
       0 0111
         0000
         0 1110
           1011
           0 1011
             1011
             0 0000
               0000
               0 0000
                 0000
                 0 0000
                   0000
                   0 000
```

Reciever side.

```
1011 | 1000101000.
       1011
       0 0111
         0000
         0 1110
           1011
           0 1011
             1011
             0 0000
               0000
               0 0000
                 0000
                 0 0000
                   0000
                   0 000
```

it accept the data.

* The bit is corrupted from LSB side at 3rd position.

1000101100

```
1011 | 1000101100
       1011
       0 0111
         0000
         0 1110
           1011
           0 1011
             1011
             0 0001
               0000
               0 0010
                 0000
                 0 0100
                   0000
                   0 100
```

CRC.

It rejects the data ∵ CRC code is not matches.

Calculate CRC code for the data frame 1101011111 with the divisor 10011

Data frame, M = 1101011111

Divisor, r = 10011

Step-1 :- Append the 0's

⇒ r-1 ⇒ length of r-1

⇒ 5-1

⇒ 4.

we need to add 4 bits of 0's to data at LSB.

M = 11010111110000.

Sender side

```
10011 ) 11010111110000
        10011
        010011
         10011
         000001
          00000
          000011
           00000
           000111
            00000
            001111
             00000
             011110
              10011
              011010
               10011
               010010
                10011
                000010
                 00000
                 00010
```

CRC code

11010111110010.

Reciever side

```
10011 ) 11010111110010
        10011
        010011
         10011
         000001
          00000
          000011
           00000
           000111
            00000
            001111
             00000
             011110
              10011
              011010
               10011
               010011
                10011
                000000
                 00000
                 00000
```

we recieve all 0's so, it accepts the data.

when the data is corrupted by 12 bit from LSB.

```
10011 ) 11110111110010
        10011
        01101 1
         10011
        010001
          10011
        000101
           00000
         001011
            00000
         010111
             10011
          001000
              00000
          010000
               10011
           000111
                00000
            001110
                 00000
             01110
```

It detects the error, rejects the data.
The CRC code dismatch.

3. Calculate the CRC code for data is given as 1001001001 with the divisor of 11010 in both transmitter and reciever.

M = 1001001001

r = 11010

Step-1 : Append the 0's $\Rightarrow r-1 \Rightarrow 5-1$
$\Rightarrow 4$.

we need to add 4 bit of 0's at LSB side

M = 10010010010000

11010 | 10010010010000

11010
0|10000
11010
0|10101
11010
0|11110
11010
0|01000
00000
0|10001
11010
0|10110
11010
0|11000
11010
0|00100
00000
0|01000
00000
0|1000.

CRC code.

10010010011000

11010
0|10000
11010
0|10101
11010
0|11110
11010
0|01000
00000
0|10001
11010
0|00111
00000
0|01010
00000
0|00000
00000
0|00000
00000
0|0000

11010 | 10010010010110 1011

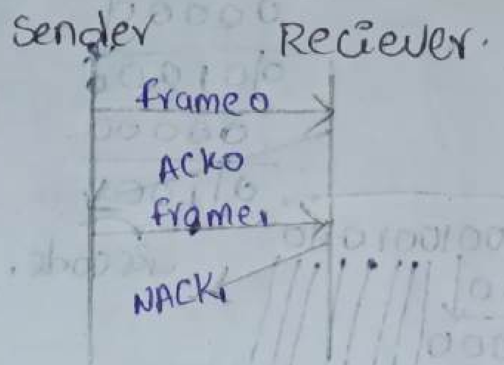
11010
0|10000
11010
0|10101
11010
0|11110
11010
0|01000
00000
0|10001
11010
0|10110
11010
0|11001
11010
0|00111
00000
0|01110
00000
0|1110

Error Correction Codes :-

It is a concept of detecting the error as well as correcting the error. Basically, it is of two ways.

1. Re-transmition.
2. Hamming Code.

1. **Retransmition** :- If any error is there in the transmitted frame. the reciever gives negative acknowledgement to the sender. If negative Acknowledgement is recieved by the sender. The sender transmits the previous frame. That is nothing but retransmition technique.



14/8/24

2. Hamming code :-

Hamming code is a method used for Error detection and correction in digital communication. Imagine you have a message made up of bits and and you want to send it across a network. However during transmission, some bits might get flipped due to noise or other issues. Hamming code helps detect and correct these errors for only one bits.

R.W hamming was developed the hamming code. Mathematically it can represented as

$2^r \geq m+r+1$. $m \rightarrow$ length of given data.
$r \rightarrow$ Redundant Bits.

1. Adding redundant Bits :-
first, you add extra bits to your message. These are called parity bits. The number of parity bits depend on the length of your message.

2. Placing Parity Bits :- These parity bits are placed in specific positions in the message i.e powers of 2. Each parity bit checks a certain set of bits in the message to ensure their sum is even (for even parity) or odd (for odd parity).

3. Error detection :- when the message recieved the reciever checks the parity bits. If the parity bits don't match the expected values, it means there was an error in transmission.

4. Error Correction :- If an error is detected, Hamming code can identify exactly which bit is incorrect. By flipping that bit back the error is corrected.

1. Calculate Hamming code for data 1100101. Determine the required redundant bits and transmit the data to the reciever assume single bit error is occured at 10th bit in recieved data.

$M = 1100101$ [Even Parity]

length of given data, $M = 7$.

$2^r \geq m + r + 1$

If $r = 1$, $2^1 \geq 7 + 1 + 1$

$2 \geq 9$ Not Satisfied.

If $r = 2$, $2^2 \geq 7 + 2 + 1$

$4 \geq 10$ Not Satisfied.

If $r = 3$, $2^3 \geq 7 + 3 + 1$

$8 \geq 11$ Not satisfied.

If $r=4$, $2^4 \geq 7+4+1$

$16 \geq 12$ satisfied.

∴ So, we have 4 Parity bits.

2^3 2^2 2^1 2^0.

P_8 P_4 P_2 P_1

1	1	0	P_8	0	1	0	P_4	1	P_2	P_1
11	10	9	8	7	6	5	4	3	2	1

	P_8	P_4	P_2	P_1
1 →	0	0	0	1.
2 →	0	0	1	0
3 →	0	0	1	1
4 →	0	1	0	0
5 →	0	1	0	1
6 →	0	1	1	0
7 →	0	1	1	1
8 →	1	0	0	0
9 →	1	0	0	1
10 →	1	0	1	0
11 →	1	0	1	1.

$P_1 \rightarrow 1, 3, 5, 7, 9, 11 \rightarrow P_1\ 10001 \rightarrow 0$

$P_2 \rightarrow 2, 3, 6, 7, 10, 11 \rightarrow P_2\ 11011 \rightarrow 0$

$P_4 \rightarrow 4, 5, 6, 7 \rightarrow P_4\ 010 \rightarrow 1$

$P_8 \rightarrow 8, 9, 10, 11 \rightarrow P_8\ 011 \rightarrow 0$.

Recieve side.

1	1	0	0	0	1	0	1	1	0	0
11	10	9	8	7	6	5	4	3	2	1

$2^0 \rightarrow 1$

$2^1 \rightarrow 2.$

$2^2 \to 4$

$2^3 \to 8$

$\boxed{2^4 \to 16}$

Total length of data,

$m + r \Rightarrow 7 + 4$

$= 11.$

$\therefore r = 4.$

$P_1 \to 1, 3, 5, 7, 9, 11 \to 010001 \to 0$

$P_2 \to 2, 3, 6, 7, 10, 11 \to 011011 \to 0$

$P_4 \to 4, 5, 6, 7 \to 1010 \to 0$

$P_8 \to 8, 9, 10, 11 \to 0011 \to 0$

P_8	P_4	P_2	P_1
0	0	0	0

If we recieves all zeros at recievers, that means we didn't get any ~~zeros~~. Errors

10th bit is corrupted.

1	0	0	0	0	1	0	1	1	0	0
11	10	9	8	7	6	5	4	3	2	1

$2^0 \to 1 > 11$ ✗

$2^1 \to 2 > 11$ ✗

$2^2 \to 4 > 11$ ✗

$2^3 \to 8 > 11$ ✗

$2^4 \to 16 > 11$ ✓

$\therefore r = 4.$

$P_1 \to 1, 3, 5, 7, 9, 11 \to 010001 \to 0$

$P_2 \to 2, 3, 6, 7, 10, 11 \to 011001 \to 1$

$P_4 \to 4, 5, 6, 7 \to 1010 \to 0$

$P_8 \to 8, 9, 10, 11 \to 0001 \to 1$

P_8	P_4	P_2	P_1
1	0	1	0

$\therefore$ Therefore we get an Error at 10th bit.

1	1	0	0	0	1	0	1	1	0	0

2\. 101100010010. Odd Parity.

length of given data, $m = 12$

$$2^r \geq m + r + 1.$$

If $r = 1$, $2^1 \geq 12+1+1$

$2 \geq 14$ Not Satisfied.

If $r = 2$, $2^2 \geq 12+2+1$

$4 \geq 15$ Not Satisfied.

If $r = 3$, $2^3 \geq 12+3+1$

$8 \geq 16$ Not Satisfied.

If $r = 4$, $2^4 \geq 12+4+1$

$16 \geq 17$ Not Satisfied.

If $r = 5$, $2^5 \geq 12+5+1$

$32 \geq 18$ Satisfied.

$\therefore r = 5$.

2^4 2^3 2^2 2^1 2^0.

P_{16} P_8 P_4 P_2 P_1

Total length of data,

$m + r = 12 + 5$

$= 17$

1	P_{16}	0	1	1	0	0	0	1	P_8	0	0	1	P_4	0	P_2	P_1
17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1

	P_{16}	P_8	P_4	P_2	P_1
$1 \rightarrow$	0	0	0	0	1
$2 \rightarrow$	0	0	0	1	0
$3 \rightarrow$	0	0	0	1	1
$4 \rightarrow$	0	0	1	0	0
$5 \rightarrow$	0	0	1	0	1
$6 \rightarrow$	0	0	1	1	0

7 → 0 0 1 1 1

8 → 0 1 0 0 0

9 → 0 1 0 0 1

10 → 0 1 0 1 0

11 → 0 1 0 1 1

12 → 0 1 1 0 0

13 → 0 1 1 0 1

14 → 0 1 1 1 0

15 → 0 1 1 1 1

16 → 1 0 0 0 0

17 → 1 0 0 0 1.

P_1 → 1, 3, 5, 7, 9, 11, 13, 15, 17 → P_1 01010101 → 1

P_2 → 2, 3, 6, 7, 10, 11, 14, 15 → P_2 0000010 → 0

P_4 → 4, 5, 6, 7, 12, 13, 14, 15 → P_4 1000110 → 0

P_8 → 8, 9, 10, 11, 12, 13, 14, 15 → P_8 1000110 → 0

P_{16} → 16, 17 → P_{16} 1 → 0

Reciever side

1	0	0	1	1	0	0	0	1	0	0	0	1	0	0	0	1
17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1

$2^0 \to 1 > 17$ ×

$2^1 \to 2 > 17$ ×

$2^2 \to 4 > 17$ ×

$2^3 \to 8 > 17$ ×

$2^4 \to 16 > 17$ ×

$2^5 \to 32 > 17$ ✓

$r = 5$.

P_1 → 1, 3, 5, 7, 9, 11, 13, 15, 17 → 101010101 → 0

P_2 → 2, 3, 6, 7, 10, 11, 14, 15 → 00000010 → 0.

$P_4 \to 4, 5, 6, 7, 12, 13, 14, 15 \to 01000110 \to 0$

$P_8 \to 8, 9, 10, 11, 12, 13, 14, 15 \to 01000110 \to 0.$

$P_{16} \to 16, 17 \to 01 \to 0$

P_{16}	P_8	P_4	P_2	P_1
0	0	0	0	0.

∴ It does not recieve any Error.

If 8th bit is corrupted.

1	0	0	1	1	0	0	0	1	1	0	0	1	0	0	0	1
17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1

$2^0 \to 1 > 17$

$2^1 \to 2 > 17$

$2^2 \to 4 > 17$

$2^3 \to 8 > 17$

$2^4 \to 16 > 17$

$2^5 \to 32 > 17$ ✓

$\therefore r = 5$

P_{16} P_8 P_4 P_2 P_1

$P_1 \to 1, 3, 5, 7, 9, 11, 13, 15, 17 \to 101010101 \to 0$

$P_2 \to 2, 3, 6, 7, 10, 11, 14, 15 \to 00000010 \to 0$

$P_4 \to 4, 5, 6, 7, 12, 13, 14, 15 \to 01000110 \to 0$

$P_8 \to 8, 9, 10, 11, 12, 13, 14, 15 \to 11000110 \to 1$

$P_{16} \to 16, 17 \to 01 \to 0$

P_{16}	P_8	P_4	P_2	P_1
0	1	0	0	0.

∴ The Error was detected at 8th bit. so, that bit was replaced with 0.

1	0	0	1	1	0	0	0	1	0	0	0	1	0	0	0	1
17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1

corrected data

19/8/24 :- Flow Control :-

* It deals with the speed Mechanism.
* The flow control tells about how much of data the sender sends before the expecting the acknowledgement from the reciever.

Congestion = Traffic.

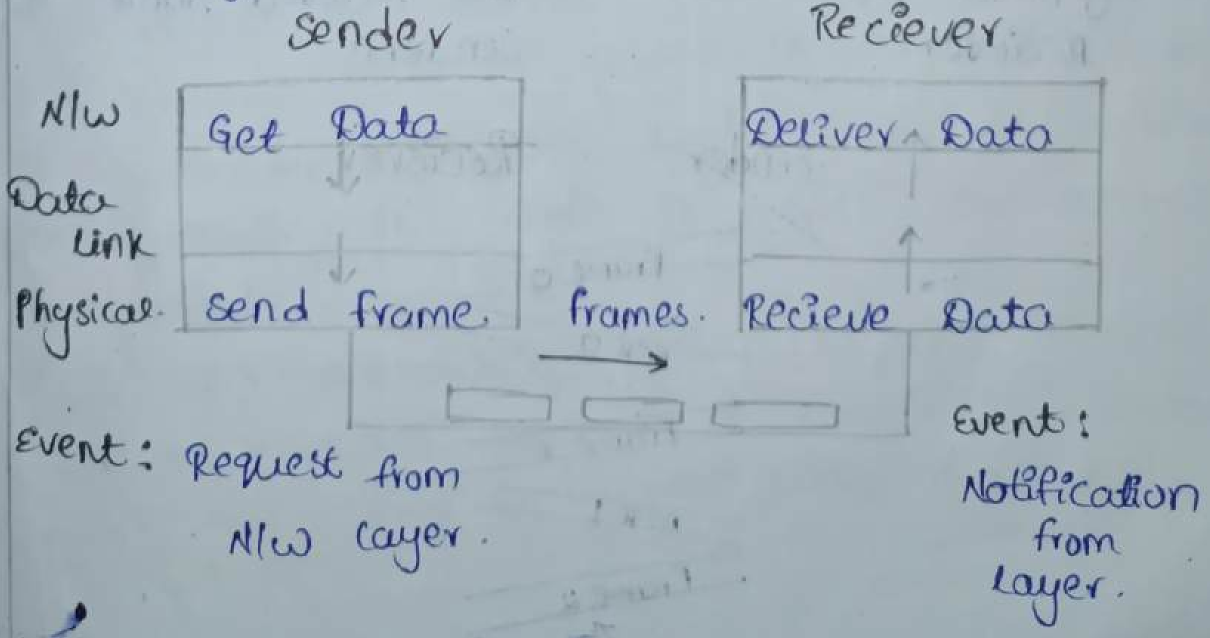
Flow Control.

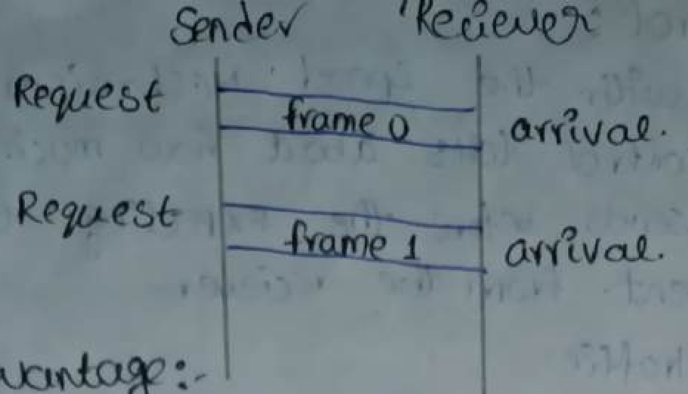
- Noiseless Channel
  - Simplest Protocol
  - Stop. and. wait. Protocol.
- Noisy Channel
  - Stop. and. wait ARQ
  - GO-BACK-N ARQ
  - Selective Repeat Protocol.

sliding window protocol.

Noiseless channel :-

1. Simplest Protocol: It is used in noiseless channel. In this the frames are transmitted in uni-direction. It will not wait for acknowledgement from the reciever. Release frames depends on availability of communication media.





Disaduantage:-

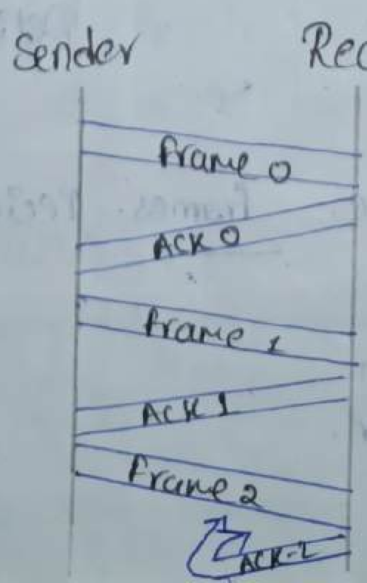
* The main draw back of simplest protocol is sender is not Expecting acknowledgement after transmission of data.

2. Stop and wait Protocol:- It is another type of noiseless - channel protocol. It provides uni-directional data - transmission with flow control facilities.

The main idea of stop and wait protococ is after transmitting a single frame the sender must wait for acknowledgement.

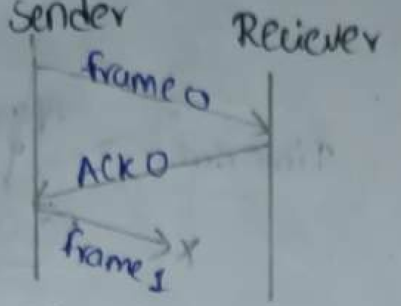
Primitives of Stop and wait Protocol:-

sender	Reciever
1. send data frame to Reciever.	1. Recieve and consume data packet from media
2. wait for acknowled-gement from the reciever.	2. send acknowledgement for the recieved frame to the sender.

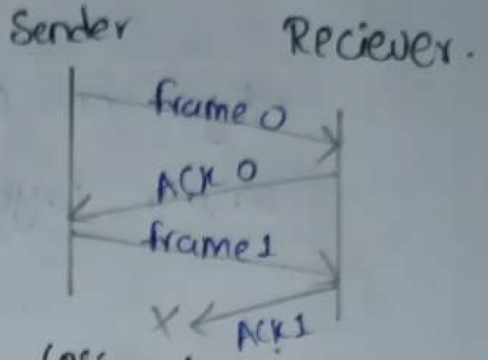


Draw back:-

The main draw back of Stop and wait Protocol is the sender waits for acknowledgement for a random amount of time and the reciever will waits for data a random amount of time for data.



Loss of frame



Loss of Acknowledgement

22/8/24

Noisy channel :-

It is a unreliable channel used to transmit different data frames.

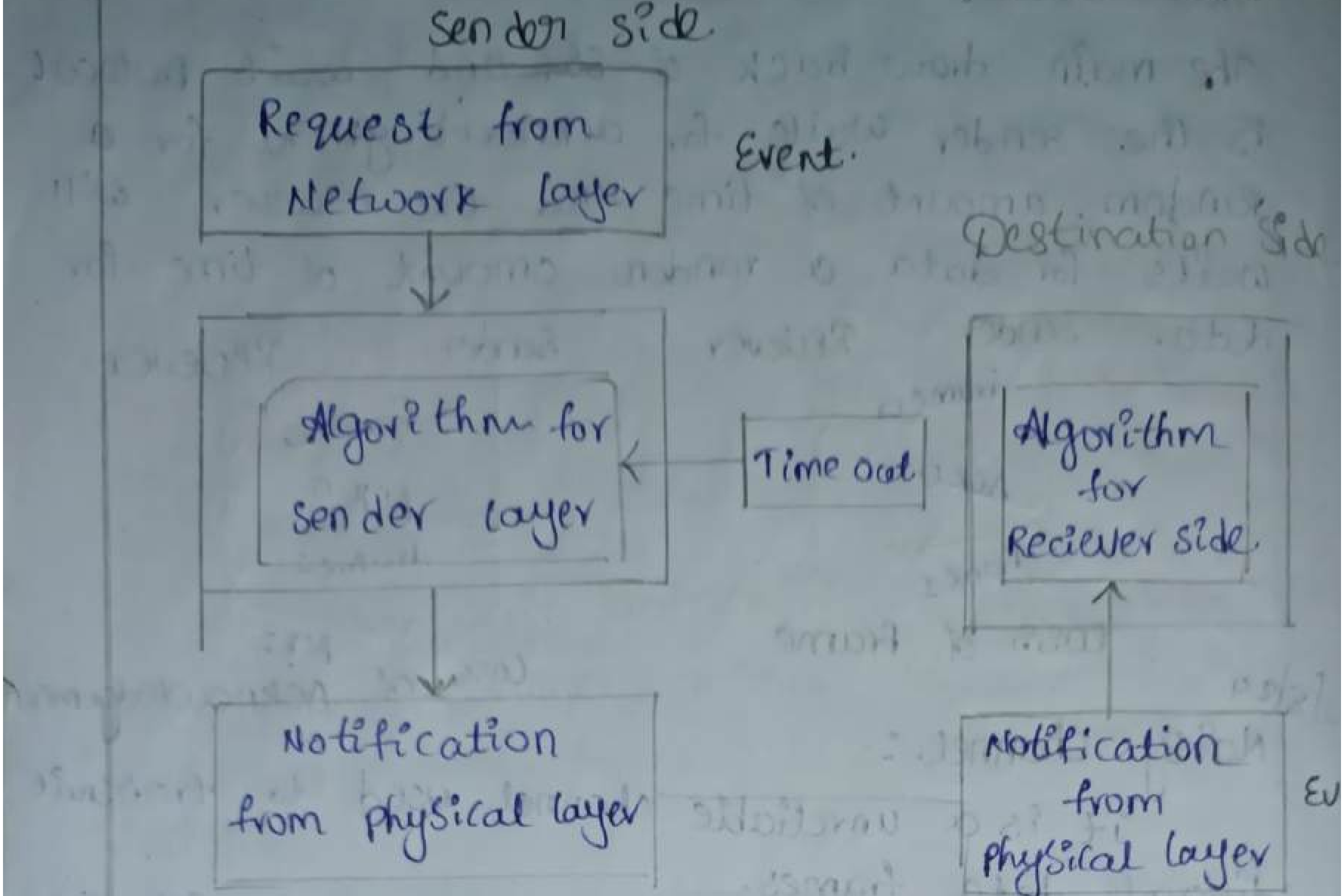
In this noisy channel we are dealing with the protocols.

They are .

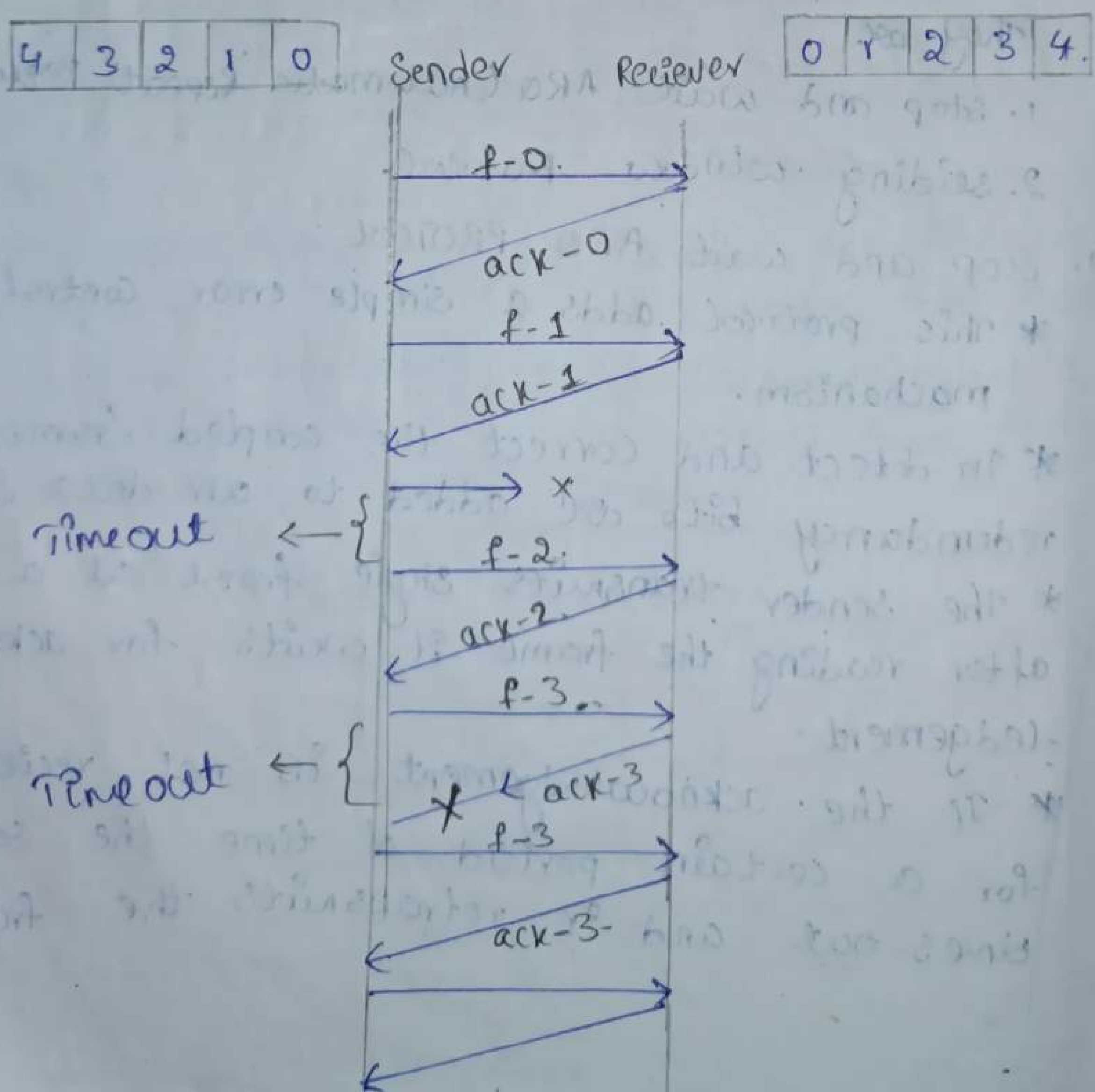
1. Stop and wait ARQ (Automatic Repeat Request).
2. sliding window Protocol.

1. Stop and wait ARQ PROTCOL :-

* This protocol adds a simple error control machanism.
* To detect and correct the coupled frames the redundancy bits are added to our data frame.
* The sender transmits single frame at a time after reading the frame it waits for acknow--ledgement.
* If the acknowledgement is not recieved for a certain period of time the sender times out and it retransmits the frame.



Working of Stop and wait Protocol:-



Sliding window Protocol :-

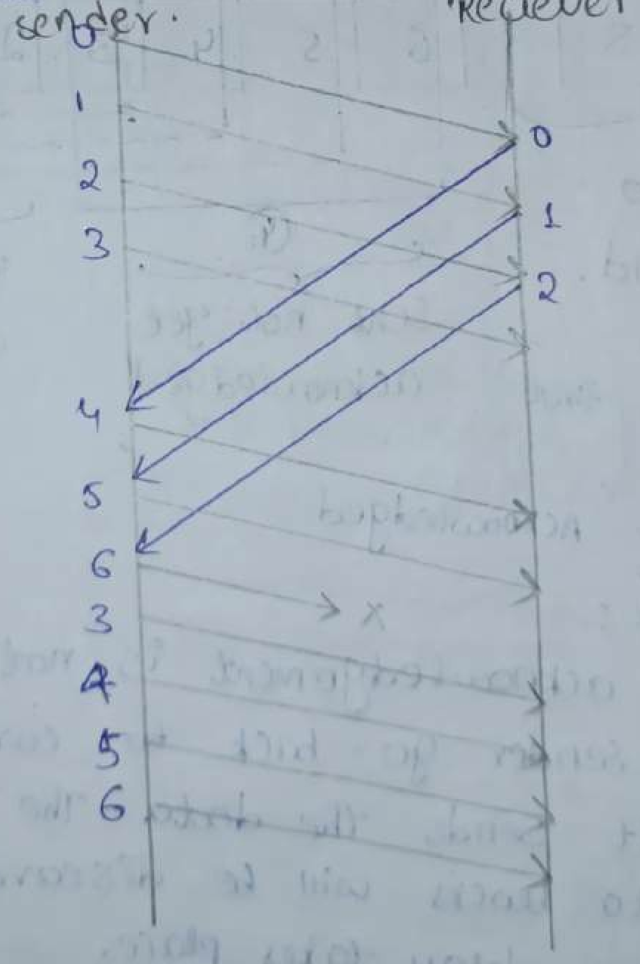
* It is a protocol that can able to transmit multiple frames at a time.
* The number of frames are to be transmitted is based on the window size (N).
* The frame are arranged in the sequence of 0 to N-1.

Ex :- If N = 3 then the number of frames are 0, 1, 2.

1. Go - Back - N ARQ Protocol
2. Selective Repeat ARQ.

Go - Back - N ARQ Protocol :-

window size (N) = 4 [Go - Back - 4].



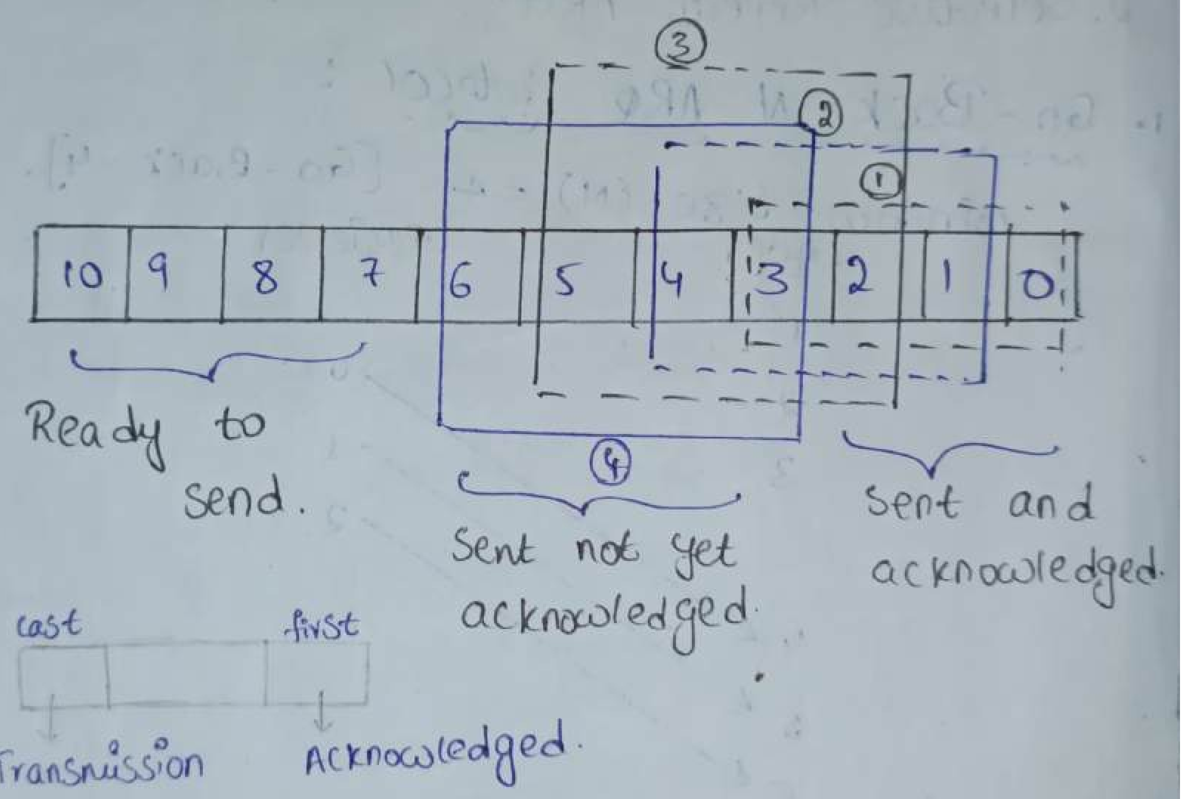
0	1	2	3	4	5	6	3	4	5	6
↓	↓	↓	×	×	×	×				

* The concept of Protocol pipelining that is the sender can send multiple frames before recieving the acknowledgement for the first frame.

* The number of frames that can be send depends on the window size of the sender.

* If the Acknowledgement of the frame is not recieved within time period. All the frames in the current window are transmitted. "2^m-1"

* where the sender window size should be 2^m-1 and the reciever window size always "1"



* Draw Back:-

If one acknowledgement is not recieved by sender, The sender go-back to current window and again it sends the data. The before recieved data blocks will be discarded. Due to this time delay takes place.

To overcome this situation, we go for selective Repeat ARQ.

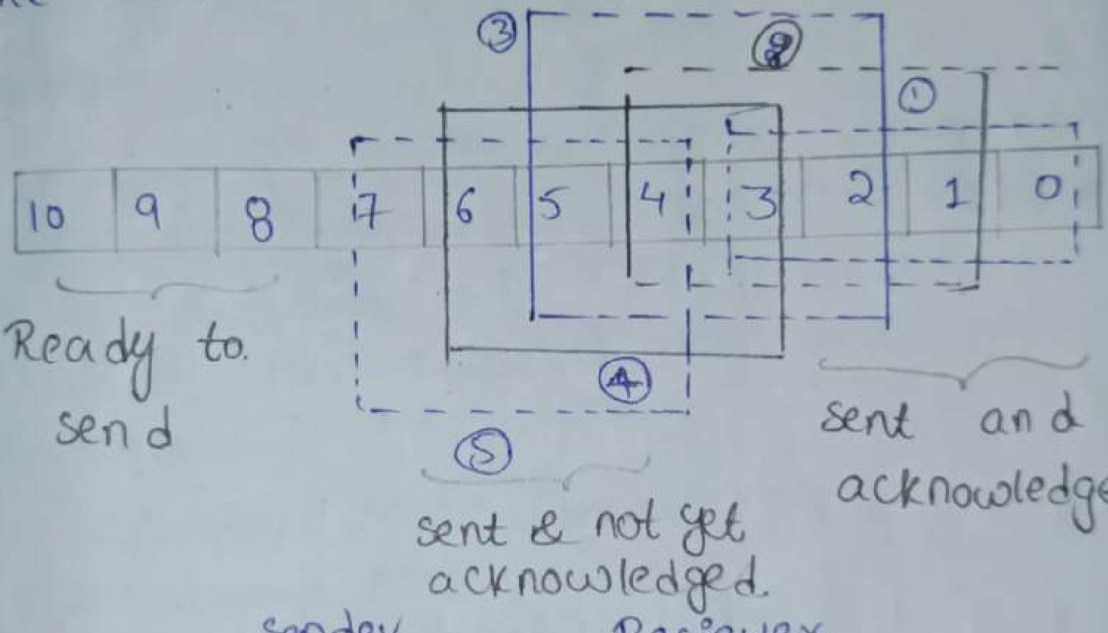
Selective - Repeat ARQ :-

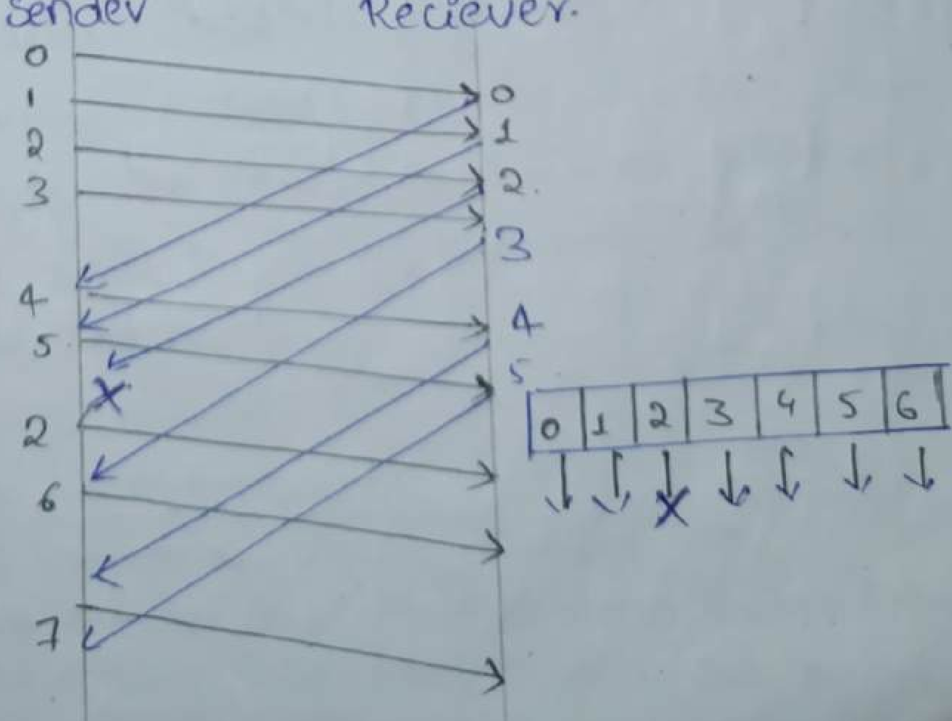
In selective Repeat ARQ only the last frames are retransmitted, while correct frames are recieved and Buffer.

The reciever keeps tracking of sequence numbers and and Buffers the frames in the memory and sends negative Acknowledgement for missing or damaged frames.

The sender will send (or) retransmitt the frame for which is negative Acknowledgement is recieved.

The sender window size is $\frac{2^m}{2}$ and the reciever window size is $2^m/2$.





7/8/24

UNIT-III

Medium Access Control Sublayer

MAC

- Random Access protocol
  - Aloha
  - CSMA
  - CSMA/CA
  - CSMA/CD
- Controlled Access Protocol
  - Reservation
  - Poling
  - Token passing
- Channelization Protocol
  - FDMA
  - TDMA
  - CDMA

Multiple Access Control Protocol :-

If the multiple stations wants to access the single channel then the multiple access protocols are used.

Multiple access protocols are divided into 3 types

They are :-

1. Random Access Control Protocol
2. Controlled Access Protocol.
3. Channelised Protocol:

1. Random Access Protocol :- All the stations have same priority. It leads to collisions. whenever the data is available within the node it places data on the channel.

Random Acess Protocol are classified into 4 types.

1. Aloha → a. Pure Alho, b. slotted Aloha.
2. CSMA → carrier sense Multiple Acess.
3. CSMA/CD → carrier sense Multiple Acess/ collision Detection
4. CSMA/CA → carrier sense Multiple Acess/ collision Avoid

2. Controlled Accessed Protocol :-

The stations cannot send any information without authorizing by another station. Controlled Accessed Protocol are classified into 3 types:

They are :

1. Reservation.
2. Polling.
3. Token Passing.

3. Channelised Protocol :-

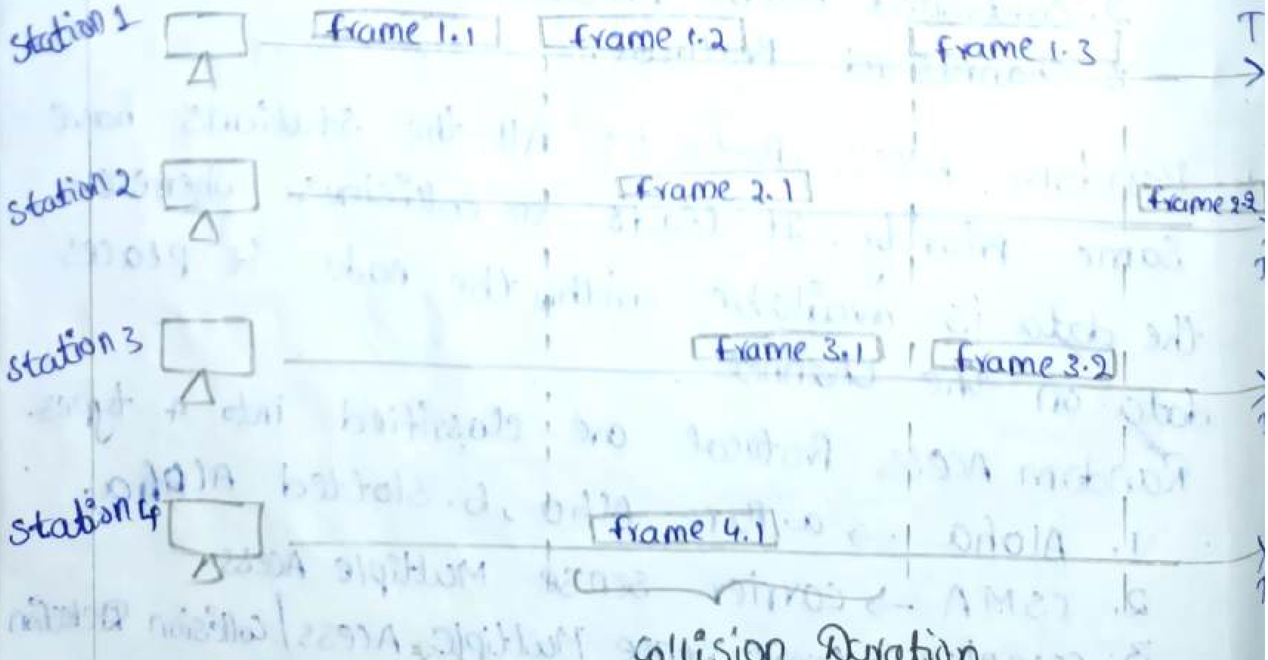
In this Protocol, the available bandwidth of media is shared in terms of time, frequency and code between different stations.

So, channelised Protocol are divided into 3 types.

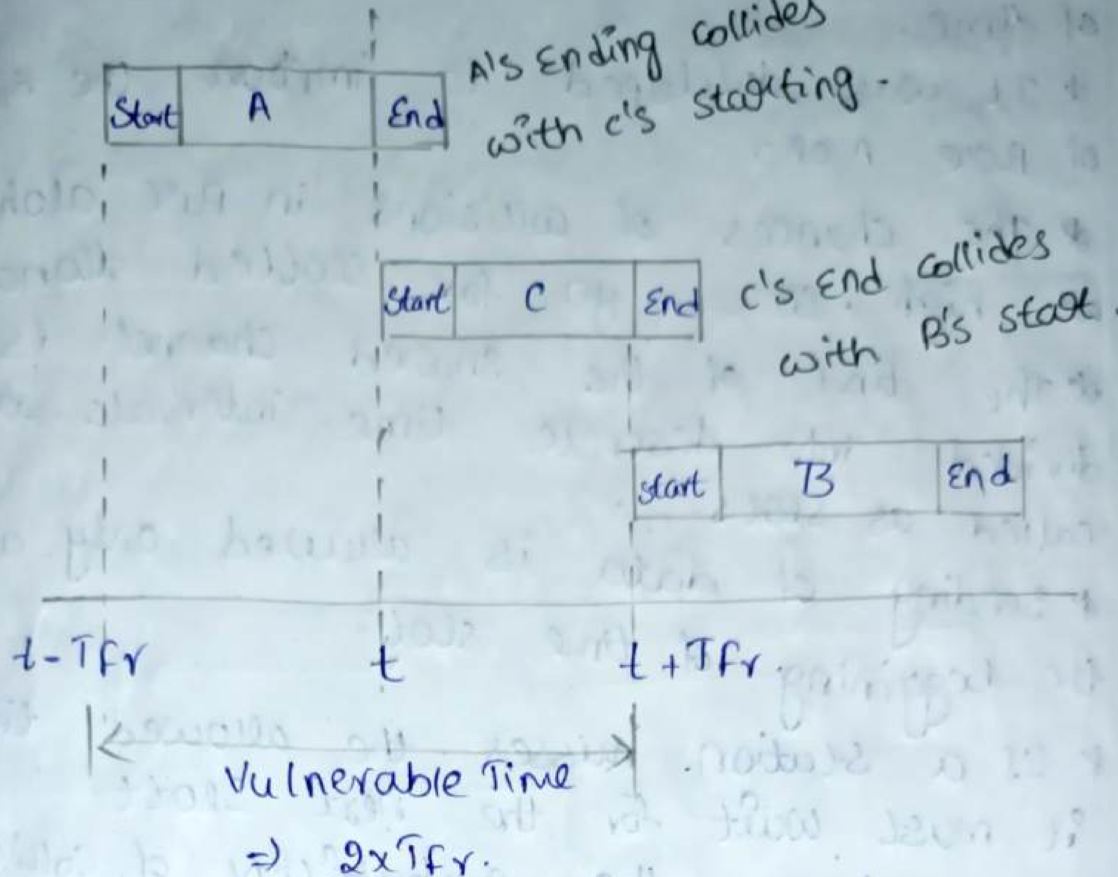
1. FDMA → Frequency Division Multiple Access.
2. TDMA → Time Division multiple Access.
3. CDMA → code Division multiple Access.

* Random Access Control Protocol :-

1. Pure Aloha :-



Vulnerable Time :



Time in which chances of collisions.

Aloha :- In computer networks, Aloha refers to a simple communication protocol for the transmission of data between devices. It was developed at University of Hawaii for wireless communication. in 1970.

Pure Aloha :- A network protocol where devices send data whenever they have data, without checking if the channel is free or not. It leads to frequent collisions.

Slotted Aloha :- It is a network protocol that improves pure Aloha by divide the channel into slots.

* Once the time slot is started we have to wait for next time slot.
* It decreases the Vulnerable time

* collision are done at a Particular slot of time.
* It was developed to improve the Effeciency of Pure Aloha.
* The chances of collisions in Pure aloha are high. So, we go for slotted Aloha.
* The time of the shared channel is divided into descrete time intervals are called as slots.
* sending of data is allowed only at the beggining of time slot.
* If a station misses the allowed time it must wait for the next slott
* It reduces the probability of collisions

* Pure Aloha Effeciency

$$S = G \cdot e^{-2G}$$

$G \rightarrow$ No. of stations want to transmit the data at a time.

$G = \frac{1}{2}$.

$S = 0.5 \times e^{-2(0.5)}$

Efficiency = 0.186

$\eta\%$ = 18.6%

* Slotted Aloha Effeciency

$$S = G \cdot e^{-G}$$

$G \rightarrow$ No. of stations want to communicate at a time.

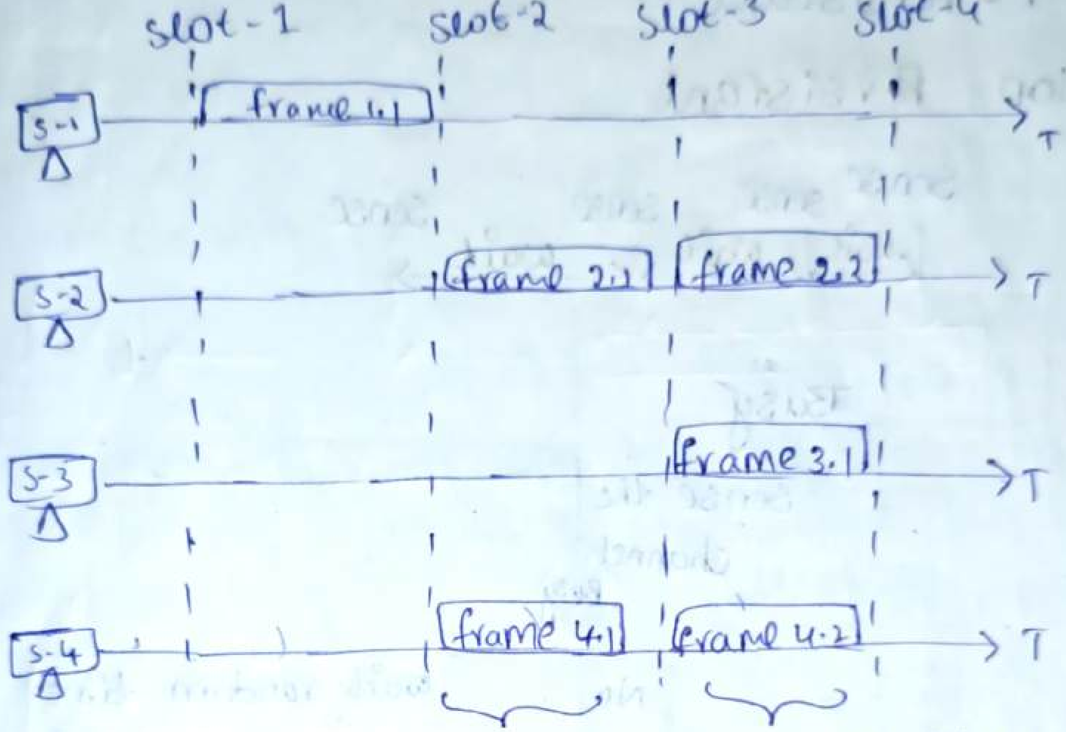
$G = 1$.

$S = (1) \cdot e^{-(1)}$

$= 0.364$

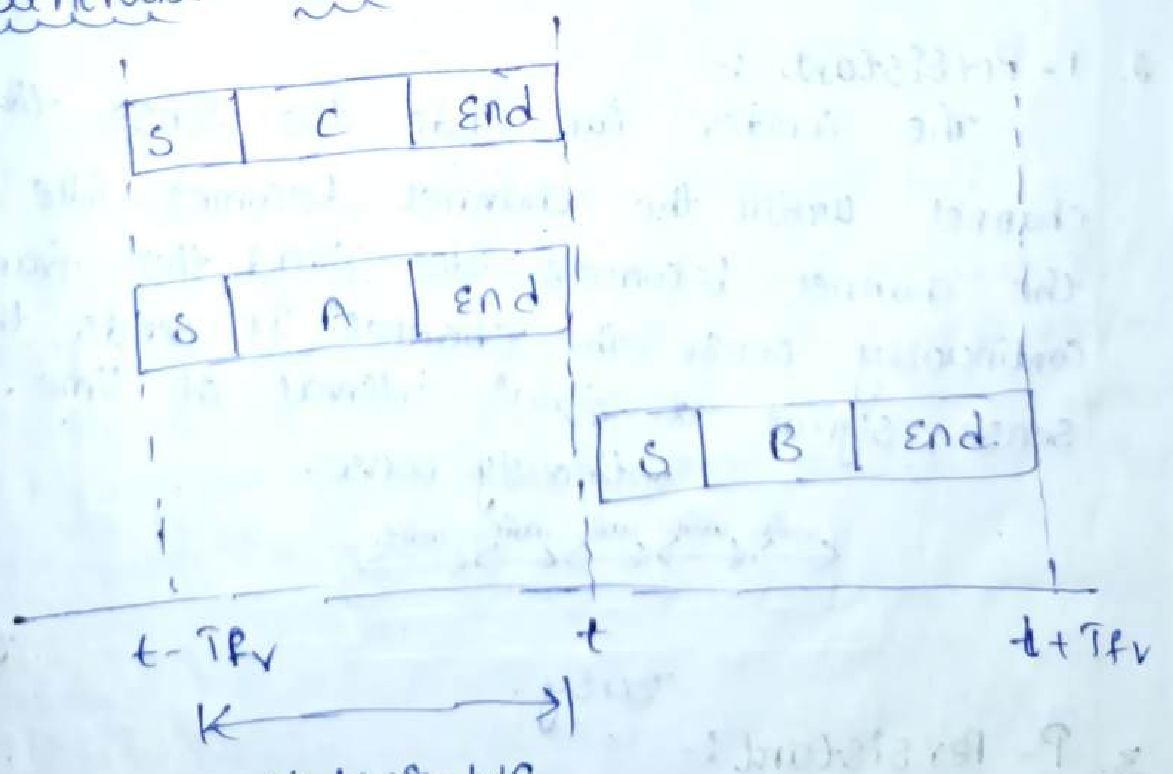
Efficiency, % $= 36.4\%$.

* Slotted Aloha



collision durations.

* Vulnerable Time :-

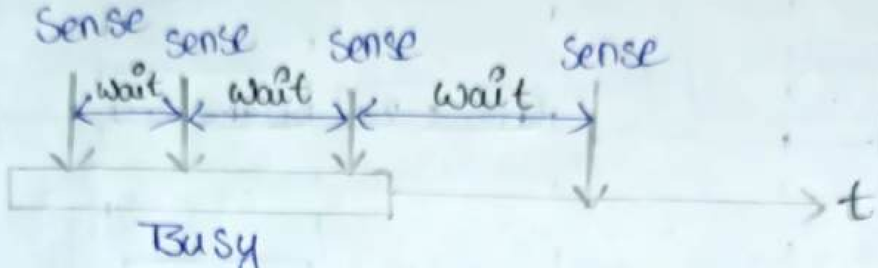


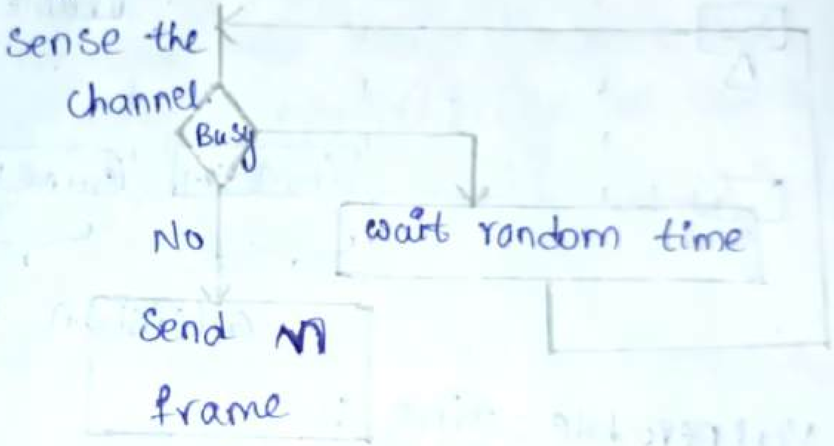
Vulnerable time = Tfr.

11/9/20 CSMA [Carrier Sense Multiple Access].

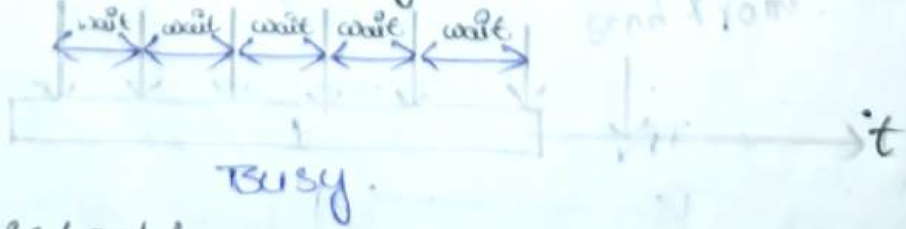
Three methods
→ Non - Perisistant.
→ 1 - Persistant.
→ P - Persistant.

1. Non - Persistant



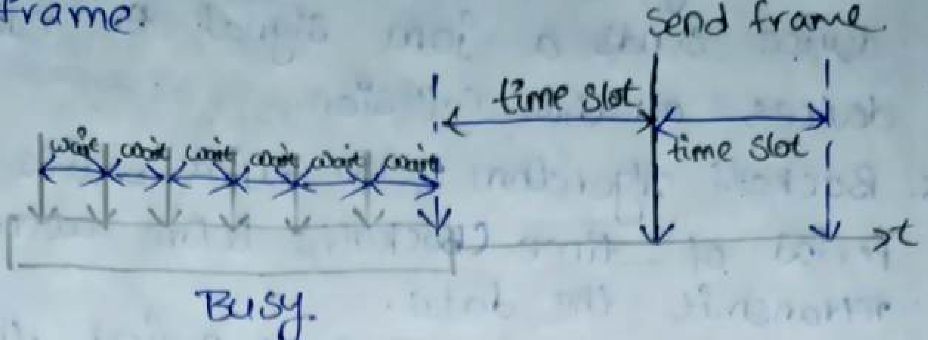


2. 1 - Persistant :-
The sender can finds the Sense the channel untill the channel becomes idle once the channel becomes idle send the frame continuosly sense the channel. It sends the sense signal at equal interval of time.

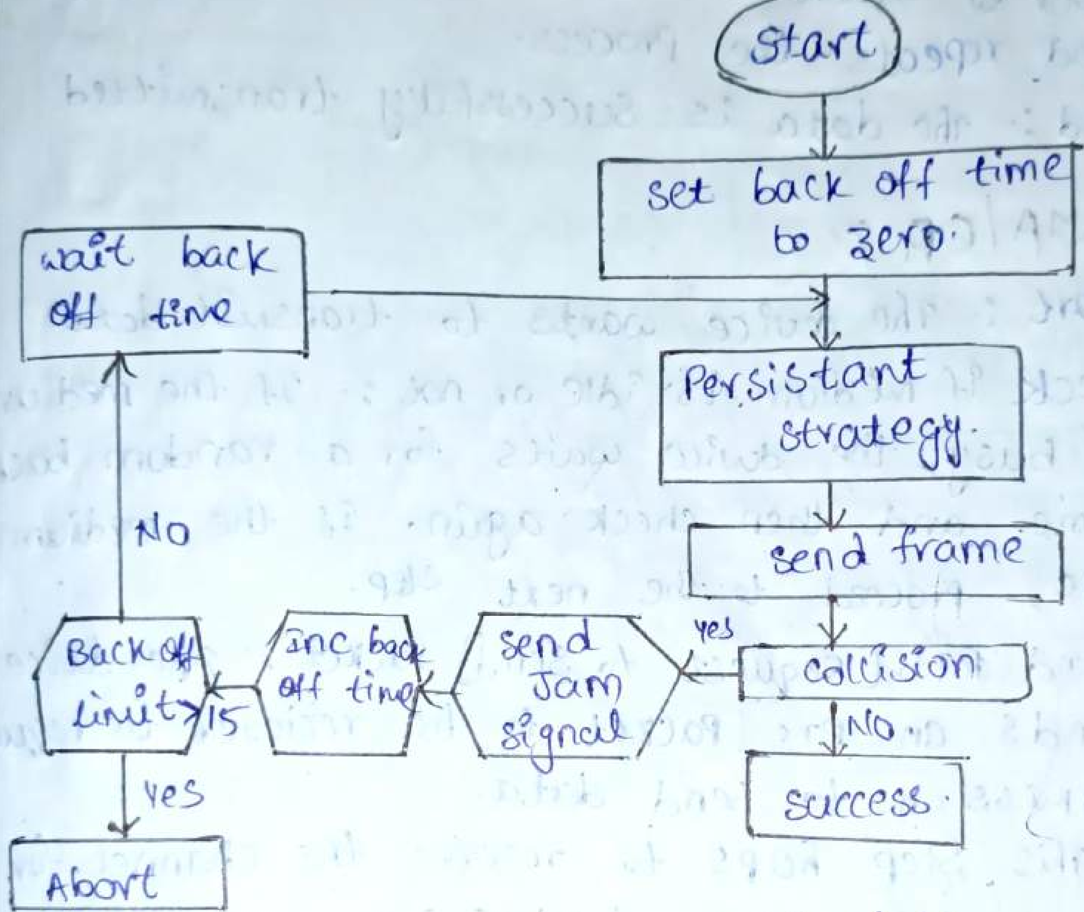


3. P - Persistant :-
The entire channel is divided into time slots. The sender continously sense the channel until the channel becomes idle. once the channel

it waits for time slot and then send a frame.



CSMA/CD :-



1. Start :- The device wants to send data.
2. check if medium is idle :- If the medium is busy, wait for a random time and check again. if the medium is idle, proceed to transmit data.
3. Transmit Data : The device starts transmitting its data over the medium.
4. collision Detection : During transmission, the device monitors the medium to detect if a collision occur.
   * If no collision is detected, the transmission completes successfully.
   * If a collision is detected, the transmission stops immediately.

5. Jam signal :- If a collision is detected, the device sends a jam signal to notify other devices of the collision.
6. Backoff Algorithm :- The device waits for a random Period of time (backoff) before attempting to retransmit the data.
7. Retry :- After the backoff period, the device goes back to check the medium if it idle again and repeats the process.
8. End :- The data is successfully transmitted.

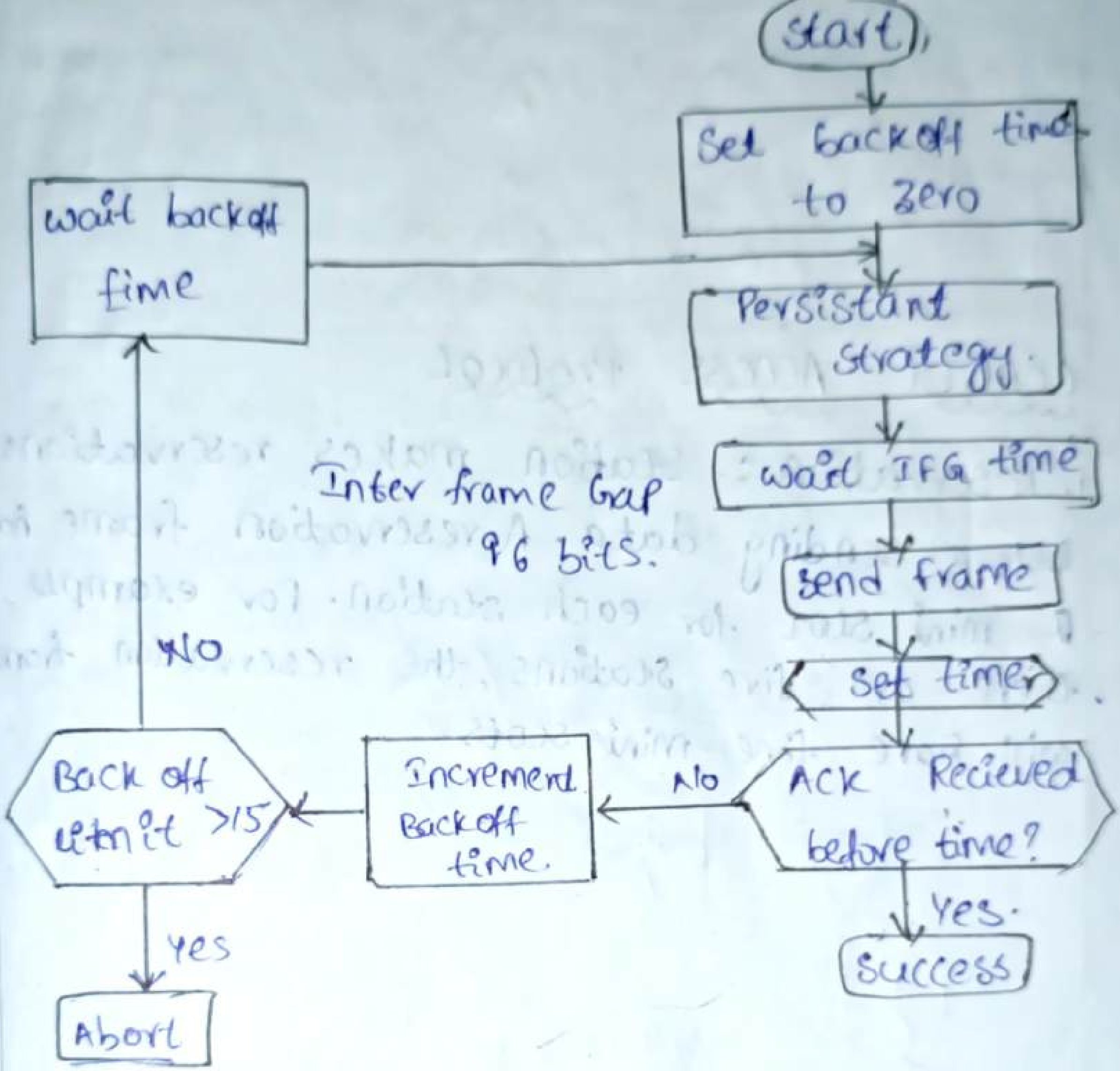
CSMA/CA :-

1. Start :- The device wants to transmit data.
2. Check if medium is idle or not :- If the medium is busy, the device waits for a random backoff time and then check again. If the medium is idle, proceed to the next step.
3. Send RTS (Request to Send) Packet :- The device sends an RTS Packet to the reciever to request permission to send data.
 This step helps to reserve the channel for the upcoming data transmission.
4. Recieve CTS (Clear to Send) Packet :- The recieving device responds with a CTS packet if it is ready to recieve data and the medium is clear.
5. Transmit Data : The device begins transmitting its data after recieving the CTS signal.
6. Collision Avoidance :- During transmission, the device checks to ensure there is no interface or collision.
 If a collision is detected, the transmission stops and the device waits for a backoff Period

before retrying.

Acknowledgement (ACK): After the data is transmitted successfully, the reciever sends an acknowledgement to confirm reciept of the data.

End : The data transmission is complete.



Ethernet

- physical [Physical media]
- Datalink. [Protocols.].

Technology -

Bridged Ethernet } Based on Configuration
switched Ethernet

Based on operating speed

Standard Ethernet : 10 MBPS

fast Ethernet 100 MBPS

Giga-bit Ethernet 1 GBPS

10-Giga-bit Ethernet 10 GBPS

→ 10-Base 5, 10-Base 2 10 Base T 10-Base F

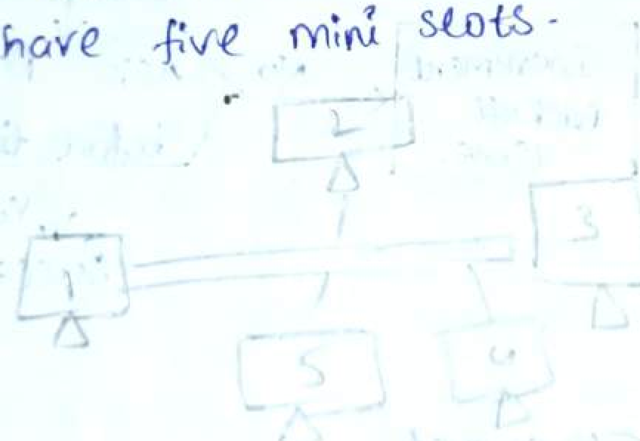
Speed(mbps type of length of

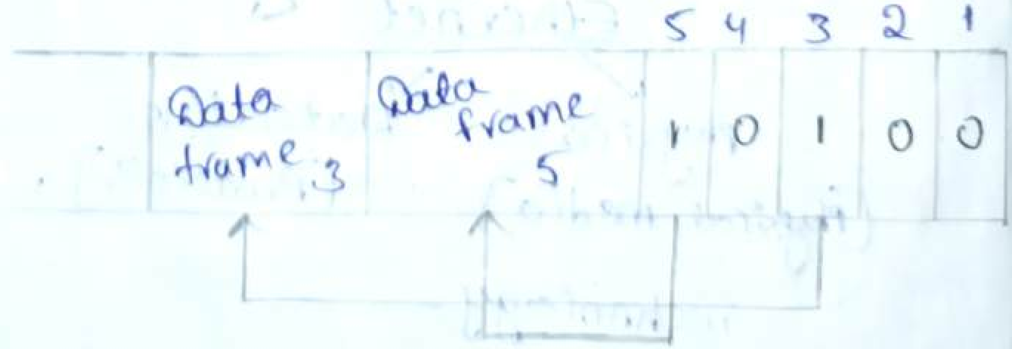
signal cable=100 m

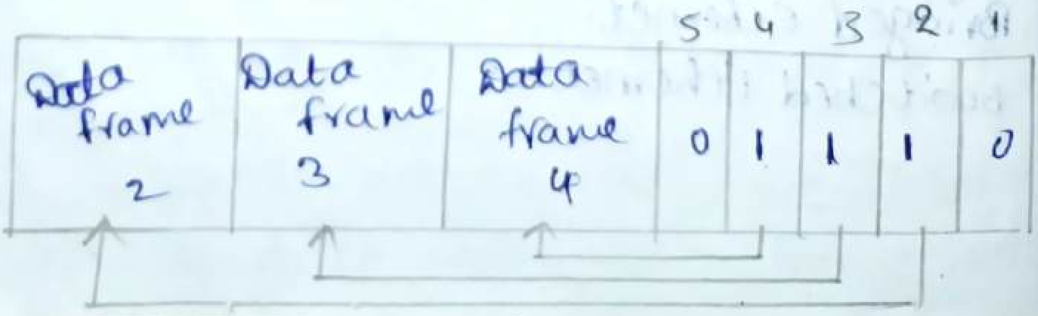
500 m

Control Access Protocol

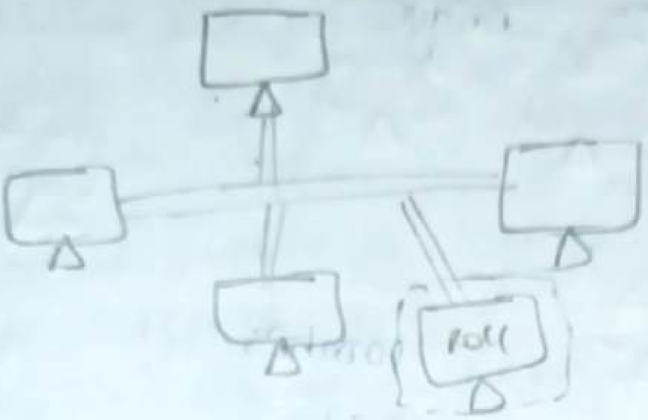
1. Reservation : Station makes reservations before sending data. A reservation frame has a mini slot for each station. For example, if there are five stations, the reservation frame will have five mini slots.





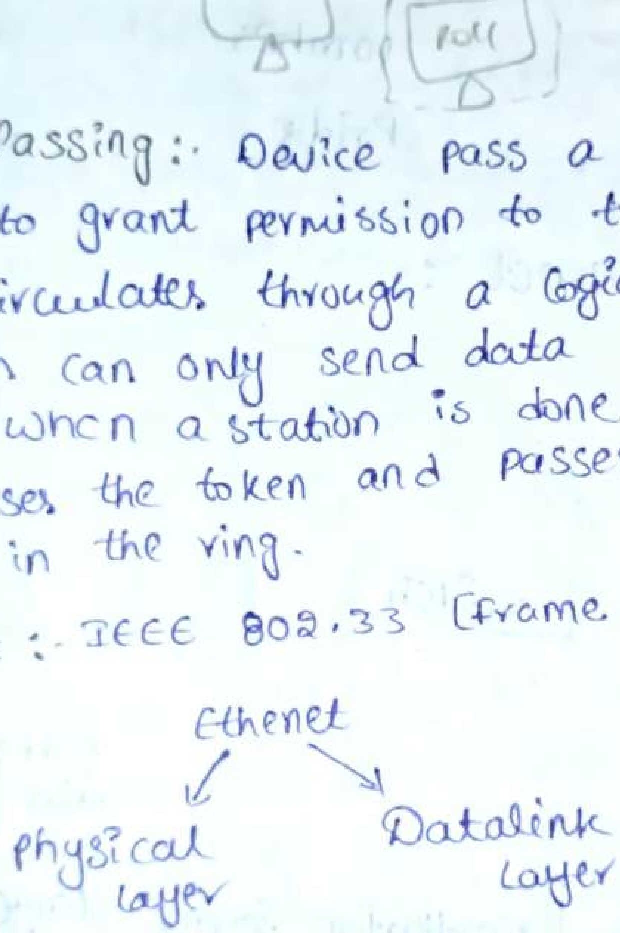


2. Polling :- A central controller ask each device if it has data to send. The controller can determine when a device has finished sending data by looking for a lack of signal on the chanel.



Token Passing :- Device pass a token to each other to grant permission to transmit data. The token circulates through a logical ring of stations A station can only send data when it has the token. when a station is done sending data, it releases the token and passes it to the next station in the ring.

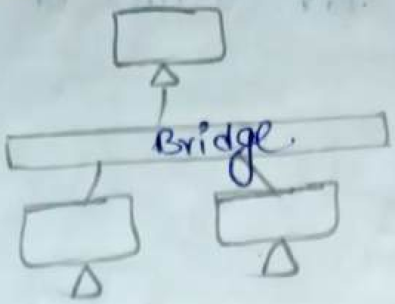
*Ethernet :- IEEE 802.33 [frame formate]

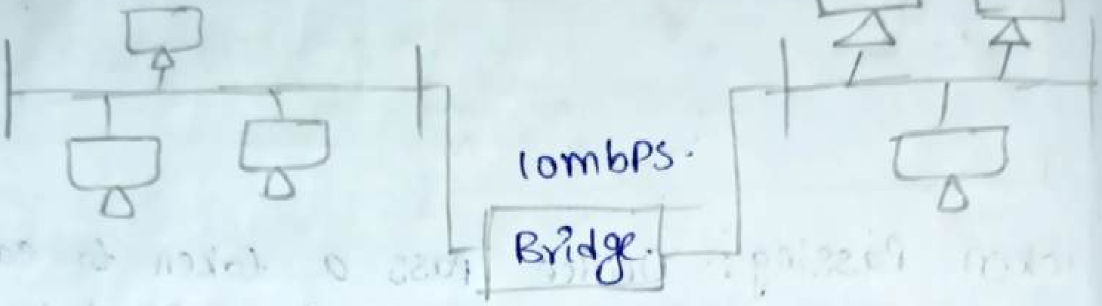


[Physical Media] Technology.   [Protocols]

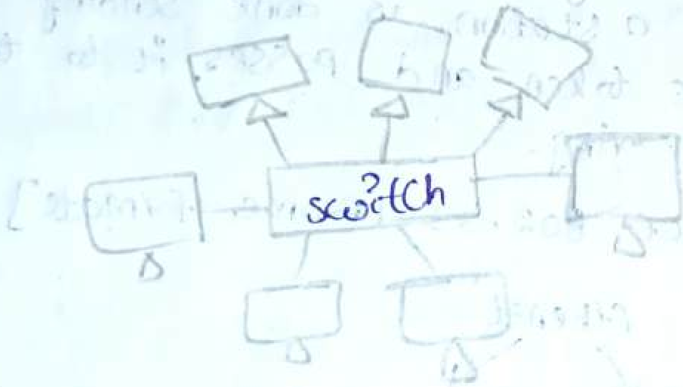
Physical Layer :-
Based on configuration.
1. Bridge Ethernet.
2 switch Ethernet

1. Bridge Ethernet :- Bridge is one of the connecting device. It operates in the lower layer in OSI reference model.





2. Switch Ethernet :-

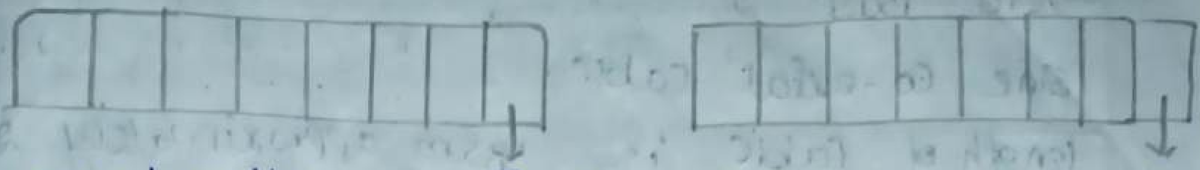


Max : 1518
min : 46

Preamble	SFD	Destination address	Source address	Lenght	Data	CRC
7 byte	1 byte	6 bytes	6 bytes	2 bytes		4 byte

* Maintain a clock or synchronization 10101010.
* SFD :- 10101011, Actual framing has going to start 10101011 = flag.
* Mack Address is nothing but Destination address MCK = Hexadecimal format.

Unicast Multicast



* Broadcasting all 48 bits are 1
* unicast → LSB bit is '0'.
* multicast → LSB bit is '1'.

Datalink Layer :-

Based on operating speed.

1. standard Ethernet: 10 MBPS
2. fast Ethernet 100 MBPS
3. Giga-bit Ethernet. 1 GBPS
4. 10-Giga-bit Ethernet. 10 GBPS.

1. Standard Ethernet.

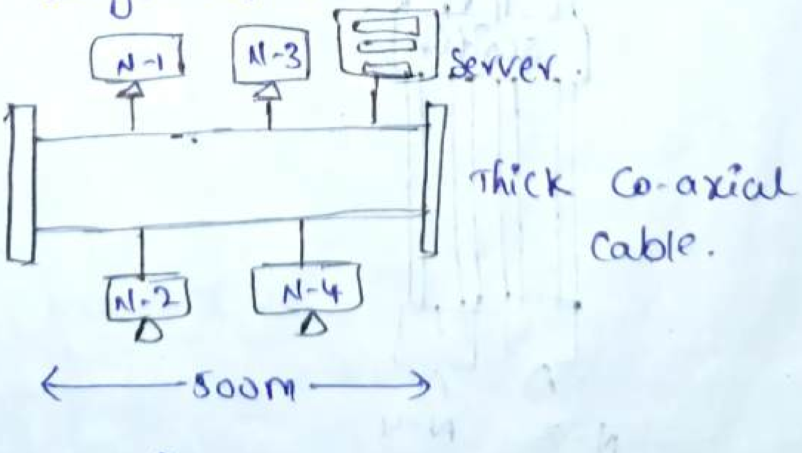
1. 10 Base 5
2. 10 Base 2
3. 10 Base-T.
4. 10 Base F.

10 → speed → 10 MBPS.

Base → Type of signal → Base Band signal.

5 → length of cable of 100 m → 5x100 ⇒ 500 m.

It is used to use the thick coaxial cable and the length of cable is 500 m.



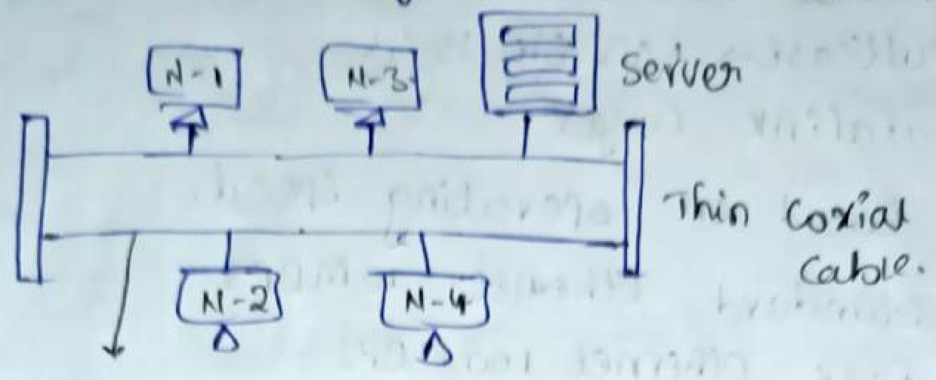
Resource sharing.

→ 10 Base 2.

Sine Co-axial cable.

length of cable is 185m approximately 200

the type of topology is bus Topology.



185 meters length.

→ 10 Base-T.

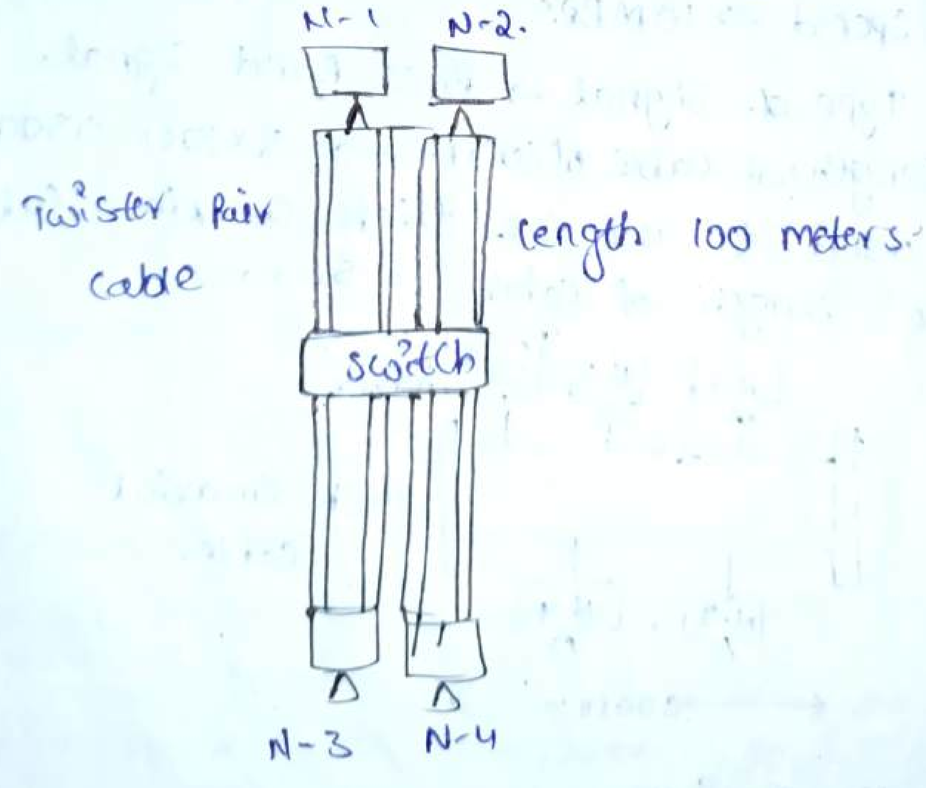
T → Twisted Pair cables

10 → speed.

length of cable is 100m.

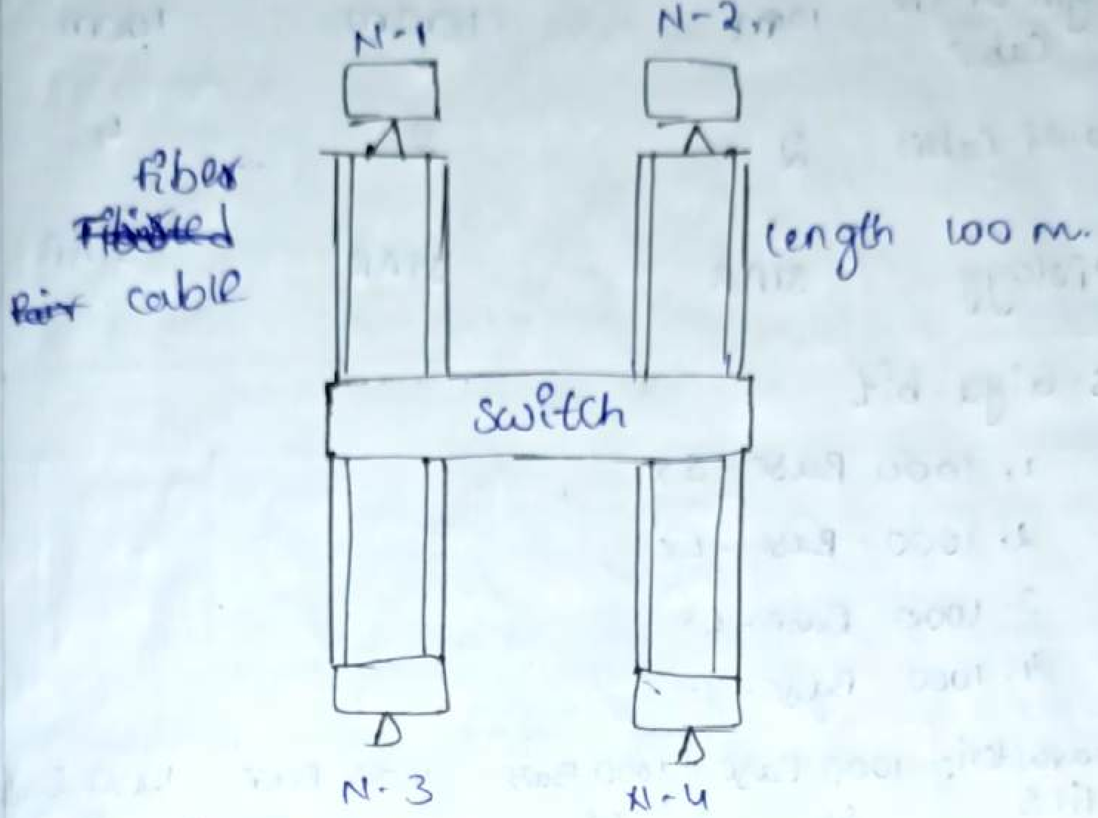
The star topology is used.

Two twisted cables are used [Transmission reciever



→ 10 Base - F.

f → fibre optical Cables.
- length of cable is 2000 m.
- Two optical cables
- Star topology is used.



CHARACTERI-STICS	10 Base 5	10 Base 2	10 Base T	10 Base f
TRANSMISSION MEDIUM	Thick coaxial cable	Thin coaxial cable	Twisted Pair cable	Fiber optical cable
LENGTH OF CABLE	500 m	200 M	100 m	2000 m.
NO. of CABLES	1	1	2	2
TOPOLOGY.	BUS	BUS	STAR	STAR

2. fast Ethernet :

It provides speed upto 100mbps.

1. 100 Base Tx
2. 100 Base Fx
3. 100 Base T4.

CHARACTERISTICS	100 Base TX	100 Base FX	100 Base T4.
Transmission media.	Twisted Pair cable [UTP]	Fiber optic cable	Twisted Pair cable [STP].
Length of the cable	100m	100m	100m.
No. of cables	2.	2.	4.
Topology	STAR	STAR	STAR

3. Giga-bit

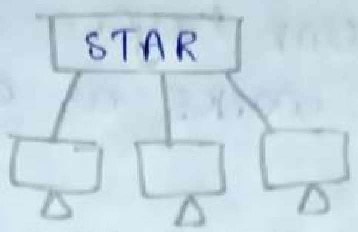
1. 1000 Base - SX
2. 1000 Base - LX
3. 1000 Base - CX
4. 1000 Base - T.

Characteristics	1000 Base SX	1000 Base LX	1000 Base CX	1000 Base T
Transmission media	Short wave optical fiber	long wave optical fiber	shielded twisted Pair Cable [STP]	Unshielded twisted Pair cable [UTP].
Length of cable	550m	5000m	25m	100m.
No. of cables	2	2	2	4
Topology	Point to Point STAR 2 STAR Hierachy Star.	Point to Point Star 2 Star Hierachy Star	Point to Point Star 2 Star Hierachy Star	Point to Point Star 2 Star Hierachy Star

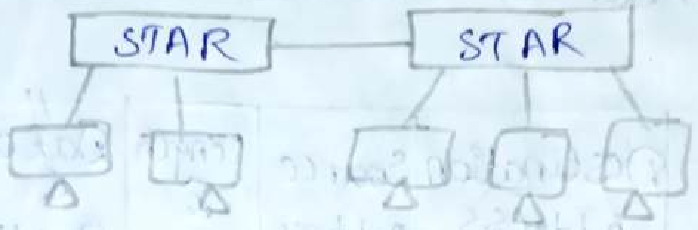
It will provides speed up to 1Gbps.

Point-to-Point Topology.

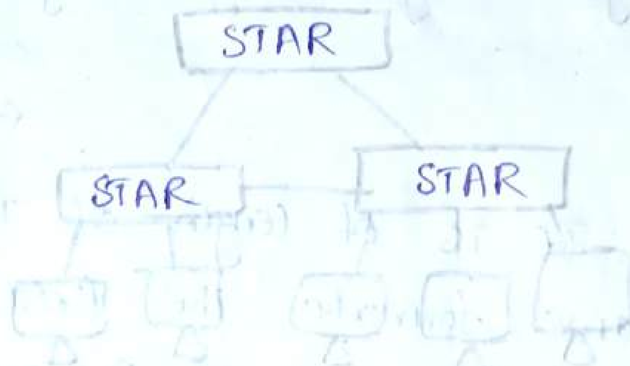
Star Topology



2 Star Topology.



Hierachy Star Topology.



10 Giga-bit.

It will provides speed up to 10 Gbps.

1. 10G Base S
2. 10 G Base L
3. 10 G Base E.
4.

Characteristics	10G Base S	10G Base L	10G Base E.
Transmission media.	Short wave 850 nm optic fiber with single mode/multi mode.	Long wave 1310 nm optical fiber with single mode	Extended 1550 nm single mode optical fiber

Length of cable	300 m	10 km	40 km
No. of cables	2	2	2.
Topology	STAR	STAR	STAR

21/9/24

Ethernet in Data link layer.

The Ethernet is works as a protocol. in datalink layer.

The IEEE standard of ethernet is 802.3

The frame format of Ethernet is as follows.

Preamble	SFD	Destination Address	Source Address	Length & Type	Data & Padding	CRC
7 byte	1 byte	6 bytes	6 bytes	2 bytes	←46→ min ←1500→ Max	4 bytes

1. Preamble :- It is of length of 7 bytes. In Ethernet frame formate, the first field is Preamble. The sequence of Preamble field consists of 56 alternate 1's and 0's. The field is used for synchronization purpose

2. SFD :- SFD stands for start frame Delimiter which is having of length is 1 byte [8-bits] According to length of 8-bits, the user can define the type of casting of data

* The LSB of SFD is 0 then it is said to be unicast Address.
* The LSB of SFD is 1 then it is said to be multicast Address.

* The entire sequence of SFD is 1s then it is said to be broad cast Address.
Ex:- TV Transmission.

* The SFD is used for synchronization purpose.
  → Physical Address - Datalink.
  → IP Address - Network.
  → Port Address - Transport.

3. Destination Address and Source Address:-
It is nothing but the physical address of particular nodes which are in between communication link. It is of length 6 byte [48-bits].
Basically, the address is denoted by using hex-decimal notation separated with colon ":"
Ex:- 47:7C:AB:64:EC:12.

4. Length and Type:- The speed is having a length of 2 bytes [16-bits]. It is used to define the data type and length of data.

5. Data and Padding:- Data and padding, which contains the original information have to be transferred. In this, padding concept is used to maintain fixed size framing.
The minimum length of data field is equal to 46 bytes.
The maximum length of data field is equal to 1500 bytes.

6. CRC:- CRC means Cyclic Redundancy check. It is of length is 4 bytes [32-bits]. It is mainly used for detect Errors by adding Extra bits.

Maximum length of Ethernet Format
[[7 +1+6+6+2+4] + [1500]] bytes = 1526 bytes
= 12,208 bits.

Minimum length of Ethernet format

$[[7+1+6+6+2+4]+[46]]$ bytes

$= 72$ bytes

23/9/24 $= 576$ bits.

Wireless LAN (Local Area Network)

Wifi = IEEE 802.11

In IEEE, we use the technology of FHSS [Frequency Hopping Sperad Spectrum]. It operates at 2.4 GHz. This 2.4 GHz are divided into 79 channels.

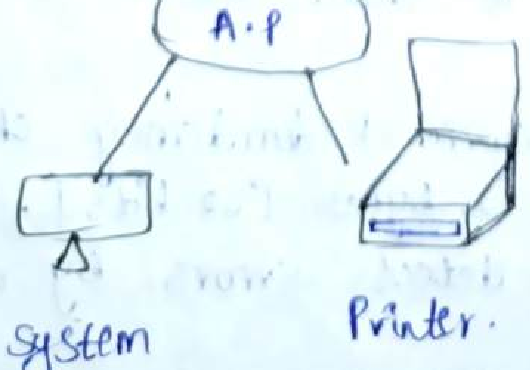
Channel → frequency bands.

we have separate each and every frequency by guard band. It jumps from one frequency band to another frequency. band.

Based on the type of the mode. There are two types of network.

1. Infrastructure Mode.
2. Ad-hoc Mode.

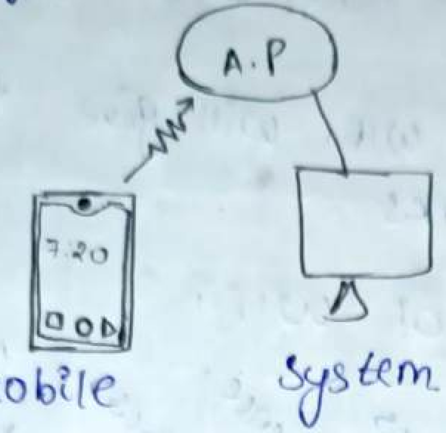
1. Infrastructure Mode :- whenever the nodes are in stationary modes that network is called as infrastructure network.



No change of network.

In this state, the systems & nodes are stable.

2. Adhoc Mode:- whenever the nodes are in mobility, that networks are called Ad-hoc network. It is the combination of stable and mobility.

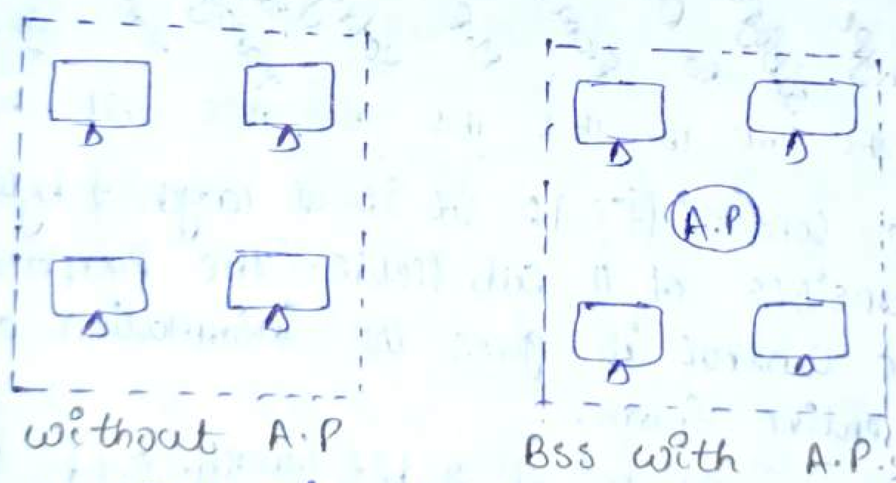


change of Network

* Architecture of 802.11 (or) WIFI

1. Basic service set (BSS)
2. Extended service set (ESS).

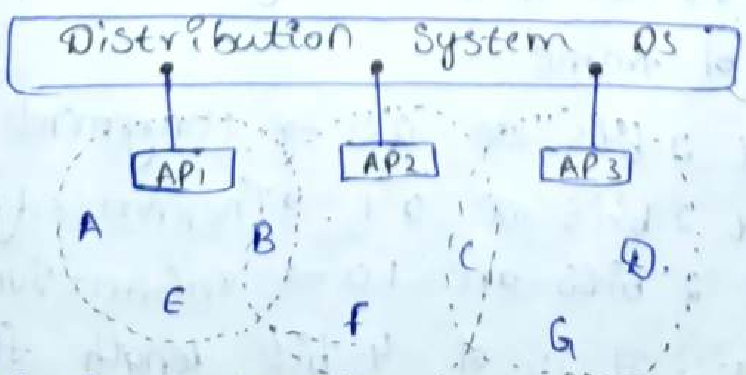
1. Basic service set :-



BSS without A.P   BSS with A.P.

Through Access point, the communication takes place without A.P there is no communication.

2. Extended service set.



Node - Node
Node - AP
AP - Node
AP - AP.

It is a collection of more than 1 BSS is called ESS.

* Scanning :- The process of selecting an Access point by a node is known as Scanning. In this we have 2 types of scannings.

1. Active Scanning :- Node selects particular Access Point.

2. Passive Scanning :- wifi will give the available Access point networks.

25/9/24

Frame Format of WIFI

frame control	Duration	address 1	address 2	address 3	sequence	address 4	Data	checksum
2 byte	3 byte	6 byte	6 byte	6 byte	2 byte	6 byte		4 byte

Version	Type	Subtype	To DS	From DS	M.F	Retry	Power management	more data	WEP	RVSD
2 bits	2 bits	4 bits	1 bit	1 bit	1 bit	1 bit	1 bit	1 bit	1 bit	1 bit

* Frame Control [FC] :- It is of length 2 byte length. It consists of 11 Sub fields. The purpose of frame control is gives the information about the entire frame.

1. Version :- It is of 2 bit length field. It gives the type of version like 802.11, 802.11a, 802.11b

2. Type :- It is of 2 bit length field. It defines the type of frame

* If first 2 bits are 0 0 → Management frame
* If first 2 bits are 0 1 → [Acknowledged] Control
* If first 2 bits are 1 0 → Information frame

3. Sub type :- It is of 4 bit length field. It defines the sub type of the frame based on field type

4. To DS :- It is of 1 bit length. Based on this field type of address is defined.
It is set to 1. Nodes send the data to Ds. while from DS is set to 0.

5. from DS :- It is of 1 bit length. Based on this field type of address is defined. While from DS is set to 1., D.S sends the data to nodes. then To DS is set to '0'.

6. M.F [More fragmentation]:- It is of 1 bit length. It is used to split the original information into Parts. If it is set to 1, the transmitted data requires more splitting. otherwise '0'.

7. Power Management :- If it is said to 1, the sender node is in the Power saving mode. otherwise it is set to '0'.

8. Retry [Retransmission] :- It is of 1 bit length. If any case, the frame is ~~reda~~ retransmitted, it is set to '1'. otherwise it will set to '0'.

9. More data :- It is of 1 bit length. It is always set to '1'. If it is a last frame, set to '0'.

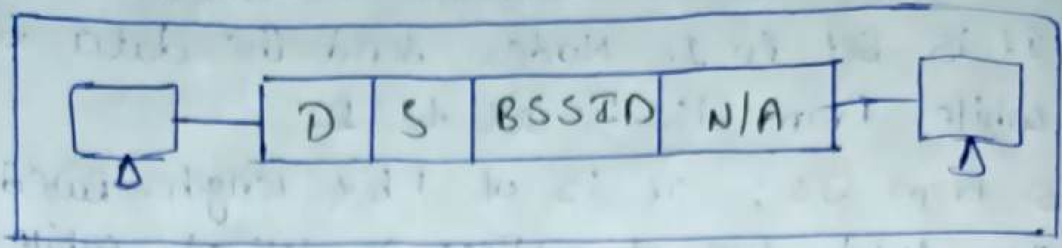
10. WEP [Wired Equivalent Privacy]:-
If WEP is set to '1', the information is related to privacy. If it is said to '0', the information is related to Public.

11. RVSD [Reserved field]:- It is of 1 bit length It is always set to '0'. It is for future adding purpose.

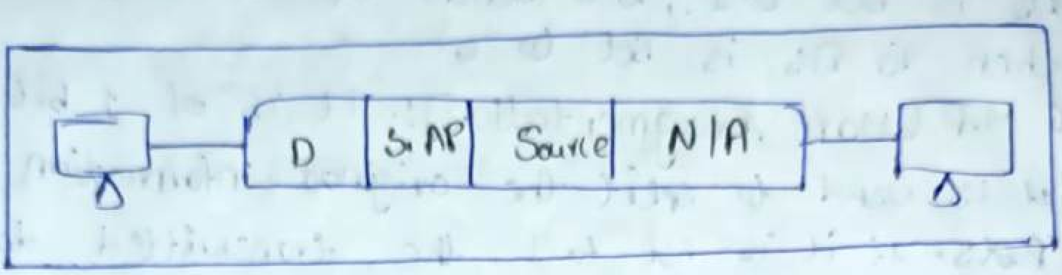
Addressing of 802.11.

To DS	from DS	address 1	address 2	address 3	address 4
0	0	Destination	Source	BSS ID	N/A.
0	1	Destination	sending A.P	Source	N/A.
1	0	Recieving A.P	Source	Destination.	N/A
1	1	Recieving A.P	Sending A.P	Destination	Source.

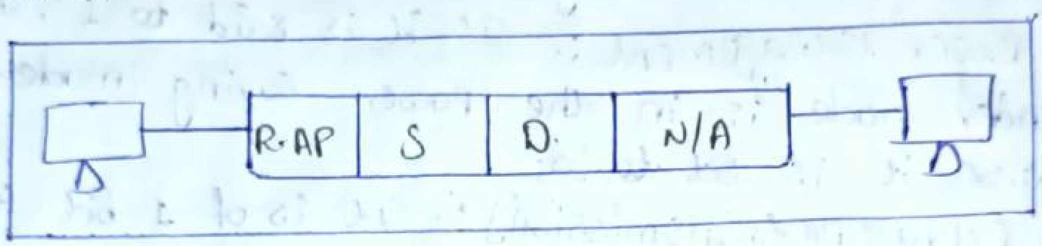
Case - i, 00



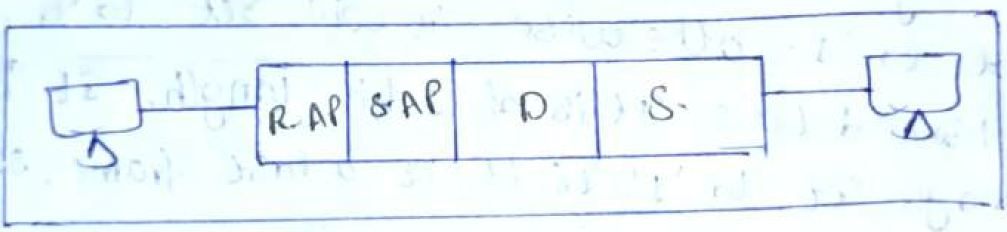
case - ii 01



Case - ii 10



case - iii, 11.



Ethernet WIFI Protocols.

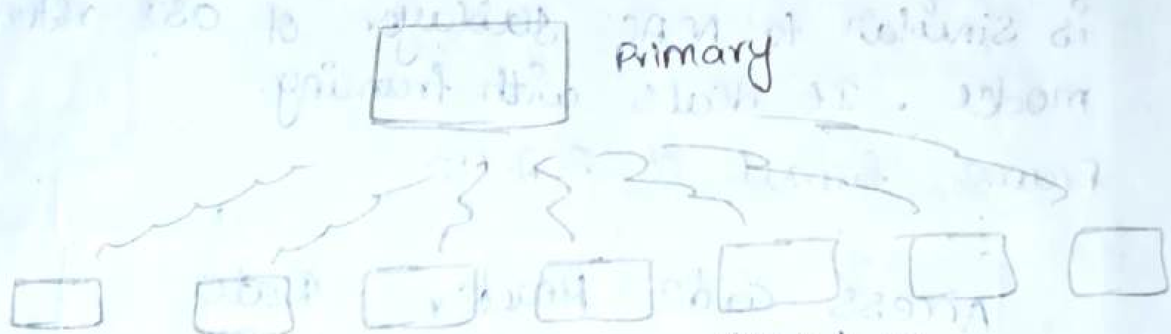
S.No	Protocol Standard	Frequency	Channel width	operating Speed
1.	802.11	2.4 GHz	20 MHz	2 MBPS
2.	802.11b	2.4 GHz	20 MHz	11 MBPS.
3.	802.11a	5 GHz	20 MHz	54 MBPS.
4.	802.11 G	2.4 GHz	20 MHz	54 MBPS.
5.	802.11 N	2.4/5 GHz	20, 40 MHz	450 MBPS.
6.	802.11 AC (wave 1)	5 GHz	20, 40, 80 MHz	866.7 MBPS.
7.	802.11 AC [wave 2]	5 GHz	20, 40, 80 & 160 MHz	1.73 GBPS.
	802.11 AX	2.4/5 GHz	20, 40, 80, 160 MHz	2.4 GBPS.

Bluetooth

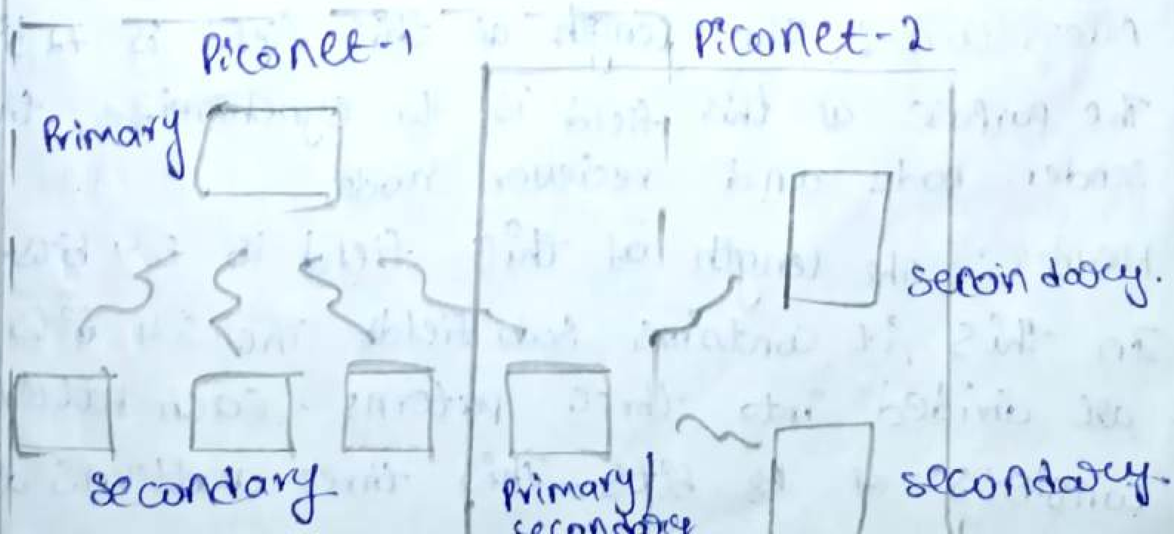
It is a wireless PAN [Personal Area Network] Technology. The IEEE standard for Bluetooth is 802.15. It operates within 10 meters. It is also have an architecture. They are two types.

1. Piconet.
2. Scatternet.

1. Piconet :- It consists of 8 nodes. one node acts as primary node. It is also known as master node. Remaining 7 nodes are acts as secondary nodes. It is also known as slave nodes. only Primary device can send or recieve the information from slave nodes.



2. Scatternet :- The Combination of more than one or two Piconets is called scatternet. If one node in Piconet can act as Secondary same node can act as Primary node in Piconet 2.



There are 4 layer in bluetooth

Application layer
LLCAP layer
Base band layer
Radio layer.

Radio layer :- It is similar to the physical layer. It deals with transmission media and synchronization. It operates at 2.4GHz and the transmission media is wireless-FHSS. The transmission media is wireless-

Base Band Layer :- The broad band layer is similar to MAC sublayer of OSI reference model. It deals with framing.

Frame format of 802.15

Access code	Header	Data
72 bits	54 bits	

Address	Type	F	A	S	HEC
3 bits	4 bits	1 bit	1 bit	1 bit	8 bits

Access code :- The length of this field is 72 bits. The purpose of this field is to synchronize the sender node and reciever node.

Header :- The length of this field is 54 bits. In this, it contains sub fields. The 54 bits are divided into three paterns. Each pattern consists of 18 bits. This three patterns are same. first pattern is original information and remaining are duplicate information.

Address :-

It is of 3-bit length field it gives the destination address to which secondary you want to transfer the data.

* If three bits are 000 → Broadcast Addressing
* If three bits are 001 → secondary one.
* If three bits are 010 → secondary two
* If three bits are 011 → secondary three.

Type field :-

It is a four bit length. It identifies the type of frame.

flow control (F):-

If it is set to '1' then the buffer of that node is full.

Acknowledgement (A) and Sequence Number :-

The bluetooth uses stop and weight ARQ If this field is set to '1'. then it is an acknowledgement.

Information → Sequence number.

Acknowledgement → Ack number

Header Error Control [HEC]:-

It is a 8-bit field. It deals with Error Controlling and here we are using checksum as a Error detection technique.

Data :-

Data is nothing but information (or) payload The length of this fied is varies.

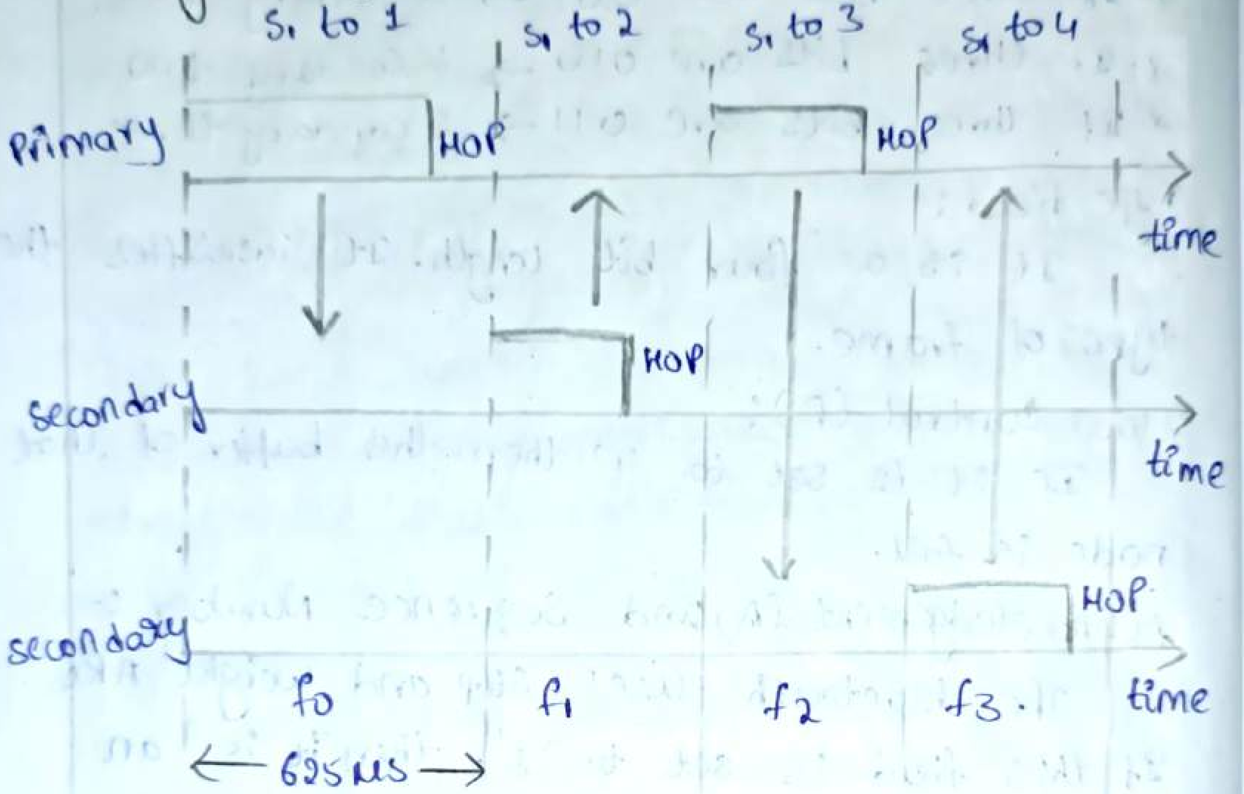
* LLCAP layer :- Logical link control adoption layer. It is a sub layer of data link layer. It deals with the Effective virtualization of

channel.

L2CAP: We are using TDMA channelization protocol.

The Entire channel is divided into time slots in one second. It changes to 1600 times. Each time slots are divided into 625 μs.

Timing Diagram:



Primary uses odd number of slot of time and secondary uses the even number of time slot.

Power Classes of Bluetooth.

Class	Power (mW)	Power DB	Range.
Class 1	100 mW	20 dB	≃ 100 m
Class 2	2.5 mW	4 dB	≃ 10 m
Class 3	1 mW	0 dB	from ≃ 10 cm to 1 m.

28/9/24

UNIT-4

NETWORK LAYER

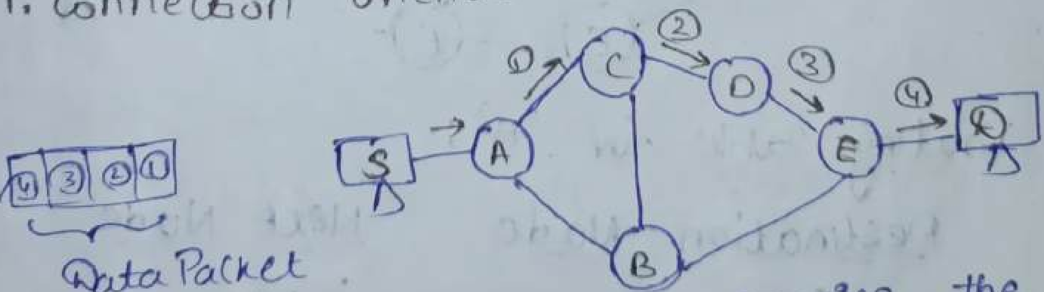
* **Network Layer:-** The basic function of network layer is to provide end to end delivary of data in between data link layer and transport layer. The data is called packets in network layer. The packet is a group of data bits that includes address. In the network layer, logical address is added addition to this packet.

The services provided by the network layer are :

1. Routing.
2. Logical Address.

1. **Routing:-** The concept of finding optimum path from source to destination is known as routing. Basically, the routing is provided by 2 services. They are:
   1. Connection oriented service.
   2. Connectionless oriented service.

1. 1. Connection oriented service:-



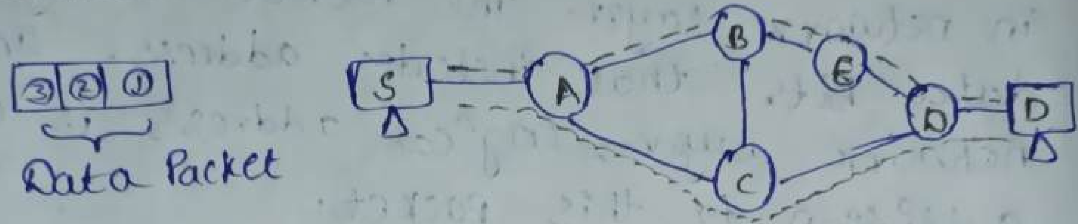
Data Packet

In the connection, oriented service, the Path is defined before the transmitting the data, during the entire transmission, the Packets are travelled through that defined Path only. The process of connection oriented services includes.

1. Connection establishment.
2. Data Transfer.
3. Termination.

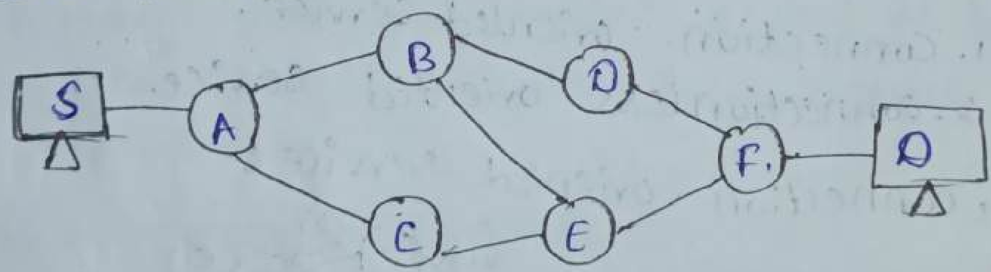
In the connection oriented service, the data Packets are called as virtual circuit Packets.

2. Connection-less oriented service :-



In connectionless oriented service, the path is selected at the time of starting of the transmission. There is no fixed path.
The connectionless services includes
1. Select a Path [when the data is available in the source.
2. Data Transfer.

In the connectionless oriented service, the data packets are known as datagram packets.



Routing Table for 'A' :-

Destination Node	Next Node
A	N/A
B	B
C	C
D	B [or C]
E	C or B
F	B (or) C

Routing Table for B:-

Destination Node	Next Node
A	A
B	N/A.
C	A /E.
D	D
E	E
F.	D/E.

Routing Table for 'C':-

Destination Node	Next Node.
A	A
B	A/E.
C	N/A
D	E/A
E	E
F	E/.

Routing Table for 'D':-

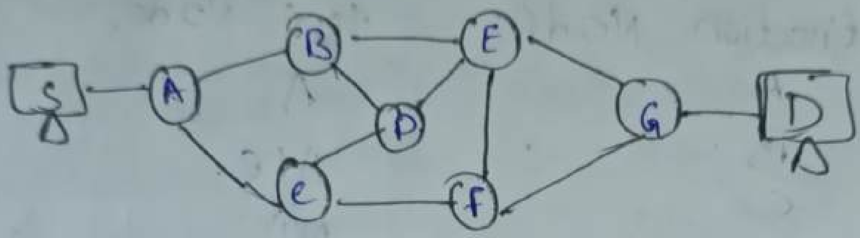
Destination Node	Next Node.
A	B
B	B
C	B/F.
D	N/A.
E	B/F.
F.	F.

Routing Table for 'E':-

Destination Node	Next Node.
A	C/B
B	B
C	C
D	F/B
E	N/A
F	F.

Routing Table for 'F':-

Destination Node	Next Node.
A	D/E
B	D/E
C	E
D	D
E	E
F	N/A.



Routing Table for 'A':.

Destination Node	Next Node.
A	N/A
B	B
C	C
D	B/C
E	C/B
F	C/B
G	C/B.

Routing Table for 'B':-

Destination Node	Next Node.
A	A
B	N/A
C	A/D
D	D
E	E
F	E/D
G	E/D.

Routing Table for 'C':-

Destination Node	Next Node
A	A
B	A/E
C	N/A
D	D ~~(or) E~~
E	D/F
F	F
G	F/D

Routing Table for 'D':-

Destination Node	Next Node
A	B/C
B	B
C	C
D	N/A
E	E
F	C/E
G	E/C

Routing Table for 'E':-

Destination Node	Next Node
A	B/F
B	B
C	D/F
D	D
E	N/A
F	F
G	G

Routing Table for 'F':-

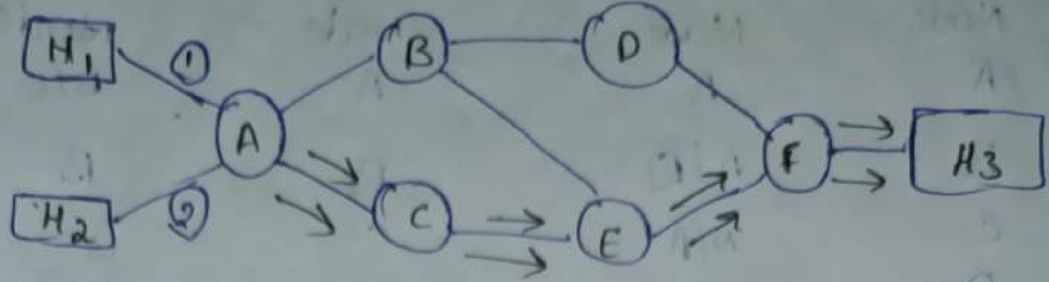
Destination Node	Next Node
A	C/E
B	E/C
C	C
D	E/C
E	E
F	N/A
G	G

Routing Table for 'G':-

Destination Node	Next Node
A	E/F
B	E/F
C	F/E
D	E/F
E	E
F	F
G	N/A

3/10/24 1. Connection - oriented

1.



Routings for 2 Path.

1. H1 - A - C - E - F → H3
2. H2 - A - C - E - F → H3

Routing Table for A's.

IN		OUT	
H1	1	C	1
H2	2	C	2

Routing Table for C's

IN		OUT	
A	1	E	1
A	2	E	2

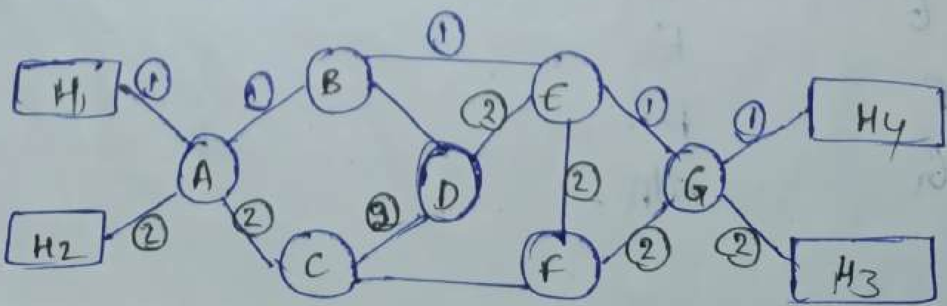
Routing Table for E's

IN		OUT	
C	1	F	1
C	2	F	2

Routing Table for F's

IN		OUT	
E	1	H3	1
E	2	H3	2

2.



Routing for 2 Paths

1. $H_1 - A - B - E - G - H_4$

2. $H_2 - A - C - D - E - F - G - H_3$.

Routing Table for A's

IN

H_1	1
H_2	2

OUT

B	1
C	2

Routing Table for B's.

IN

A	1
-	-

OUT

E	1
-	-

Routing Table for C's.

C	2

IN

D	2

OUT.

Routing Table for D'S

IN

C	2

OUT

E	2

Routing Table for E's.

IN

B	1
D	2

OUT

G	1
F	2

Routing Table for F's

IN

E	2
-	-

OUT

G	2
–	-

Routing Table for G's

IN

E	1
F	2

OUT

H_4	1
H_3	2

4/10/24 Routing Algorithm.

finding the optimal path from sourc to destination is called routing. The routi can be classified into 2 types-

1. Static Routing.
2. Dynamic Routing.

Static Routing :- It is also called non-adaptive routing. The route of the subnet does not changes with topology and Traffic (flow of data) is known as static routing.

Dynamic Routing:- It is also called adaptive routing. The route of the subnet changes with topology and Traffic (flow of data) of the network is called Dynamic routing.

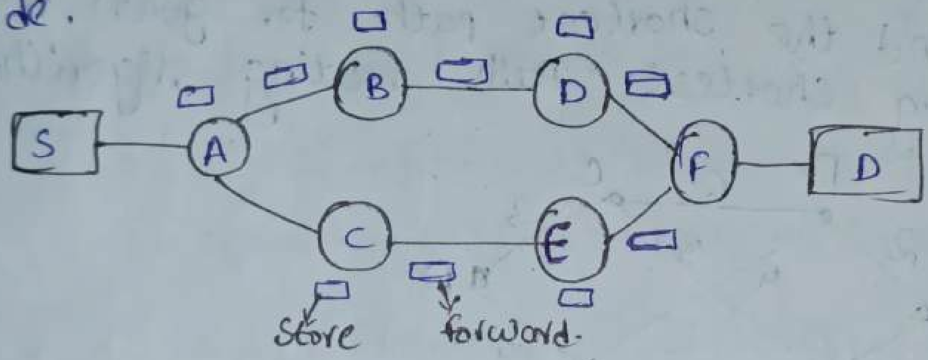
Differences between Static & Dynamic Routin

Static Routing.	Dynamic Routing
① In static routing, it set-up the optimal path manually from source to destination.	① It uses dynamic proto-cols to update the routing Tables and find the optimal path for between source & destinatio
② The router uses static Algorithm and do not have any control mechani-sm if any failures in the routing path.	② It uses dynamic Algorithms and identifie failure routers to find optimal path
③ It is suitable for very small networks.	③ It is suitable for large networks.

④ Static Routing is simplest way of routing to transmit the data from Source to Destination.	④ Dynamic routing uses complex algorithms for routing and transmit the data from source to destination
⑤ It has an advantage that it requires minimum of memory.	⑤ It requires more memory and depends on the type of the algorithm.

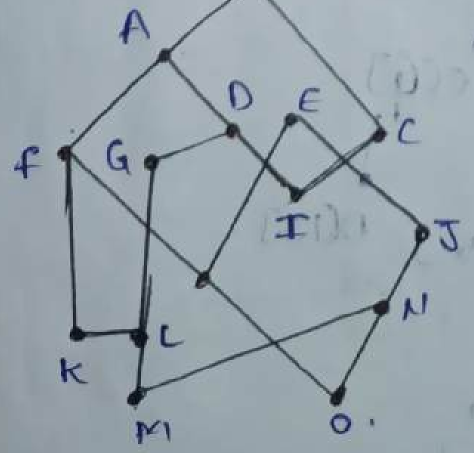
* Store and Forward Packet Switching

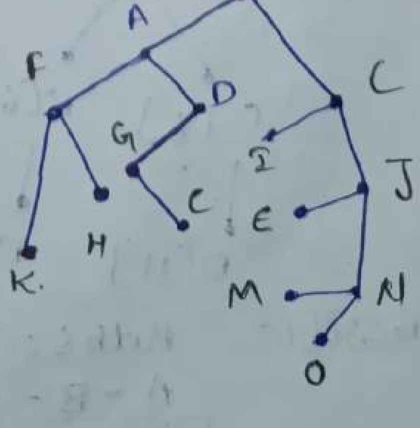
In this each node having a memory element It stores every Packet and forwards to the next node.

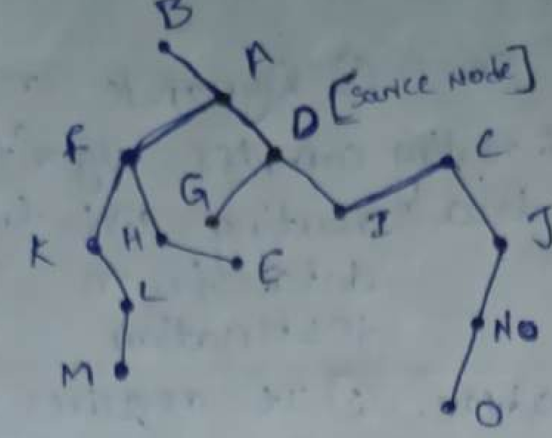


* optimality Principle :-

It states that the optimal route from all the Sources to a given destination and to form a 3 routed at a destination. This is called sync tree.







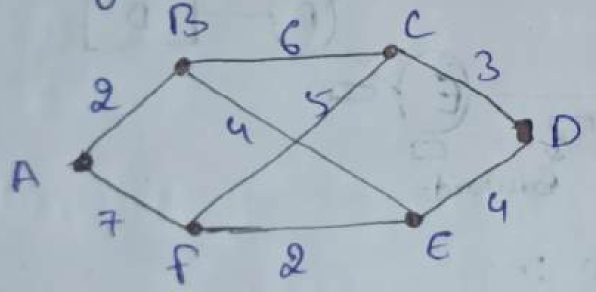
3/10/24

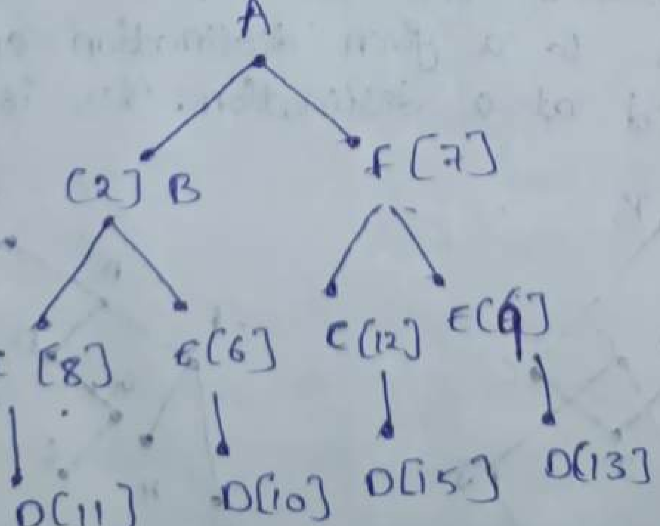
Routing Algorithm, classifications:-

1. Shortest Path routing Algorithm
2. Hierarichal routing Algorithm
3. Link - state routing Algorithm.
4. Distance - Vector routing Algorithm.

1. Shortest Path routing Algorithm :-

* find the shortest path for given network by using shortest Path routing Algorithm.





Possible Paths :-

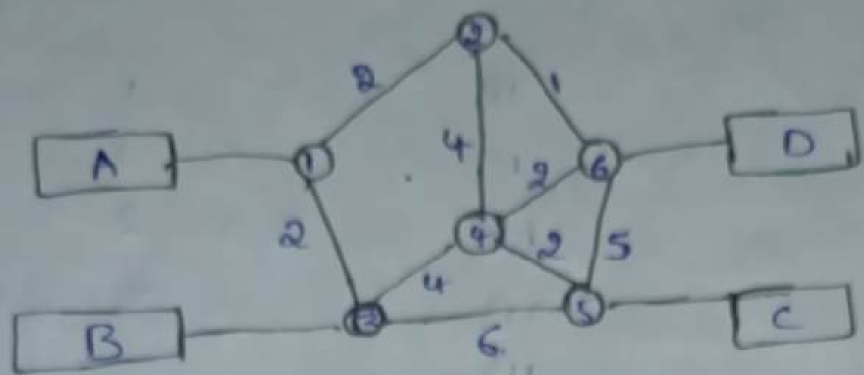
A - B - C - D = 11

$\boxed{A - B - E - D = 10}$

A - F - C - D = 15

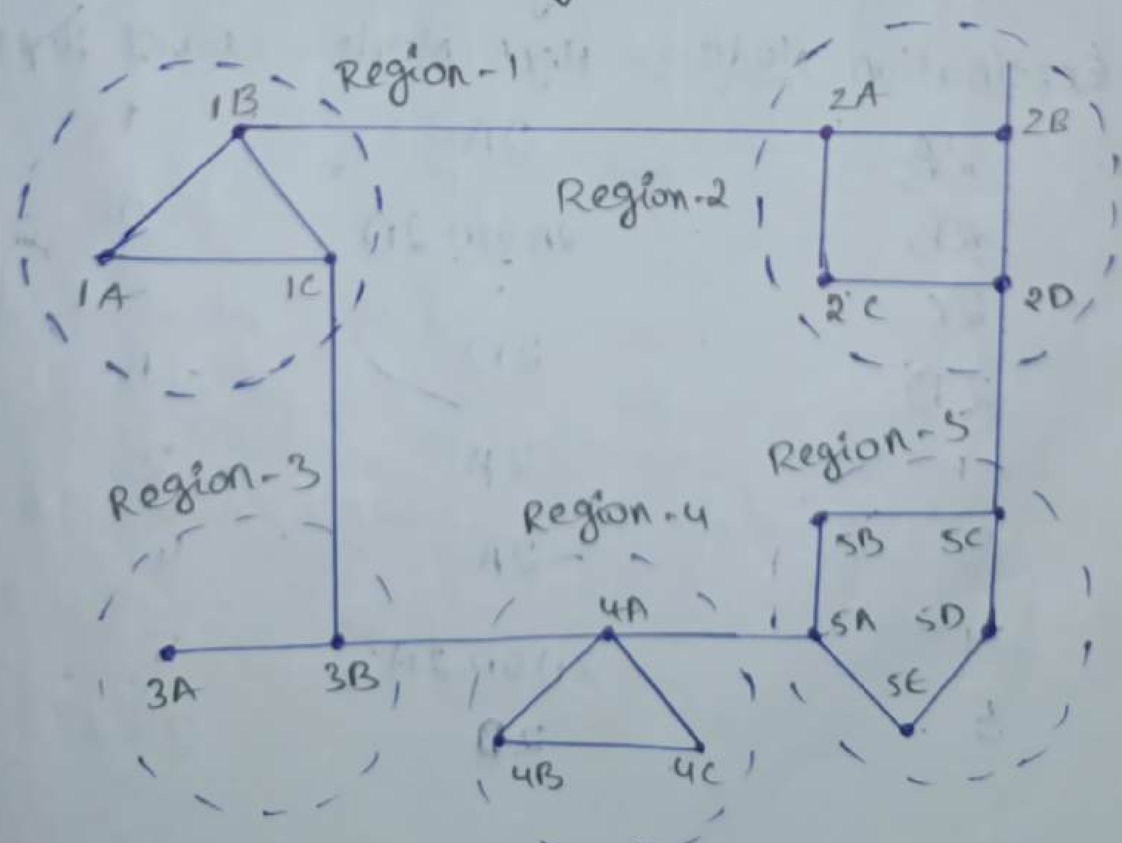
A - F - E - D = 13

→ Dijkstra's Algorithm :-



Iteration	Nodes	C(2) Path	C(3) Path	C(4) Path	C(5) Path	C(6) Path.
1.	{1}	2, {1-2}	2{1-3}	∞, -	∞, -	∞, -
2.	{1, 2}	2{1-2}	2{1-3}	6{1-2-4}	∞, -	3{1-2-6}
3.	{1, 2, 3}	2{1-2}	2{1-3}	5{1-2-6-4}	7. {1-2-6-4-5}	3 {1-2-6}
4	{1, 2, 3, 4}	2{1-2}	2{1-3}	5{1-2-6-4}	7{1-2-6-4}	3{1-2-6}
5	{1, 2, 3, 4, 5}	2{1-2}	2{1-3}	5{1-2-6-4}	7{1-2-6-4}	3{1-2-6}
6	{1, 2, 3, 4, 5, 6}	2{1-2}	2{1-3}	5{1-2-6-4}	7{1-2-6-4}	3{1-2-6}

2. Hierarichal Routing algorithm:-



Source Node is "1A" [Connection less oriented]

Destinat Node	Next Node	No. of Hopes.
1A	-	-
1B	1B	1
1C	1C	1
2A	1B	2
2B	1B	3
2C	1B	
2D	1B	4
3A	1C	3
3B	1C	2
4A	1C	3
4B	1C	4
4C	1C	4
5A	1C	4
5B	1C	5
5C	1B	5
5D	1B	6
5E	1C	5

Hierarchal Routing Tables for 2'C.

Destination Node	Next Node	No.of Hopes
2A	2A	1
2B	2A (or) 2D	2
2C	-	-
2D	2D	1
1	2A	2
3	2A	4
4	2D (or) 2A	5
5	2D	2

Hierarichal routing for 1C.

Destination Node	Next Node	No. of Hopes
1A	1A	1
1B	1B	1
1C	–	–
2	1B	2
3	3B	1
4	3B	2
5	3B	3.

Hievarichal Routing for 3B.

Destination Node	Next Node	No. of Hopes.
3A	3A	1
3B	–	–
1	1C	1
2	1C	3
4	4A	1
5	4A	2.

Heivarichal Routing for 4C

Destination Node	Next Node	No. of Hopes.
4A	4A	1
4B	4B	1
4C	–	–
1	4A	3
2	5A/4A	5
3	4A	2
5	4A	2

Hierarichal Routing for 5C.

Destination Node	Next Node	No. of Hopes
5A	5B	2
5B	5B	1
5C	–	–
5D	5D	1
5E	5D	2

1	2D	4
2	2D	1
3	5B	4
4	5B	3.

Hierarichal Routing for 1'A.

Destination Node	Next Node	No. of Hope
1A	-	-
1B	1B	1
1C	1C	1
2	1B	2
3	1C	2
4	1C	3
5.	1C	4

Hierarichal Routing for 2'A.

Destination Node	Next Node	No. of Hopes.
2A	-	-
2B	2B	1
2C	2C	1
2D	2B(or) 2C	2.
1	1B	1
3	1B	3
4	1B	4
5.	2B(or) 2C	3.

Hierarichal Routing for 3'A

Destination Node	Next Node	No. of Nodes.
3A	-	
3B	3B	1
1	1C	1
2	1C	3.

4	4A	1
5.	4A	2.

Hierarichal Routing for 4'A:

Destination Node	Next Node	No. of Hopes.
4A	-	
4B	4B	1
4C	4C	1
1	3B	2
2	3B (or) 5A	4
3	3B	1
5	5A	1.

Hierarichal Routing for 5'A.

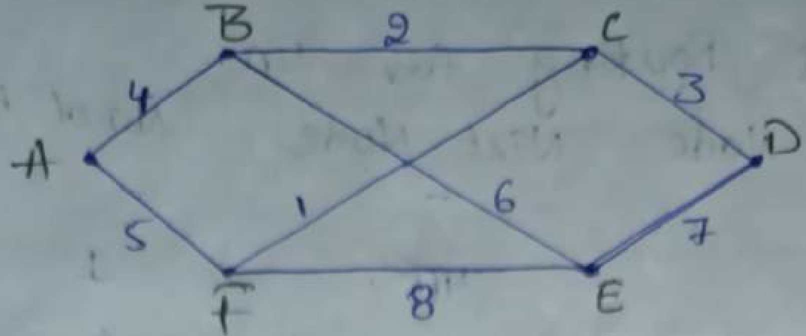
Destination Node	Next Node	No. of Hopes.
5A	-	
5B	5B	1
5C	5B	2
5B	5E	2
5E	5E	1.
1	4A	3.
2	5B	3
3	4A	2
4	4A	1.

14/10/24

3. Link-state Routing Algorithm:

Link-state Routing Algorithm is the concept of adding additional field to the routing table based on age and sequence of the packet.

* find shortest path for given network by using link-state Routing algorithm?



A's Routing.

Sequence no. of Packet	
Age of the Packet	
B	4
F	5

B's Routing.

Sequence No. of Packet	
Age of the Packet	
A	4
C	2
E	6

C's Routing.

Sequence No. of Packet	
Age of the Packet	
B	2
D	3
F	1

D's Routing.

Sequence No. of Packet	
Age of the Packet	
C	3
E	7

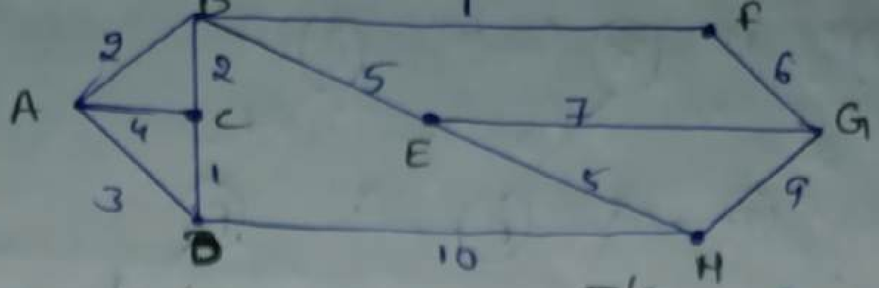
E's Routing

Sequence No. of Packet	
Age of the Packet	
B	6
D	7
F	8

F's Routing.

Sequence No. of Packet	
Age of the Packet	
A	5
C	1
E	8

2.



* A's

Sequence No. of Packet	
Age of the Packet	
B	2
C	4
D	3

* B's

Sequence No. of Packet	
Age of the Packet	
A	2
C	2
E	5
H	1.

* C's

Sequence No. of Packet	
Age of the Packet	
A	4
B	2
D	1

* D's.

Sequence No. of Packet	
Age of the Packet	
A	3
C	1
H	10

* E's

Sequence No. of Packet	
Age of the Packet	
B	5
G	7
H	5

* F's

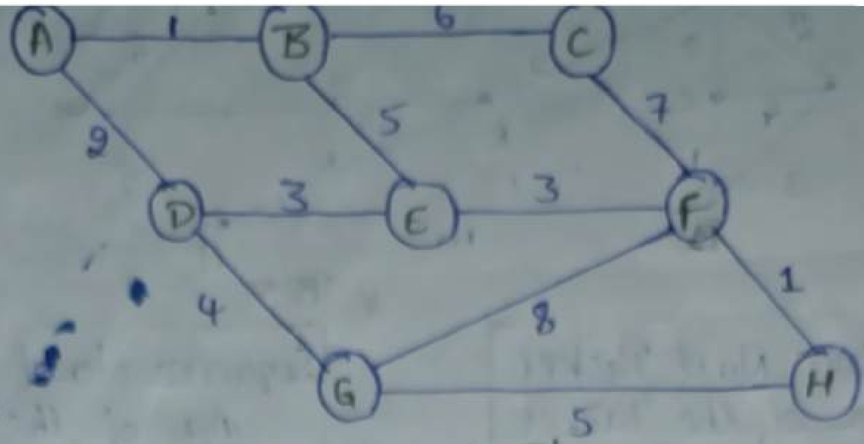
Sequence No. of Packet	
Age of the Packet	
B	1
G	6

* G's

Sequence No. of Packet	
Age of the Packet	
E	7
F	6
H	9

* H's.

Sequence No. of Packet	
Age of the Packet	
D	10
E	5
G	9



* A'S

Sequence No. of Packet	
Age of the Packet	
B	1
D	2

* B's.

Sequence No. of Packet	
Age of the Packet	
A	1
C	6
E	5

* C'S

Sequence No. of Packet	
Age of the Packet	
B	6
F	7

* D'S.

Sequence No. of Packet	
Age of the Packet	
A	2
E	3
G	4

* E'S

Sequence No. of Packet	
Age of the Packet	
B	5
D	3
F	3

* F'S

Sequence No. of Packet	
Age of the Packet	
E	3
G	8
H	1
C	7

* G'S

Sequence No. of Packet	
Age of the Packet	
D	4
F	8
H	5

* H'S

Sequence No. of Packet	
Age of the Packet	
F	1
G	8.

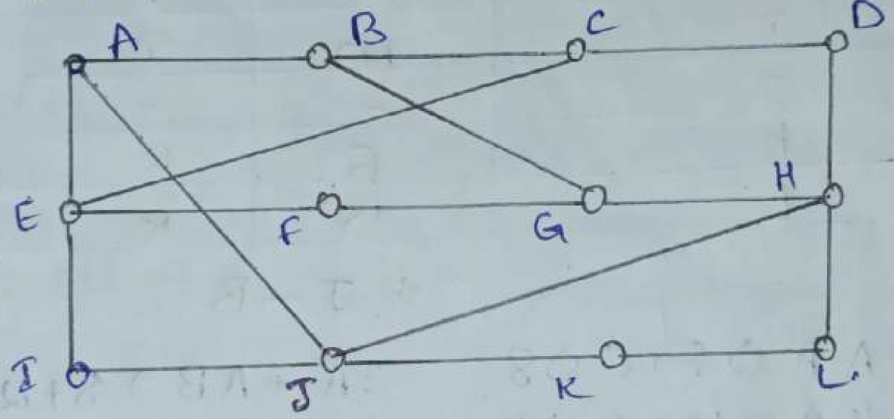
Distance vector Routing Algorithm.

* The router transmits distance vector of the own router to other neighbours.
* Each router recieves and stores the recieved distance vector from each of its neighbours.
* A router recalculates its distance vector only when

i, It recieves a distance vector from a neighbour containing different information.

ii, It discovers that a link to neighbour has disconnect.

* The distance vector calculation is based on minimum cost of possible combinations.



	A	I	H	K
A	0	24	20	21
B	12	36	31	28
C	25	18	19	36
D	40	27	8	24
E	14	7	30	22
F	23	20	19	40
G	18	31	6	31
H	17	20	0	19
I	21	0	14	22
J	9	11	7	10
K	24	22	22	0
L	29	33	9	9

The delay for J to A → 8

The delay for J to I → 10

The delay for J to H → 12
The delay for J to K → 6

J's Routing Table

Destination Node	Delay	Next Node
A	8	A
B	20	A
C	28	I
D	20	H
E	17	I
F	30	I
G	18	H
H	12	H
I	10	I
J	–	–
K	6	K
L	15	K

* J–A

JA + A.A ⇒ 8 + 0 ⇒ 8
JI + IA ⇒ 10 + 24 ⇒ 34
JH + HA ⇒ 12 + 20 ⇒ 32
JK + KA ⇒ 6 + 21 ⇒ 27

* J – B

JA + A.B ⇒ 8 + 12 ⇒ 20
JI + I.B ⇒ 10 + 36 ⇒ 46
JH + HB ⇒ 12 + 31 ⇒ 43
JK + K B ⇒ 6 + 28 ⇒ 34

* J – C

JA + AC ⇒ 8 + 25 ⇒ 33
JI + IC ⇒ 10 + 18 ⇒ 28
JH + HC ⇒ 12 + 19 ⇒ 31
JK + KC ⇒ 6 + 36 ⇒ 42

* J – D.

JA + AD ⇒ 8 + 40 ⇒ 48
JI + ID ⇒ 10 + 27 ⇒ 37
JH + HD ⇒ 12 + 8 ⇒ 20
JK + KD ⇒ 6 + 24 ⇒ 30

* J – E

JA + AE ⇒ 8 + 14 ⇒ 22
JI + IE ⇒ 10 + 7 ⇒ 17
JH + HE ⇒ 12 + 30 ⇒ 42
JK + KE ⇒ 6 + 22 ⇒ 28

* J – F

JA + AF ⇒ 8 + 23 ⇒ 31
JI + IF ⇒ 10 + 20 ⇒ 30
JH + HF ⇒ 12 + 19 ⇒ 31
JK + KF ⇒ 6 + 40 ⇒ 46

* J-G

JA + AG ⇒ 8 + 18 ⇒ 26
JI + IG ⇒ 10 + 31 ⇒ 41
JH + HG ⇒ 12 + 6 ⇒ 18
Jk + kG ⇒ 6 + 31 ⇒ 37

* J-H.

JA + AH ⇒ 8 + 17 ⇒ 25
JI + IH ⇒ 10 + 20 ⇒ 30
JH + HH ⇒ 12 + 0 ⇒ 12
Jk + kH ⇒ 6 + 19 ⇒ 25

* J-I

JA + AI ⇒ 8 + 21 ⇒ 29
JI + II ⇒ 10 + 0 ⇒ 10
JH + HI ⇒ 12 + 14 ⇒ 26
Jk + kI ⇒ 6 + 22 ⇒ 28

* J-K

JA + Ak ⇒ 8 + 24 ⇒ 32
JI + Ik ⇒ 10 + 22 ⇒ 32
JH + HK ⇒ 12 + 22 ⇒ 34
JK + KH ⇒ 6 + 0 ⇒ 6

* J-L

JA + AL ⇒ 8 + 29 ⇒ 37
JI + IL ⇒ 10 + 33 ⇒ 43
JH + HL ⇒ 12 + 9 ⇒ 21
Jk + kL ⇒ 6 + 9 ⇒ 15

* Congestion :-

The heavy flow of data packets is known as congestion.. It is also known as traffic of network. Because of the traffic of any network, the information has to be trans-mitted may not send accurately. To transmit accurate information we need to reduce the traffic.

Basically, the traffic is mainly occured in data link layer, Network layer and Transport Layer of OSI model.

General Policies of Congestion Control.

Data link layer:

1. Retransmission Policy.
2. Acknowledgement Policy.
3. Flow control Policy.
4. out of order detection Policy.

Network Layer

1. Virtual circuit subnet and Datagram circuit Subnet.
2. Packet queue and Packet discard Policy.
3. Routing Algorithms. policy.
4. Packet lifetime Management Policy.

Transport layer.

1. Retransmission Policy.
2. Timeout determination Policy.
3. Acknowledgement Policy.
4. flow control Policy.

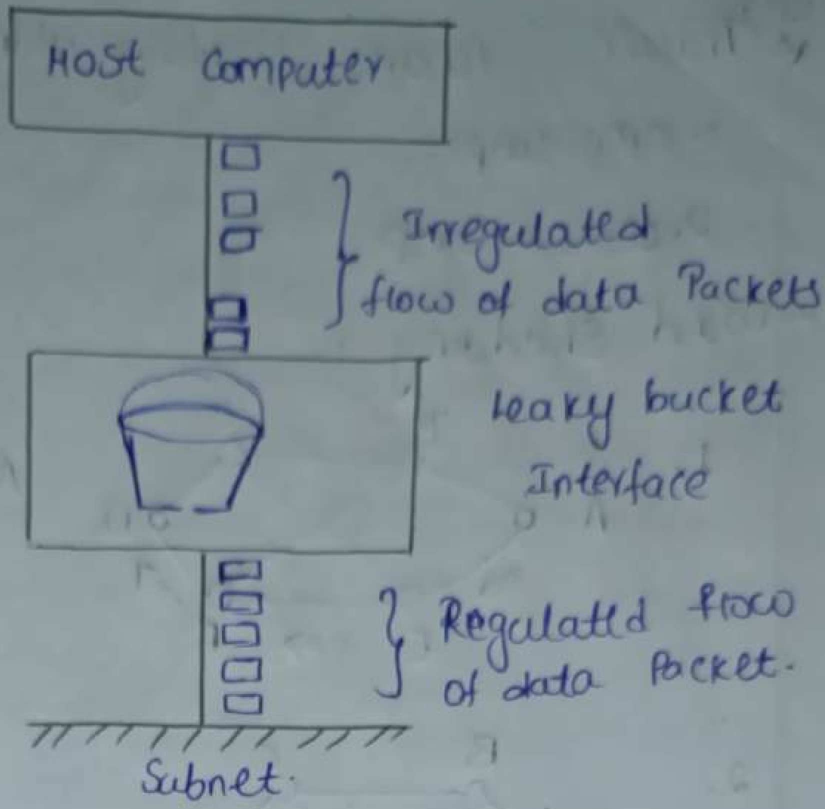
* Congestion Control Algorithms :-

To control the flow of data packets (or) traffic in a network. we need to use Congestion control Algorithms. There are 2 types

1. Leaky Bucket Algorithm.
2. Token bucket Algorithm.

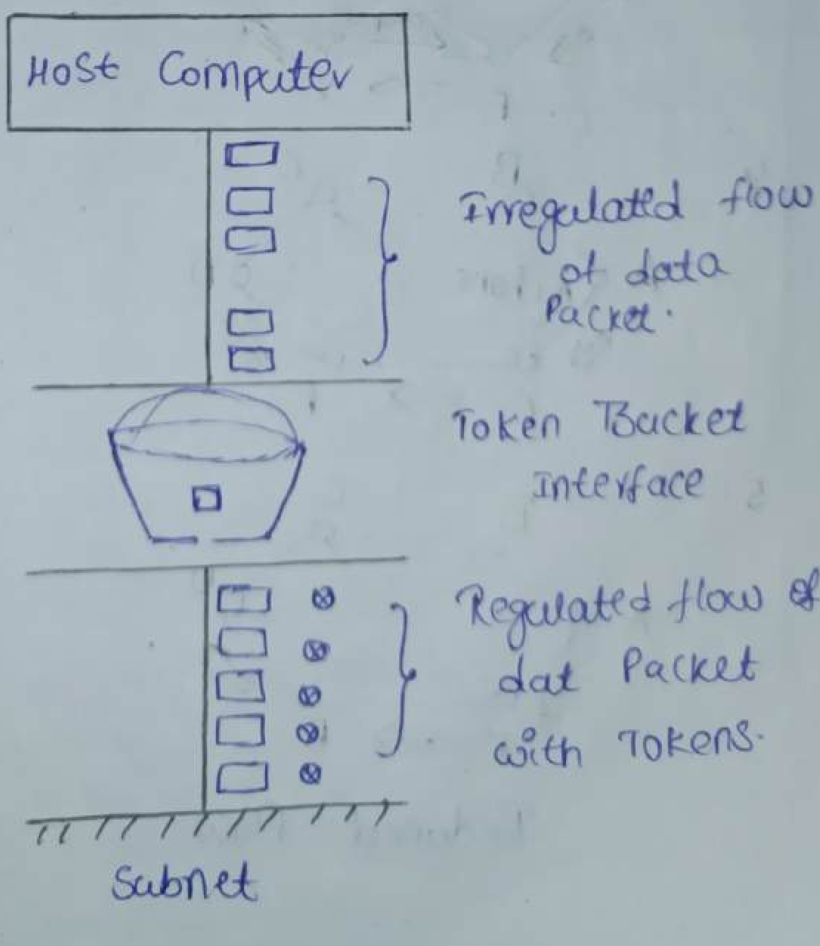
1. Leaky Bucket Algorithm :-

It converts irregulated flow of Packets into regulated flow of packets.



2. Token Bucket Algorithm :-

The interface generates the token for every ~~Packet at~~ second by adding this tokens to the packets we are providing security to the data.



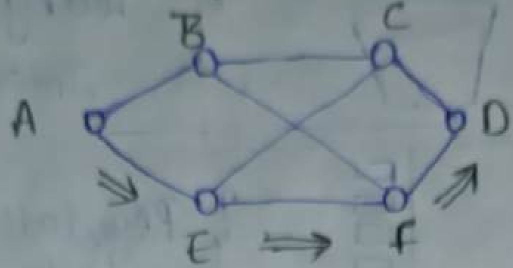
21/10/24

* Traffic Aware Routing Admission Control

1. Open loop.
2. Closed loop.

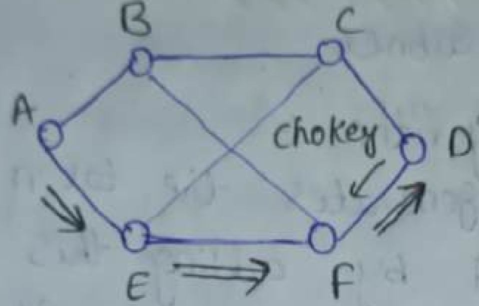
→ Load Shedding :

1.

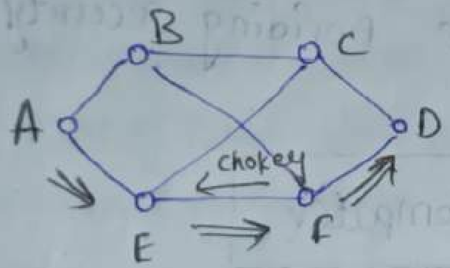


Next msg choke Packet

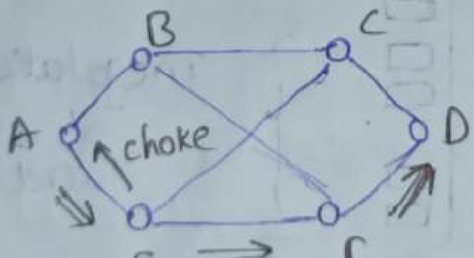
2.



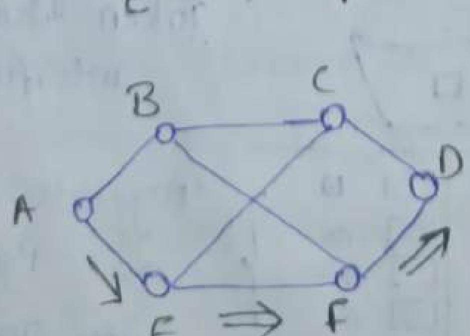
3.



4.

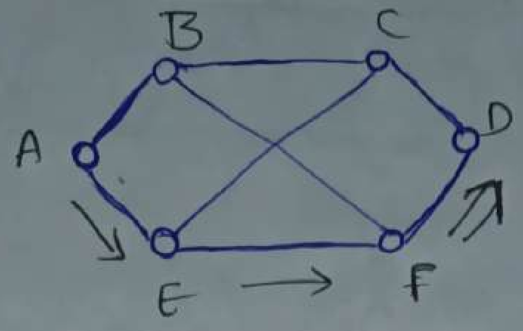


5.



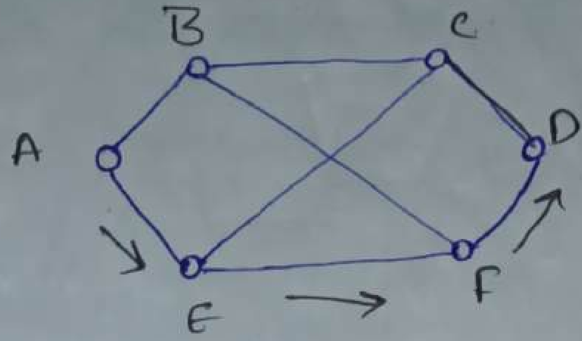
Reduced flow.

6.



flow is still reduced & maximum rate

7.



flow is totally reduced.

→ Admission Control.

1. flow of data
2. Buffers
3. Discard Packet.

24/10/24

UNIT-5

Transport Layer and Application Layer

* **Transport layer** :- Transport layer is responsible for Process to process deeivary of data. It is also responsible for End to End delivary of data.

Services of Transport layers :-

→ Port Addressing.
→ Connection Control.
→ Segmentation.
→ Reassembeing.
→ Error control.

Protocols

→ TCP Transmission Control Protocol connection oriented
→ UDP user Datagram protocol. Connectionless oriented

TCP Header Format.

<table>
<tr><td colspan="4">Source Port Number [16 bits]</td><td colspan="5">Destination Port Number [16 bits]</td><td rowspan="6">20 bytes</td></tr>
<tr><td colspan="9">Sequence Number [32 bits]</td></tr>
<tr><td colspan="9">Acknowledgement Number [32 bits]</td></tr>
<tr><td>Header Length 4 bits</td><td>Reserve 6 bits</td><td>URG</td><td>SYN</td><td>RST</td><td>ACK</td><td>PSH</td><td>FIN</td><td>window size 16 bits.</td></tr>
<tr><td colspan="8">optional (variable)</td><td>check sum 16 bits</td></tr>
<tr><td colspan="8">Data [variable]</td><td>urgent Pointer 16 bits</td></tr>
</table>

← 32 bytes →

Header :- The length of the header is 20-60 bytes minimum length is 20 bytes and maximum of 60 bytes.

Source Port Address :-

The length of this field is 16 bits here we represent the source port numbering.

Destination Port Address :

The length of this field is 16 bits here we represent the destination port numberi

Sequence Numbering :

The length of this field is 32 bits. It is the serial number of frame (or) small dat units. [→ x]

Acknowledgement Number :-

The length of this field is 32 bits It identif -es Acknowledgement [→ x+1]

Header length :-

The length of this field is 4 bits. It gives the information about the length of the header The minimum length of header is 20 bytes & maximum length of header is 60 bytes.

Reserved :-

The length of this field is 6 bits. They are used for future purpose.

URG :- Urgent Pointer

The URG flag is set to be '1'. When we have highest priority. otherwise it is '0'.

ACK :- Acknowledgement

If that frame having acknowledgement then it will be set to '1'. [delivary report]

PSH :- Push the Data immediate transfer to source.

RST :- Reset the connection.

SYNC :- Synchronisation.

FIN :- finish (or) Terminate the Condition

These 4 flags are used in Connection establishment in TCP.

Window Size :- The length of this field is 16 bits. The total length of the TCP frame. It Consists of both header length and data length. Both are combined, we can get window size [capable of Rx].

Check sum :-

The length of this field is 16 bits. It is used for controlling of Errors.

URG :- Urgent Pointer. [priortised]

The length of this field is 16 bits. The range of highest priority data depend on URG.

$Sx \rightarrow Rx$

SYN = 1

ACK = 0

$Sx \leftarrow Rx$

SYN = 0

ACK = 1.

Protocol	Description	Port Number
DNS	Domain Name server/system	-
HTTP	Hyper Text Transfer Protocol	-
FTP	file Transfer Protocol	-
ECHO	Echo is recieved datagram back to sender	7
DISCARD	Discard any datagram that is recieved	9
USERS	Active users	-
DAYTIME	Returns the Day and Time	[illegible]

Protocol	Description	Port Number
CHAR. GEN	Returns the string characters	-
Name SERVER	Domain Name Service.	53.
BOOT STRAPPING	server Port to download bootstrap information	67
BOOT PC	claim Port to download boot strap	68
TFTP	Trival five Transfer Protocol	69
RPC	Remote Procedure Call	111
SNMP	Simple Network Managem-ente Protocol	161
NTP	Network Time Protocol	123.

28/10/24

UDP Header format

Source Port Number [16 bits]	Destination Por Number [16 bits]
Length of Header [16 bits]	Check sum [16 bits]

Protocol	Description	Port Number
DISCARD	Discard any data gram that is recieved	9
TFTP	Trival file transfer Protocol	69
RPC	Remote Processor Call	111
BOOST STRAPPING	Server Port to download Boost's strap information	67

Protocol	Description	Port Number
BOOT PC	Client Port Boot strap information.	68
NAME SERVER	Domain Name service	53
NTP	Network Time Protocol.	123.
SNMP	simple Network manage--ment Protocol	161

Source Port address :- The length of this fied is 2 bytes → 16 bits . It gives source Port Number.
Destination Port address :- The length of this field is 2 bytes . It gives destination Port Number
length of Header :- The length of header field is 2 bytes i.e 16 bits . It gives the length of header and length of data
Check sum :- Thength of this field is 2 bytes i.e 16 bits . It is a error controlling technique.

Application Layer :-

In this layer, it is used to decrypting the encrypted data. In this layer, the data is converted to user readable data.

Responsibilities :-

1. Email service
2. FTMA
3. Directory service.

$\underset{\text{world wide web}}{\underline{www}}$: $\underset{\text{domain}}{\underline{APPSC}}$. $\underset{\text{sector}}{\underline{gov}}$. $\underline{in}$ country.

Last to first → In searching process.

web server stores web site.

Domain Name service [DNS] :- It is used to search the information from server. Domain Name service system is translated to an IP address.

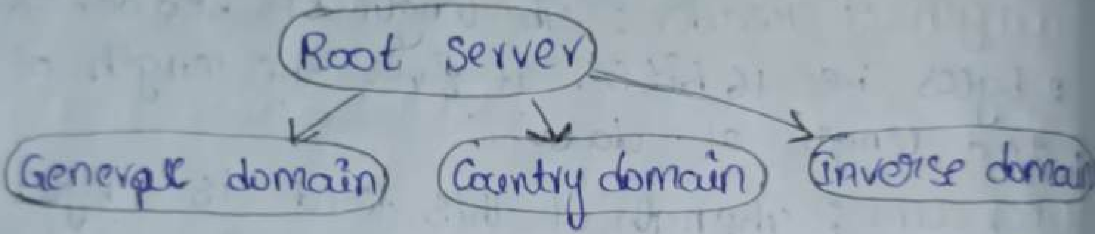
There are 4 types of services in DNS.

1. Domain Name.
2. Name server
3. Name space
4. Name resolver.

1. Domain Name :-

It is unique, easy to remember name that is mapped with web IP address.

It is classified into 3 types.



All 3 character domain comes under General domain.

S.No	Label	Description
1.	.com	commercial organisation
2.	.org	Non Point organisation.
3.	.edu	education organisation.
4.	.gov	Government organisation
5.	.Int	International organisation.
6.	.mil	military group.
7.	.NET	Network support counters.

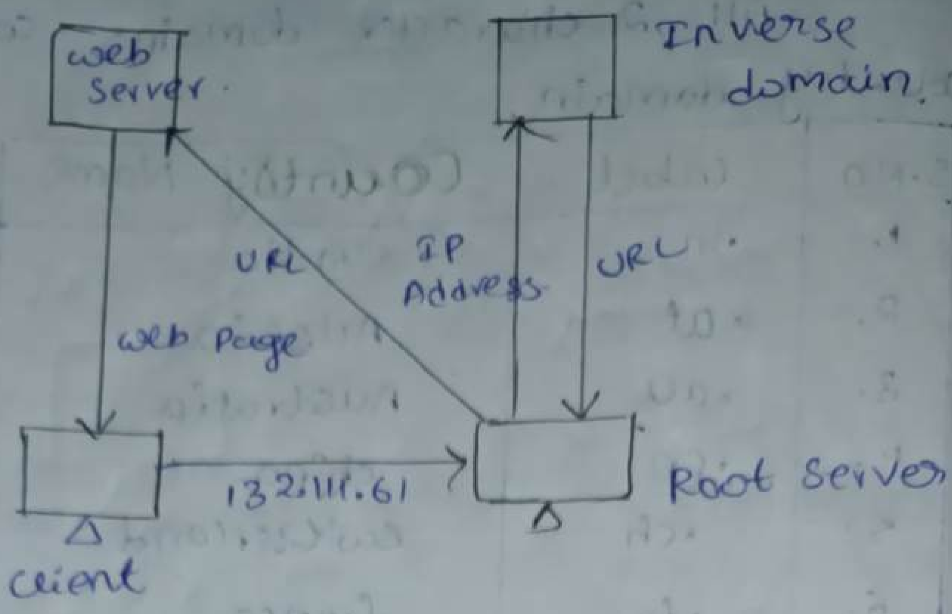
2. Country Domain :-

All 2 character domains comes under country domain.

S.NO	label	Country Name
1.	.in	India
2.	.at	Austria
3.	.au	Australia
4.	.cn	china.
5.	.ch	switzerland
6.	.fr	france
7.	.hk	Hongkong
8.	.be	brazil
9.	.ca	canada
10.	.de	Germany.
11.	.es	Spain.
12.	.il	israel.
13.	.it	Italy.
14.	.jp	Japan
15.	.kr	South korea.
16.	.nz	Newzland.
17.	.pt	Portugal
18.	.qa	qatar.
19.	.ru	Russia
20.	.us	united states
21.	.uk	united kingdom.
22.	.zw	Zimba

Inverse Domain :-

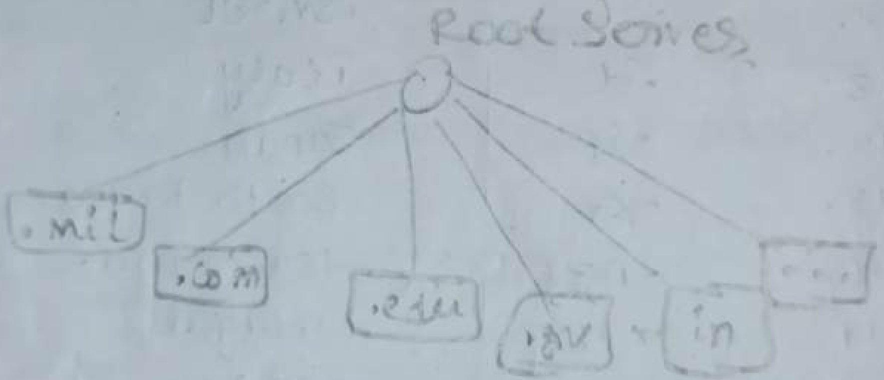
The purpose of inverse domain is to map ip address with name. The list contains only ip address. The server sends query to the inverse domain and request for mapping addresses to name.



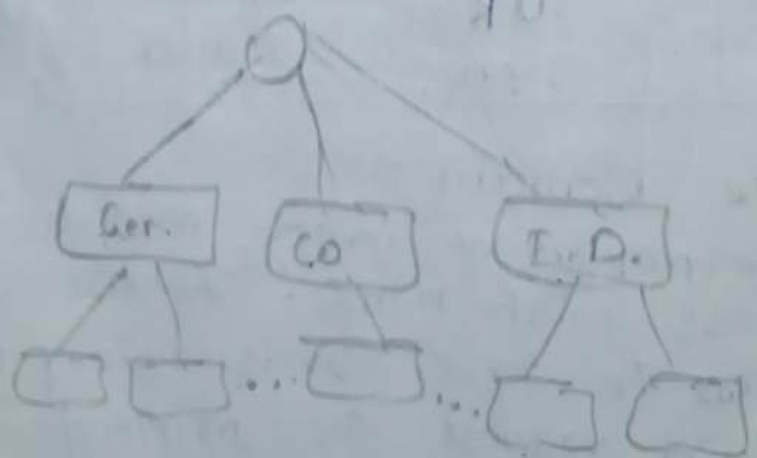
2. Name Space

1. flat Name Space
2. Hierarical Name Space.

1. flat Name Space.



2. Hierarichal Name Space.



There are 2 types. of Name spaces.

1. flat Name Space :-

In flat name space, all the domains are directly connected with the root server

2. Hierarichal Name Space :-

In the hierarichal name space, root server is connected with TOP level domains and it is connected with secondary level domain

3. Name Server

Primary server:- It keeps a file about the zone for which it is responsible and have authority.

secondary server:- It loads all information from primary server.

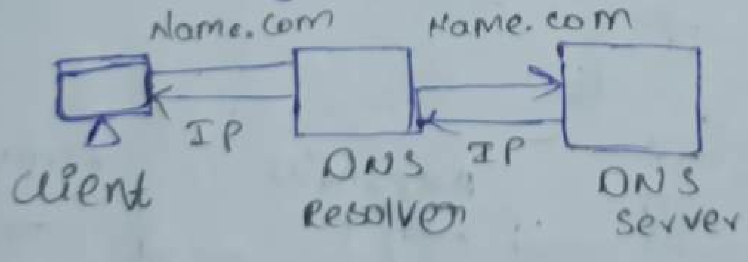
→ Address Records

→ Name Records.

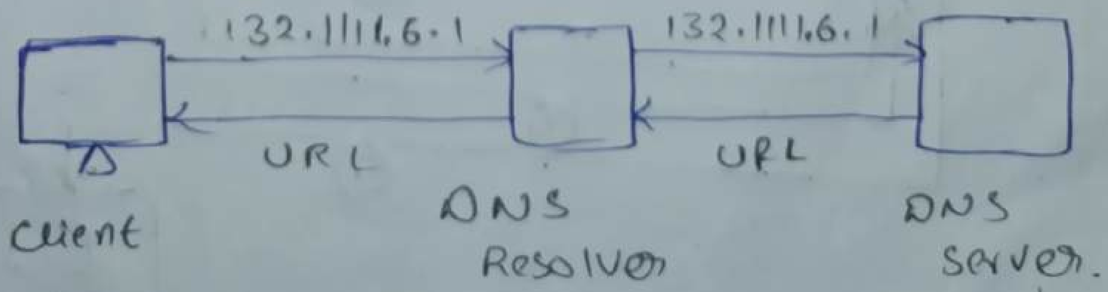
→ Email Exchange Record.

4. Name Resolver.

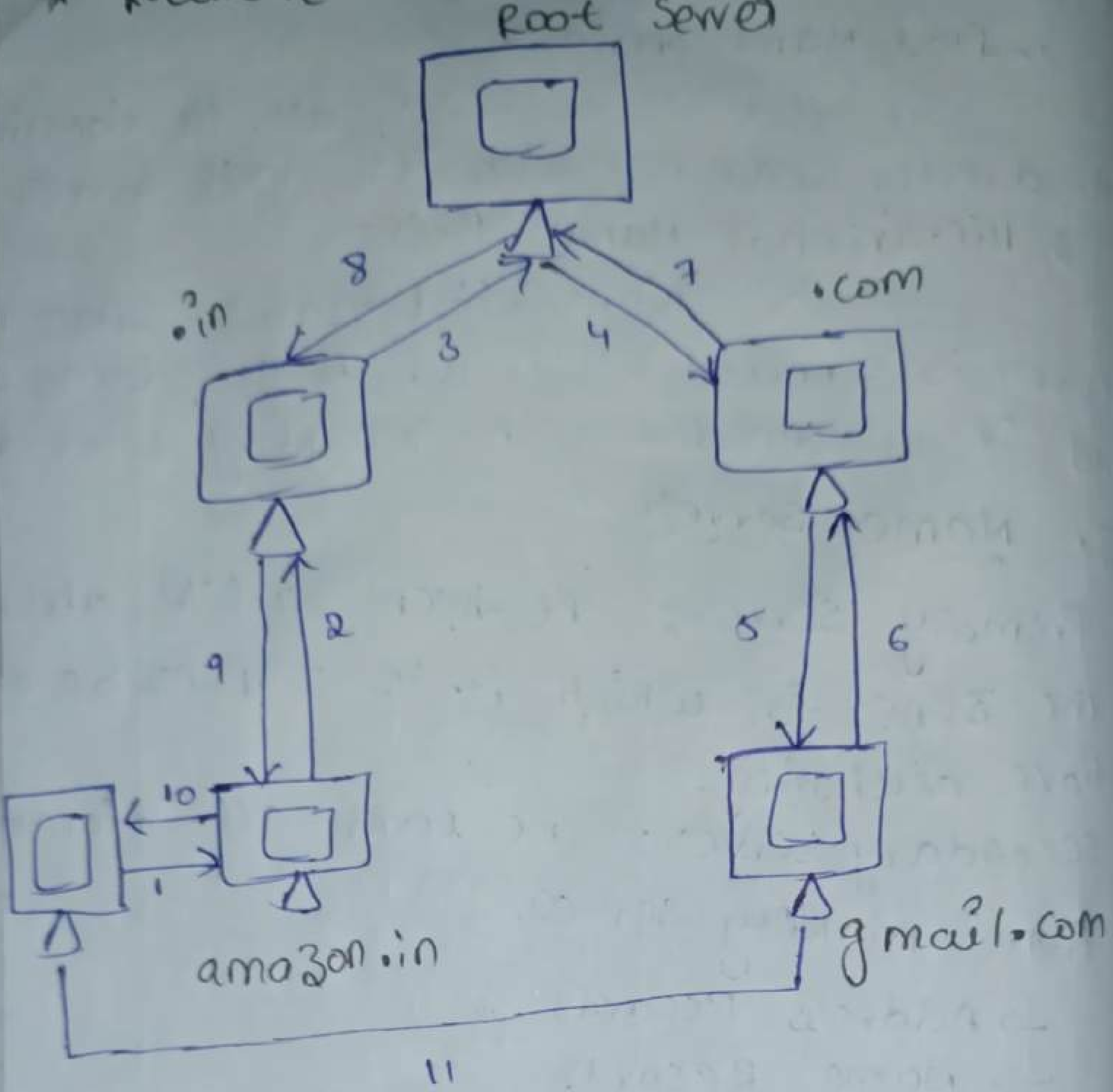
mapping Name to IP Address



mapping IP to Name.



* Recursive Method.



* Iteration Method.